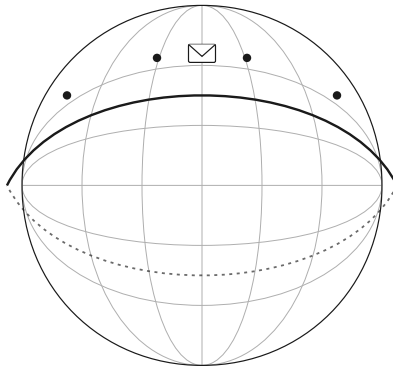


THE CURIOUS FAMILY LIBRARY

How the World Got Wired

A BRIEF HISTORY OF THE INTERNET

— *for young readers* —



MEIRA DAVIDSON

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For the Jessica family kids —
be curious, stay open, be kind.

How the World Got Wired

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Prologue — A Sunset in Da Nang



A grandmother filming a ten-second sunset on a beach in Da Nang.

Before I tell you how the internet works, I want to tell you about something that happens every day, in a way that would have seemed impossible only thirty years ago.

It begins with your grandmother, on a beach, on the other side of the planet.

She is on vacation in a city called Da Nang, in Vietnam, on the other side of the planet from where you are right now. She's standing on a beach at sunset. The sky is pink. There are little boats on the water. She holds up her phone and presses a button, and the phone starts filming.

She films for about ten seconds. A wave comes in. A child laughs somewhere off-camera. The pink in the sky gets a little deeper. Then she stops filming.

She taps her phone a few more times, and uploads the video to Facebook so her family back home can see it.

Now. Stop and think about what just happened.

Your grandmother is standing on a beach on the other side of the planet from you. Between her phone and your phone there is an entire ocean. There are mountains. There are deserts. There are eleven time zones. The video she just took — a real video, with light and color and sound, ten whole seconds of the world — needs to somehow cross all of that and arrive in your bedroom.

In my childhood, it could not have arrived. Not in ten seconds, not in ten days, not in ten weeks. There was no way to do it. If your grandmother had wanted to share a sunset on a beach in Vietnam with her family in another country in 1990, she would have taken a photograph with a camera that used film, waited until she got home, brought the film to a shop to be developed into pictures, picked them up a few days later, put one of them in an envelope, written an address on the envelope, taken it to a post office, paid for a

stamp, and waited three weeks for it to arrive. The sunset would already be a month old by the time anyone saw it. And only one person would see it, because there was only one photograph.

You are about to watch the same sunset, in motion, with sound, less than one second after your grandmother stopped filming. So will her sister in Toronto. So will her cousin in Sydney. So will an old friend of hers in Paris. All at once. All for free.

How?

That is the question this book is going to answer.

The short version is this: your grandmother's phone, the moment she pressed that button, did something a little strange. It took her ten-second video and chopped it up into thousands of tiny pieces. Each piece is called a *packet*. Each packet is numbered, like a page in a book: *I am piece number 47 out of 12,330*. Each packet is also stamped with an address — not a street address, but a numerical address that tells the internet where to send it. The address belongs to a giant computer that Facebook owns, sitting in a building somewhere in America.

Then her phone threw all those numbered pieces into the air.

Well, not quite *the air*. Her phone sent them, by radio, to a little box in her hotel room called a *router*. The router sent them, through a wire, to a bigger router somewhere in Da

Nang. That router sent them to an even bigger one in Ho Chi Minh City. From there, the packets traveled along a fiber-optic cable — a strand of glass thinner than a hair, with laser light pulsing through it — out of Vietnam, under the South China Sea, through Singapore, and then *down*. Down to the bottom of the Pacific Ocean, onto a cable lying on the sea-floor that stretches almost six thousand miles to California.

The packets crossed the entire Pacific Ocean in less than a tenth of a second.

They surfaced in California. They were sorted by another router. They were sent across the United States — through Nevada, through Utah, through Wyoming — and arrived at the Facebook building, which is in Oregon, or maybe Iowa, or maybe Virginia, depending on which one was least busy. The building is the size of an aircraft carrier. It has no windows. It hums.

Inside the building, a computer caught all 12,330 of the packets, checked the numbers to make sure none of them got lost on the way (a few always do — they go back and ask Vietnam to send the missing ones again), and stitched the pieces back together into a ten-second video.

Then it waited.

When you opened Facebook on your phone, *your* phone sent its own packets — little notes that said *hello, I am here, do you have anything for me?* The Facebook computer said *yes, your grandmother just sent something*. And then the whole journey happened again, in reverse, in a different direction. The

video was chopped back into packets, sent across America, dropped into a cable on the floor of the Atlantic Ocean (or the Pacific, if you live in California), and pulled up onto land near you. Then through a router in your city. Then through a router in your neighborhood. Then through the little box in your living room. Then through the air, by radio, to your phone.

You tapped the video. It played. There was your grandmother. There was the sunset. There was the wave.

It took less than two seconds.

In those two seconds, the packets that made up your grandmother's video traveled, in total, somewhere between fifteen and twenty thousand miles. They passed through dozens of computers owned by dozens of companies in at least four countries. They dove to the bottom of an ocean and came back up. They were checked, sorted, and reassembled by machines that have no idea what a grandmother is, or what a sunset is, or why you might want to see one.

Nobody planned your grandmother's video specifically. Nobody at Facebook woke up that morning and said *we'd better get ready, she's about to film something*. The whole enormous system was just sitting there, the way a highway sits there, ready for anyone who shows up. She showed up. The system carried her sunset across the world.

How did we build a system like that?

Who paid for the cables on the seafloor? Who owns them? Who decided that a video should be cut into 12,330 numbered pieces, instead of just sent in one big chunk? Who invented the idea that computers in different countries could talk to each other at all?

Most of those questions have answers. The answers involve a war, a satellite, a man named Licklider, a 22-year-old in Illinois, a milkman from Mississippi, a cable ship in the North Atlantic, a Canadian scientist who didn't want to take the Navy's money, and a lot of other people you've never heard of.

That's where we're going next.

Chapter 1 — A Memex and a Satellite

IN THE SUMMER OF 1945, AN AMERICAN SCIENTIST NAMED Vannevar Bush sat down to write an essay.

The Second World War had just ended. Bush had spent the previous five years running the most important science office in the United States, called the Office of Scientific Research and Development. He had been the man who told President Franklin Roosevelt, in a famous meeting in 1940, that America needed to start a serious atomic research program — and when Roosevelt agreed, Bush helped make it happen. The bomb that ended the war was, in a sense, his bomb. Or at least, he had helped pay for it.

Now the war was over. Bush was tired. And he was worried.

He was worried about a particular problem that most people did not think was a problem at all. The problem was that science was producing too much information.

Every year, in laboratories around the world, scientists wrote tens of thousands of papers. Each paper described a small new discovery — a new chemical, a new equation, a new way of doing something. The papers were published in journals. The journals piled up in libraries. There were so many papers, in so many journals, in so many libraries, that no human being could possibly read them all. A chemist in Boston might spend years working on a problem that a chemist in Berlin had already solved, and never know it, because the Berlin chemist's paper was sitting on a shelf in a library nobody had visited.

Bush thought this was a tragedy. So in July of 1945, he wrote an essay called *As We May Think*, in a magazine called *The Atlantic Monthly*, and in it he described a machine.

He called the machine the *memex*.

THE MEMEX



Vannevar Bush's imagined memex (1945) — a desk that could link any document to any other.

The memex would be the size of a desk. You would sit at it. Inside the desk, on tiny strips of film, would be stored every book, every article, every photograph, every letter you had ever wanted to keep. You would press a button, and the machine would find you any document in a second. That was useful — but it was not the magical part.

The magical part was that you could *connect* documents to each other.

If you were reading an article about birds, and the article reminded you of a poem you had read years before, you could press a button and *link* the article to the poem. From then on, anyone using your memex could jump from the art-

icle to the poem instantly. Then, if the poem reminded you of a painting in another book, you could link the poem to the painting. Soon you would build up *trails* — paths of connected ideas, running through everything you had ever read.

You could share your trails with other people. A scientist could share her trail through the chemistry of the human body. A historian could share his trail through a century of letters between two presidents. A child could share her trail through the books that had made her cry.

Knowledge, Bush wrote, would stop being a pile of separate things. It would become a *web*.

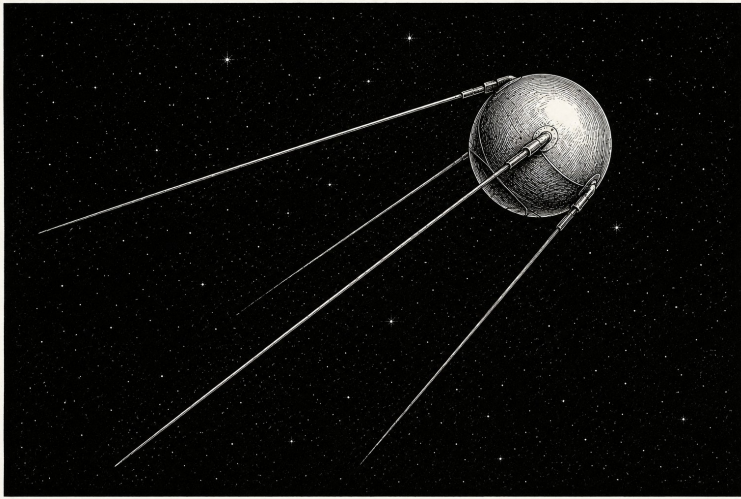
Vannevar Bush never built the memex. The technology of 1945 was not nearly good enough. He was describing a machine that wouldn't really be possible for another fifty years. But he had drawn a picture of it, in words, in a magazine that a lot of curious young people read. They remembered it. They grew up. Some of them became scientists. And when the time came to build something that actually worked the way Bush had imagined, they remembered the picture.

One of the young people who read Bush's essay was a man named J.C.R. Licklider.

You'll meet him in the next chapter.

But before we get to Licklider, we have to talk about a beach ball.

THE BEACH BALL



Sputnik, the polished metal sphere whose beep panicked an entire country.

On the night of October 4, 1957, a small metal sphere — polished to a mirror finish, about the size of a beach ball, weighing about 184 pounds — separated from the top of a rocket high above the Earth. It had four whip-like radio antennas trailing behind it. It was now in orbit, which meant it was going to circle the planet, over and over, until something pulled it down. On a clear evening, in the right kind of twilight, you could actually look up and see it — a tiny moving star, much faster than any real star should move.

It was called Sputnik. The word means *fellow traveler* in Russian.

The Soviet Union had launched it.

Inside Sputnik was a simple radio transmitter, designed to send out a beep on a frequency that almost any shortwave radio could pick up. *Beep. Beep. Beep.* Every fraction of a second, for three weeks, until the batteries ran out.

Sputnik's signal was on a frequency that almost any short-wave radio could pick up. All over the world — kids with little radios, men in basements, scientists in observatories — people turned their dials, found the right frequency, and listened to the Russian beach ball announcing itself as it passed over their countries every 96 minutes.

In America, this was a catastrophe.

Not because Sputnik itself was dangerous. Sputnik was just a beeping ball. It couldn't do anything except beep. But the *rocket* that had put Sputnik into orbit was the same kind of rocket that could carry a nuclear bomb. The Soviet Union had just demonstrated, in front of the whole planet, that they could launch a heavy object into space and send it precisely to where they wanted.

If they could do that to a beach ball, they could do it to a city.

Worse — much worse, in the minds of many Americans — the Soviets had done it *first*. The United States had been working on its own satellite. It wasn't ready. The Russians had beaten them. And the Russians were the people America was most afraid of, because the world after the Second World War had divided itself into two camps that did not trust each other at all, and the Russians ran the other one.

The American newspapers went wild. Politicians gave panicked speeches. Schools were criticized for not teaching enough math and science. Parents wondered whether the country was raising the wrong kind of children. President Eisenhower, who was usually a calm man, tried to reassure everyone, and mostly failed.

Less than four months later, in February 1958, the United States government created a new agency.

ARPA

The new agency was called the Advanced Research Projects Agency. People called it ARPA.

ARPA's job was strange. Most government agencies have a clear thing they do. The Post Office delivers mail. The Department of Agriculture handles farms. ARPA's job was almost the opposite. ARPA's job was to handle *whatever the regular system couldn't*. It was supposed to fund science projects that were too weird, too long-term, too risky, or too unproven for any normal organization to pay for. Its purpose, in a sentence, was to make sure America was never surprised by another Sputnik.

It had a lot of money. It had very few rules. And — this is the important part — it was allowed to give that money to almost anyone.

Universities. Private companies. Individual scientists with strange ideas. ARPA didn't need to know exactly what the re-

search would produce. It just needed to believe that the people doing the research were smart, and that what they were trying to do, if it worked, would matter.

This was a new kind of government. For most of American history, when the government wanted scientific work done, it built a laboratory and hired scientists to work in it. ARPA didn't build laboratories. ARPA wrote checks.

A lot of the checks went to projects that failed completely. ARPA paid for nuclear-powered airplanes that never flew. For psychic experiments that produced nothing. For weapons that didn't work. That was fine. The whole point of ARPA was that most of its bets would lose, as long as a few of them won enormously.

A few of them did win. Enormously.

One of the bets ARPA made, in the early 1960s, was on a man at MIT who had read Vannevar Bush's essay about the memex when he was young, and never quite gotten over it. His name was J.C.R. Licklider, and he had recently written a paper called *Man-Computer Symbiosis*, in which he argued that computers and people would one day work together so closely that they would think as a single mind.

In October of 1962, ARPA hired Licklider to run a brand-new office, called the Information Processing Techniques Office. He was given a budget. He was told he could spend it more or less however he liked, as long as it was on computers.

He had some ideas about what to do.

He started writing memos to a small list of scientists around the country, addressing them — half-jokingly, half-seriously — as *Members and Affiliates of the Intergalactic Computer Network*.

That is where the internet begins.

But Licklider's story is the next chapter.

Chapter 2 — Licklider's Dream

IN THE SPRING OF 1957, A SCIENTIST AT MIT DID SOMETHING unusual.

He took out a notebook, and for several days he wrote down every single thing he did at work.

He wrote down when he sat at his desk. He wrote down when he stood up to get a book. He wrote down when he drew a graph, when he made a phone call, when he searched for a document, when he did a calculation on paper, when he stared at the ceiling thinking. He wrote everything down.

His name was J.C.R. Licklider — everyone called him Lick — and he wasn't tracking himself because he was bored. He was a psychologist by training, which meant he liked to study how human beings actually behaved, including himself. He had a hunch about how he spent his days. He wanted to test it.

When he added everything up, his hunch was right, and it was worse than he had expected.

About 85 percent of what he did at work was not actually *thinking*. It was clerical. Looking things up. Filing. Plotting

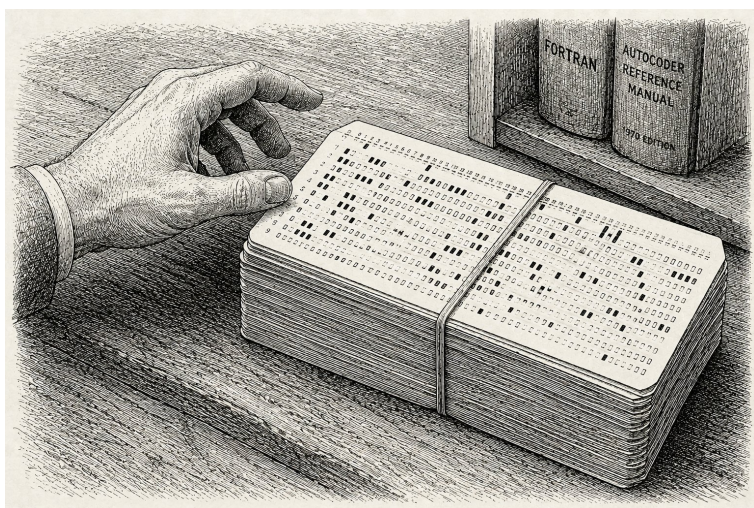
numbers onto charts. Doing arithmetic. Finding the right paper in a drawer. Copying figures from one document to another. The things he was paid for — the real ideas, the actual scientific work — took up only about one hour in every six. The rest was busywork that any reasonably careful clerk could have done in his place.

Lick stared at his notebook. He thought: *what a waste.*

And then he thought something stranger: *a computer could do most of this.*

This was, in 1957, an unusual thought.

HOW COMPUTERS WORKED THEN



A deck of punched cards — the way a person ‘talked’ to a computer in the 1950s.

To understand why Lick's thought was strange, you have to understand what computers were like in the 1950s.

There were not many of them. A single computer was the size of a large room — bigger than your bedroom, sometimes bigger than your entire apartment. It cost as much as a small office building. It used as much electricity as a city block. Universities and big companies might have one. The United States government had a few. Most people had never seen one in person.

And nobody *talked* to a computer.

If you wanted a computer to do something for you, this is what you did. You wrote your instructions out on paper. Then you went to a special room and punched those instructions onto cardboard cards, using a machine that looked like a typewriter. Each card could hold one line of instructions. A normal program might take a few hundred cards, stacked into a deck and bound with a rubber band.

You then carried your deck of cards to the computer room and handed it to an operator — a person whose job was to feed cards into the computer. The operator would put your deck into a queue behind everyone else's deck. You would go home.

The next morning, you would come back. The operator would hand you your cards, plus a stack of paper. The paper was the computer's answer to your program. You would read the answer. Almost always, there was a mistake somewhere in your cards — a typo, a missing instruction, a misplaced

comma. So you would fix the cards and hand them back in. You would go home again.

Twenty-four hours later, you would come back. And so on. A scientist might spend weeks on a problem that, if she could have just *talked* to the computer in real time, she could have solved in an afternoon.

This was called *batch processing*, because the computer ran one batch of cards at a time, all alone, with no humans interfering with it. The reason it was done this way was simple: computers were so expensive that you didn't want them to ever sit idle, waiting for some slow human to type. You wanted them running at full speed, all the time. Humans were the slow part. So humans had to wait their turn.

Lick thought this was exactly backwards.

MAN—COMPUTER SYMBIOSIS

Here was Lick's argument. The expensive thing about a scientist's workday wasn't the computer's time. It was the *scientist's* time. A computer could do arithmetic millions of times faster than a person. A computer never got tired. A computer never had to go home for dinner. The computer was not the bottleneck. The bottleneck was the human brain in front of it, trying to think.

If you could let the human and the computer work together — in real time, in the same room, the computer doing the clerical part while the human did the thinking part — you

would not be wasting the computer. You would be saving the *human*. And humans were more expensive than computers, when you really thought about it. Humans were the rare resource.

In 1960, Lick wrote up his idea in a short paper called *Man-Computer Symbiosis*. The word *symbiosis* comes from biology. It describes two different creatures that live together so closely that they depend on each other — the way a clownfish and a sea anemone do, or the way a tree and the fungi on its roots do. Lick was suggesting that humans and computers should become like that. Two very different kinds of mind, working as one.

He wrote, in that paper, that he could imagine — within ten or fifteen years — a kind of “thinking center” that would combine the functions of a library with new ways of storing and finding information, so that anyone who needed an answer could find it quickly.

He was, without quite saying so, describing Google. He was describing it in 1960. He would not live to see Google built — he died in 1990 — but a lot of the people who eventually built it had read his paper when they were young, and remembered.

Two years after he published *Man-Computer Symbiosis*, in October of 1962, Lick got a phone call. It was from ARPA — the agency you met at the end of the last chapter. Remember: ARPA was the panicked-after-Sputnik agency, with a lot of

money and very few rules, looking for smart people with big ideas.

ARPA wanted to fund research on computers. It wanted somebody to run a new office. Would he come to Washington and do it?

Lick said yes.

THE INTERGALACTIC COMPUTER NETWORK

When Lick arrived at ARPA, he found that there were already several big computers being paid for by his new agency. They were sitting in different universities around the country — MIT in Massachusetts, Berkeley in California, Carnegie in Pittsburgh, and others. Each university used its computer for its own research. The computers did not know about each other. They did not talk to each other. There was no reason they should have.

But Lick had been thinking.

Each of those computers was very expensive. ARPA had bought them because the research couldn't happen without them. But every time a new research group came to ARPA wanting to do computer work, ARPA had to buy *another* computer for them, because the existing computers were already being used by other people. It was getting expensive. Lick's budget could only stretch so far.

What if, Lick thought, the computers could share?

What if a scientist in Pittsburgh could, when her own computer was busy, send her program *over a wire* to the computer in Berkeley, run it there, and get the answer back? The Berkeley computer might be sitting idle at that moment — three in the morning, California time. The Pittsburgh scientist could use it without anyone in Berkeley even noticing. ARPA would get twice the science for the same money.

This was, on the face of it, a budgeting idea. A way to stretch a dollar. But Lick saw something larger. If computers could share work, they could also share *information*. A library at MIT could be read by a student in Pittsburgh. A weather model in Berkeley could be used by a scientist in Boston. Researchers around the country, around the world, could be connected to each other through their machines.

In April of 1963, Lick wrote a memo. It was addressed, with a wink, to *Members and Affiliates of the Intergalactic Computer Network*.

The memo was about a practical problem — how to get all the computers ARPA was funding to use a common system, so they could eventually talk to each other. But the title was a joke that turned out not to be a joke. Lick was sketching, in that memo, the shape of every network that would ever come after: a galaxy of computers, far apart, talking.

He addressed it to twelve people.

WHAT LICK BUILT, AND WHAT HE DIDN'T

Here is the part of the story that surprises a lot of people. Lick did not build the internet. He didn't even build the small version of it that came first, which was called the ARPANET.

He stayed at ARPA for less than two years. In 1964, he went back to academia. He had set up the office, hired some early staff, written some memos, and convinced everyone he worked with that connecting computers was important. Then he handed the keys to someone else and left.

The person who took over after Lick was named Ivan Sutherland, and after him a man named Bob Taylor, and after Taylor a man named Larry Roberts. Taylor was the one who, in 1966, walked into the office of his boss at ARPA and said something like: *We need to build a network*. His boss asked how much it would cost. Taylor said about a million dollars. His boss said yes. The whole meeting, according to Taylor's memory, took about twenty minutes.

Roberts was the one who actually built the thing. Three years later, in October of 1969, the first message would be sent on the network Roberts built. You'll hear about that in Chapter 4.

But none of it would have happened the way it did without Lick. He was the person who first said, clearly and in writing, that computers were not just calculators — they were a new kind of medium for human thought. He was the

person who first said that the right way to use a computer was to talk to it, in real time, the way you talk to a friend. He was the person who first imagined a network of them, stretching across the world, sharing what they knew.

A lot of important things in the world are built like this. One person has a vision and writes it down. Another person reads what was written and builds part of it. A third person builds another part. A fourth person turns the whole thing into something the rest of us can use. By the time you and I see the result — a phone in your hand, a video from your grandmother — the original person is often long gone, and most people have no idea their name was ever attached to it.

J.C.R. Licklider died on June 26, 1990. The very same year, in a physics laboratory in Switzerland, a young British scientist who had read Lick's papers as a graduate student turned on the first web server in history. The thing most of us mean when we say "the internet" — the world wide web — was being born in that laboratory while Lick was dying.

He never saw it.

His memos sit in archives now. You can read them online, if you want, on the internet that he imagined before there was an internet to imagine it on.

The next part of the story is about how packets — those numbered little pieces of grandma's video, in the prologue — were actually invented. The idea did not come from MIT, or from ARPA, or from any university computer lab. It came

from a man at a research company in California, whose real worry was not science.

His real worry was nuclear war.

That's Chapter 3.

Chapter 3 — Packets

IN 1960, THE UNITED STATES AIR FORCE GAVE AN ENGINEER named Paul Baran a strange and difficult problem to solve.

The problem was this. If the Soviet Union ever launched a nuclear attack on the United States — and almost everyone in the American government, at that time, thought this would happen sooner or later — then most of the country's bombs and missiles would be destroyed in the first hour. So would most of its cities. So would most of its phone lines. And so, very probably, would the President.

But the American military wanted to be able to fight back, even after all of that. To fight back, the parts of the military that survived — a submarine under the Atlantic, a bomber over Canada, an officer in an underground bunker in Nebraska — would need to be able to talk to each other. They would need to send messages. They would need a network.

The problem the Air Force gave Paul Baran was: build us that network. Build us a network that still works, after the bombs.

This is an unpleasant problem to think about, and you might wonder why I'm telling you about it. I'm telling you about it because the way Paul Baran solved this problem is the reason your grandmother's video can reach your phone.

The internet you use every day was, in its original design, a survival mechanism for nuclear war.

I want to be honest about that, because I think it's important. Almost everything in this book has two stories sitting on top of each other. There's the story of what people *meant* to build — a way to fight a war, a way to share a computer, a way to publish a physics paper. And there's the story of what they *actually* built, which usually turns out to be different and bigger and more peaceful than what they had in mind. Paul Baran was trying to help America survive an apocalypse. He ended up helping your grandmother share a sunset.

So. Let me explain his problem, and his solution.

WHY THE PHONE SYSTEM WAS FRAGILE

When Paul Baran started his work in 1960, the only big communication network in the world was the telephone system. If you wanted to call your aunt in another city, you picked up your phone, told the operator who you wanted to talk to, and the operator plugged a wire into a hole on a big board that connected your phone to a wire that led to your aunt's phone. Later, the wire-plugging was automated, but the idea

was the same. A call was a *circuit* — an unbroken wire, all the way from you to your aunt — held open for the whole conversation.

This system worked. But it had a weakness Paul Baran could not stop thinking about.

To make the system work efficiently, the telephone company had built a small number of enormous *switching centers*, where most of the wires came together. Almost every long-distance phone call in America passed through one of about a dozen of these centers. They were huge buildings, full of switches and operators, and they were the heart of the network.

The trouble was, if you wanted to destroy the American phone system, you didn't have to destroy every phone or every wire. You only had to destroy those dozen buildings. Twelve bombs and the country would go silent.

The Soviet Union knew where the buildings were. So did everyone.

Baran sketched this out on paper. He drew a picture of three different kinds of network.

The first kind he called *centralized*. It looked like a star — one hub in the middle, with lines going out to every other point. Destroy the hub and the whole network was dead.

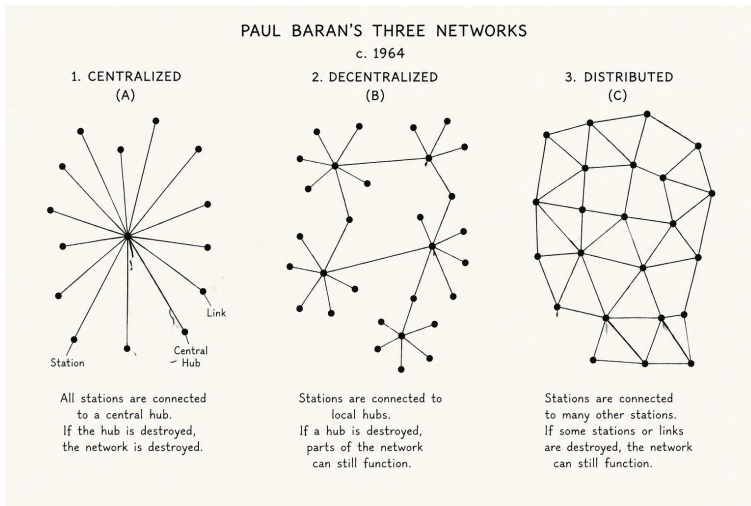
The second kind he called *decentralized*. It looked like several stars joined together — a few hubs, each connecting to many points, with the hubs connected to each other. Destroy

enough hubs and most of the network was dead. This was, roughly, the American phone system.

The third kind he called *distributed*. It looked like a fishing net, or the cracks in old pavement, or the veins on the back of your hand. There was no center at all. Every point was connected to several of its neighbors, and from any point you could reach any other point by many different paths. If part of the net was destroyed, messages could simply go around the hole, taking a different route. There was no critical building you could bomb.

The distributed network was the answer to the Air Force's question. But there was a second part to Baran's idea, and it was the part that turned out to matter the most.

THE PIECES



Paul Baran's three sketches (c.1964). The third — the distributed mesh — became the internet.

The problem with sending messages through a distributed network was that messages — especially long ones — were not really designed to find their own way. A voice call was a continuous stream. You couldn't just split it across many paths.

Baran thought about this for a long time, and eventually he had a strange idea.

What if you didn't send the message as one stream? What if you chopped it into small pieces — short, fixed-size chunks — and sent each piece *separately* through the network, by whatever path was fastest at that moment? Each piece would have its own little label saying *I'm piece number 3 of 47, going*

to this address. At the other end, the pieces would arrive — maybe out of order, having taken different routes — and the receiving computer would put them back together.

If the network was damaged, the pieces would just go around the damage. If one piece got lost, you could ask for it to be re-sent without having to re-send the whole message. If one path was busy, the pieces would take a different path. The network would be alive and adaptive in a way that the old phone system, with its fixed circuits, could never be.

Baran called these little pieces *message blocks*. The name didn't stick. In a few years, an Englishman would come up with a better word, and we'll get to him in a minute. But the idea — the thing the word names — was Paul Baran's, in 1960, in an office in Santa Monica, while he was trying to figure out how to keep America talking after a nuclear war.

He spent the next four years filling out the idea. He wrote an enormous report — eleven volumes long — called *On Distributed Communications*. He sent it to the Air Force. He sent it to engineers all over the country. He went to talk to the people who would actually have to build the thing.

The people who would actually have to build the thing were AT&T, the giant company that ran the American phone system. And AT&T said no.

“IT WON’T WORK”

The conversation between Paul Baran and AT&T is one of the great stories in the history of technology, and I want to tell it carefully, because you will see this pattern again and again in your life.

Baran went to AT&T’s engineers and explained his design. A distributed network. Messages chopped into small pieces. Each piece finding its own path. The pieces reassembled at the other end. He had the math worked out. He had the diagrams. He had simulations showing it would work.

The AT&T engineers told him it would not work.

They told him it would not work, essentially, because they were AT&T and he wasn’t. They had been building communication networks for almost a century. They were the experts. Networks were *circuits*. You set up a path from one end to the other, and you kept that path open. That was what a network was. What Baran was describing — pieces of a message running around independently, the network deciding paths on the fly — wasn’t a network. It was chaos. It couldn’t possibly work.

Baran asked them to at least *try* a small experiment. They wouldn’t. They didn’t want to. They had a perfectly good network business that worked the way they had always done it, and a young man from a think tank in California, with no experience running a phone company, was telling them they had it all wrong. They politely showed him the door.

By 1965, after years of trying, Paul Baran gave up. He couldn't get anyone to build his network. He set the eleven volumes aside and went on to other projects. He was thirty-nine years old. He probably thought his big idea was dead.

It wasn't. It was sitting in an office across the Atlantic Ocean, being thought up all over again, by a completely different person, for completely different reasons.

MEANWHILE, IN ENGLAND

His name was Donald Davies. He worked at a place called the National Physical Laboratory — a British government research center — outside London.

Davies, in 1965, was not worried about nuclear war at all. He was worried about something much smaller. He was worried that interactive computing — the thing J.C.R. Licklider had been promoting, the idea that you should be able to *talk* to a computer in real time instead of handing it cards — was going to be too expensive to ever become normal. If every person who wanted to use a computer had to keep their own phone line open to it for hours at a time, the phone bills alone would be more than most people could afford.

There had to be a more efficient way. And Davies, sitting in his office in England, working on this purely civilian problem, invented essentially the same thing Paul Baran had invented five years earlier in California.

He chopped messages into pieces. He let the pieces find their own way through a network. He let them be reassembled at the other end.

But Davies did one thing Baran hadn't done. He picked a better name.

The little pieces of a message, Davies decided, should be called *packets*. As in: a small package, a parcel, a bundle. Each piece was its own little package, with its own address label, traveling on its own.

The name was perfect. It stuck. And from that moment on, this whole technique — chopping messages up and sending the pieces independently — has been called *packet switching*.

Davies, unlike Baran, was actually able to build a small version of his network. The British government gave him a budget. By 1967 he had a tiny working prototype running inside his laboratory in England. It worked.

In October of that year, Davies's team gave a public talk about their work at a conference in the mountains of Tennessee. An American researcher sat in the audience. After the talk, one of Davies's colleagues mentioned, almost as an aside, that an engineer at RAND in California had worked out the same ideas a few years earlier — had anyone in the room heard of Paul Baran?

Almost nobody in the room had. Davies had not heard of Baran. Baran had not heard of Davies. They had invented the

same thing, on opposite sides of the world, for completely different reasons, almost a decade apart.

CONVERGENCE

The American researcher who heard about Baran in that mountain conference room was named Larry Roberts. He worked for ARPA. He had just been hired to actually build the network J.C.R. Licklider had imagined. He had been struggling. He knew he needed to connect ARPA's computers, but he didn't quite know how the messages would actually travel between them.

When Roberts heard about packet switching, something clicked. He went home from the conference and sent for Baran's eleven volumes. He read them. He read Davies's papers. And he saw, in those two independent piles of work, exactly the design he had been looking for.

The network ARPA built — the one that became the internet — would use packet switching. The technique invented by an American engineer trying to survive a nuclear war, and an English engineer trying to make computing affordable, and given its name by the Englishman, would become the way every piece of information in the world eventually traveled.

Including, sixty years later, a ten-second video of a sunset in Da Nang.

This is the part of this chapter I want you to hold onto, even if you forget everything else.

When you watch your grandmother's video, it doesn't arrive on your phone in one piece. It arrives in thousands of pieces — packets — each with its own little label, each having taken its own little path through the world. Some of them probably went one way. Some of them probably went a different way. A few of them probably got lost, and your phone quietly asked for them to be re-sent, and you never noticed. The whole video came back together on your screen in less than a second, out of a thousand tiny independent journeys.

The reason it works that way is that, in 1960, somebody was trying to figure out how messages could survive an atomic bomb.

The reason the world is what it is, very often, is not the reason you would guess.

In the next chapter, we'll meet the people who actually plugged Baran's ideas and Davies's ideas into Licklider's dream, and built the first real network. The first message ever sent on it was supposed to be the word *login*.

The computer crashed after two letters.

That's Chapter 4.

Chapter 4 — The First Message

ON THE NIGHT OF OCTOBER 29, 1969, A 21-YEAR-OLD STUDENT at the University of California in Los Angeles sat down at a computer terminal and tried to log in to another computer about 350 miles away.

His name was Charley Kline. He was working late — it was around ten o'clock at night. The lab around him was mostly empty. He had a phone in his hand. On the other end of the phone was a man named Bill Duvall, who was sitting in front of a different computer, at a research institute called SRI, near Stanford University.

Kline's job, that night, was very simple. He was going to type the letters L-O-G-I-N on his terminal. The letters would travel — for the first time in history — out of his computer, across phone lines, through a brand-new piece of equipment about the size of a refrigerator, and onto Duvall's screen in Stanford. Duvall would read the letters out loud, over the phone, so Kline could be sure they had arrived.

Kline typed L. Duvall said: *I see the L.*

Kline typed O. Duvall said: *I see the O.*

Kline typed G. Duvall said nothing.

The entire system had crashed.

For a few minutes, on the night of October 29, 1969, the very first message ever sent on the network that would become the internet was the word *lo*.

It was, accidentally, a beautiful first word. *Lo* — as in *behold*, or *look*. The very old way English speakers used to begin a story: *Lo, I will tell you what happened in this place*.

Kline did not write anything dramatic in the lab logbook that night. He didn't realize he had done anything historic. He scribbled, at 10:30 p.m., a short entry that said only: *Talked to SRI Host to Host*.

Then he and Duvall fixed the bug. About an hour later, Kline tried again. This time *login* went all the way through.

This chapter is about how he got to that desk.

LARRY ROBERTS HAS A PLAN

At the end of Chapter 2, I told you that ARPA hired a man named Larry Roberts to build the network J.C.R. Licklider had imagined. At the end of Chapter 3, I told you that Roberts had heard about packet switching at a conference in Tennessee and gone home with copies of Paul Baran's eleven-volume report.

It is now early 1968. Roberts is sitting in an office in Washington, D.C., trying to figure out how to actually do this thing.

He had a plan. The plan was: start small. Connect four computers, at four universities, in four different states. Make them able to send messages to each other. See what happens.

The four universities he picked were the University of California in Los Angeles (UCLA); the Stanford Research Institute (SRI), near San Francisco; the University of California in Santa Barbara; and the University of Utah. Three were in California. One was in the mountains. They had nothing in particular in common, except that ARPA was already paying for computer research at all four of them, and they were all willing to try something strange.

Roberts wrote up the plan and sent it out to companies, asking them to bid on building the equipment. This is how the American government often funds technology — it doesn't build the equipment itself, it pays a private company to build it.

The big famous companies — IBM, AT&T, the giants — looked at Roberts's request and either turned it down or proposed solutions that wouldn't have worked. They didn't really think it could be done. (Remember AT&T from the last chapter. The pattern repeats.)

The contract eventually went to a small company in Cambridge, Massachusetts, called BBN — Bolt, Beranek and Newman. BBN was not a famous company. Most people had never heard of it. But it had something the big companies didn't have: a few brilliant young engineers, and the willingness to attempt something almost nobody believed could work.

THE IMP

The thing BBN had to build was a special kind of computer. Roberts called it an *Interface Message Processor*, or IMP for short. (The internet world has always loved its acronyms.)

The IMP's job was to be the network's hands and feet. The actual computers at the universities — the big mainframes that scientists used for research — would not talk to each other directly. They were all different brands, different sizes, different shapes, using different software. Trying to teach them all to talk to each other directly would have been an impossible mess.

Instead, every university would get a small IMP standing next to its big computer. The IMP would handle all the network business — chopping messages into packets, sending them across phone lines to other IMPs, receiving packets from other IMPs, putting them back together, and handing the finished message to the big computer next door. The big computers wouldn't have to know anything about networking at all. They would just talk to their local IMP, and let it deal with everything else.

The IMP was, in other words, the world's first *router*. The same kind of device, in spirit, as the little box that probably sits in a corner of your living room right now connecting your home to the internet. The one in your living room is the size of a paperback book. The first IMP was the size of a re-

frigerator and weighed about 900 pounds. But it did, in essence, the same job.

At BBN, the engineer in charge of designing the IMP was a young man named Bob Kahn. He will have a much larger role in this story later — he is one of the two people you'll meet in the next chapter who invented the rules the internet still runs on today. For now, you just need to know that Bob Kahn and his small team had less than a year to build something nobody had ever built before, and that almost no one in the rest of the country believed could be built at all.

They built it.

LABOR DAY WEEKEND, 1969

On the Saturday of Labor Day weekend, 1969 — August 30, to be exact — a truck pulled up to a building called Boelter Hall at UCLA, and unloaded a refrigerator-sized metal box. The box was painted a battleship gray. It had cables coming out of one side. It was the first IMP.

The students and professors who had been waiting for it gathered around. They had been preparing for months. The IMP was wheeled into a room on the third floor and plugged into the wall. It was plugged, by a thick cable, into UCLA's big mainframe computer next door. It was plugged, by a phone line, into the rest of the world.

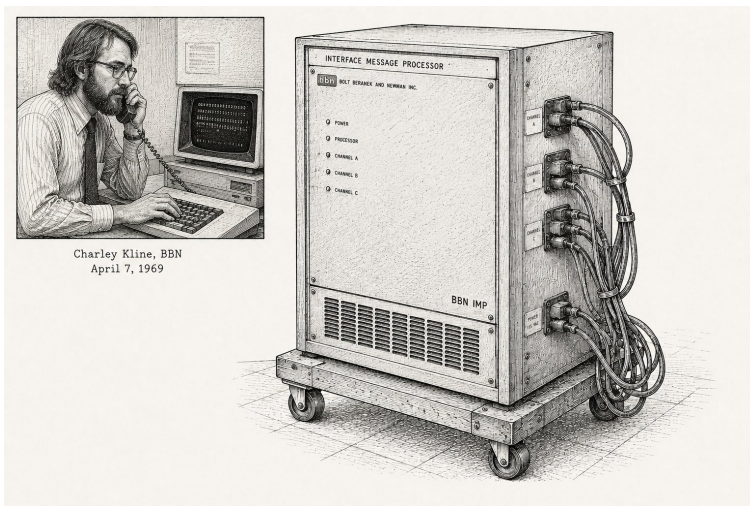
It hummed.

And then, for almost two months, it did nothing interesting. There was nothing for it to talk to. UCLA's IMP was the only IMP in the world.

The second IMP arrived at SRI, near Stanford, on the first of October. As soon as it was installed and the phone line between Los Angeles and the Bay Area was switched on, the two IMPs could, for the first time, exchange messages.

And on the night of October 29, 1969, Charley Kline sat down to try.

THE MOMENT



Charley Kline at UCLA, and the BBN Interface Message Processor — the first router.

I told you what happened. Kline typed L. Duvall confirmed L. Kline typed O. Duvall confirmed O. Kline typed G. The system crashed.

It crashed because of a bug in the way SRI's big computer handled incoming text. When Kline typed G, SRI's computer tried to be helpful — it tried to guess that the rest of the word would be *GIN* (since *LOGIN* is what people usually typed) and finish typing for him. But the way it tried to do this overflowed a tiny part of its memory, and the whole thing froze.

Kline and Duvall fixed the bug. By midnight, *login* worked end to end.

A few weeks later, the third IMP was installed at the University of California in Santa Barbara. In December, the fourth one went in at the University of Utah. By the end of 1969, four computers in four states were talking to each other over a tiny new packet-switched network. The thing had a name now. It was called the ARPANET.

If you had walked into that room at UCLA on the night of October 29, 1969, you would not have seen anything dramatic. A young man typing. A phone call. A blinking light on a gray metal box. There was no party. There were no photographers. The next day's newspapers said nothing about it. Almost nobody outside of ARPA and a few universities even knew the network existed.

This is something else I want you to remember about how the world gets built. Most of the moments that turn out to

matter most do not feel like big moments when they are happening. They feel like a graduate student typing at 10:30 at night, getting a bug, and writing one boring sentence in a notebook. The bigness only comes later, when you look back.

WHAT HAPPENED NEXT

The ARPANET grew. By 1971 it had fifteen nodes. By 1973 it had connected its first computers outside the United States — one in Norway, then one in London, through a satellite link and an undersea cable. Researchers and graduate students at the connected universities started using it for things its designers had not really planned for. They sent each other notes. They argued about software. They shared files. They played games.

The first thing that became wildly popular on the ARPANET was something the engineers had thrown together almost as an afterthought, in 1971. A man at BBN named Ray Tomlinson wrote a small program that let one user on the network send a short text message to another user on a different computer. He needed a way to separate the user's name from the computer's name in the address, so he reached for an unusual character on his keyboard — one that wouldn't be confused with any normal letter or number. He picked @.

He had just invented email. And he had just decided, more or less on a whim, that every email address in the world would forever have an @ sign in the middle of it.

By the mid-1970s, more than half of all the traffic on the ARPANET was email. The network that had been designed to share expensive computers was instead being used, mostly, by scientists writing each other letters. The technology was carrying something its designers had not predicted: human conversation.

But there was a problem.

The ARPANET worked beautifully. It used Paul Baran's packet switching, Donald Davies's word for it, Larry Roberts's design, BBN's hardware, and a set of rules — *protocols* — that the universities had worked out together for how their computers should talk to each other.

The rules, though, were the rules of the ARPANET, and only the ARPANET. They worked when every computer on the network was the same kind of node, connected through the same kind of line. They did not work for networks that were built differently — packet radio networks the military was experimenting with, satellite networks linking ships at sea, small networks growing inside other companies and other countries.

By the early 1970s, it was clear that the ARPANET was not going to be the only network in the world. There were going to be many networks. Some would be in the air. Some would

be under the ocean. Some would be inside a single university. Some would span the planet.

How could all these different networks ever talk to *each other*?

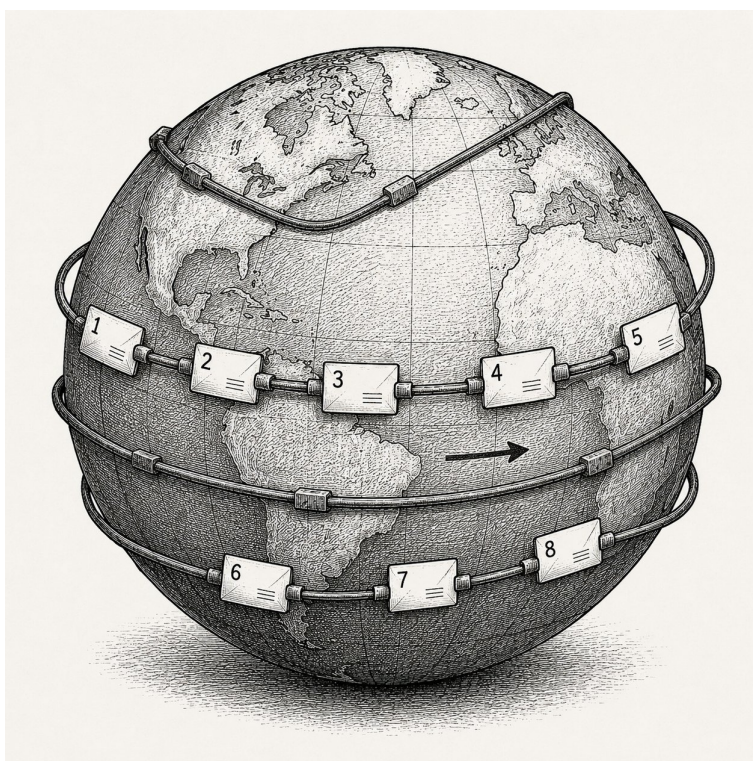
Two men sat down to answer that question in 1973. One of them was Bob Kahn, the engineer who had built the IMP. The other was a young Stanford professor named Vint Cerf, who had been a graduate student at UCLA on the very night Charley Kline typed *lo*.

What they invented is the reason any computer in the world can talk to any other computer in the world today. It is the reason your grandmother's video, leaving her phone in Vietnam, can pass through dozens of completely different networks owned by dozens of different companies in dozens of different countries, and still arrive, intact, on your screen.

The thing they invented is called TCP/IP, and it is the closest thing the internet has to a constitution.

That's Chapter 5.

Chapter 5 — The Constitution



A video, chopped into numbered packets, travels around the planet on its own.

In the spring of 1973, two American engineers met at a conference in San Francisco. After the conference ended, they

stayed an extra day, rented a hotel room at the airport, and tried to solve the hardest problem in computing.

One of them was Bob Kahn, who you met in the last chapter — the engineer who had built the first IMP at BBN. He had recently left BBN and joined ARPA, where he was thinking about a new question.

The other was Vint Cerf. Cerf had been one of the graduate students at UCLA on the night Charley Kline typed *lo* in 1969. He had moved on, after his PhD, to become a young professor at Stanford. He had a neat beard, a careful way of speaking, and an unusual habit of always wearing a three-piece suit, even when nobody else in computer science was wearing one. He still wears them today, in his eighties.

Kahn had brought a problem to Cerf. They had been talking about it for months. Now, in this hotel room near the airport, they had a few days to actually solve it.

The problem was that the ARPANET was no longer the only network in the world.

ONE NETWORK, THREE NETWORKS, MANY

In the last chapter, you saw how the ARPANET grew, in just four years, from four computers in 1969 to dozens of computers in 1973. By that point, scientists in the United States, Norway, and England were all connected to it. It worked. People loved it. They used it for email more than for anything else.

But while the ARPANET was growing, the ARPA engineers had been quietly building two *other* networks for the military, and these other networks were not at all like the ARPANET.

The first was called *packet radio*. Instead of running on telephone lines, it used radios. The idea was that a soldier in a jeep, or a pilot in an airplane, or a sailor on a ship, could carry a small computer and a radio antenna, and send messages through the air to other computers, without needing any wires at all. ARPA was already experimenting with this near San Francisco. They had vans with radios in them driving around the hills.

The second was called *packet satellite*. It used a satellite in orbit to bounce signals between ships at sea and stations on land. A submarine in the Atlantic Ocean could, in theory, send a message up to a satellite, which would bounce it back down to a base in Virginia, faster than any cable on the seafloor could carry it.

Each of these networks worked. Each one used packet switching, the technique you read about in Chapter 3. But each one was built differently. They used different equipment. They used different rules. The ARPANET's packets had a certain shape and size. The packet radio network's packets had a different shape and size. The satellite network's packets had yet another shape.

A computer on the ARPANET could talk to other computers on the ARPANET. But it could not talk to a packet

radio computer or a satellite computer. It was like having three different languages, each one perfectly good for its own group, with no way to translate between them.

Kahn had been turning this over in his mind for months. The military wanted all three networks to work together. A general in Washington should be able to send a message to a jeep in the field — through the ARPANET, then onto the packet radio network, then to the jeep — without anyone in between having to translate. But how?

This was the problem he brought to Cerf in San Francisco.

THE INSIGHT

Here is what they realized, sitting in that hotel room in the spring of 1973.

The standard idea, up to that point, was to make all the networks more alike. If they all spoke the same language — the same kind of packets, the same kind of equipment — they could talk to each other directly. This was the obvious solution. It was also impossible. Each network had been built for a different purpose, with different physical equipment. You couldn't make the satellite network into a copy of the ARPANET. You couldn't make the packet radio network into a copy of the satellite network. They had to stay different.

So Cerf and Kahn flipped the problem.

What if, they thought, you didn't try to make the networks the same? What if, instead, you let each network be whatever

it wanted to be — different equipment, different rules, different speeds — and you added one new thing on top of all of them: a single, simple language that *every computer in every network* would agree to speak when it wanted to talk to a computer in a different network?

You wouldn't need to change any of the existing networks. You would just need a translator. A common language, sitting above all the local languages, used only when packets had to cross from one network to another.

This is what we now call *internetworking*. The word *internet* comes from this idea: a network *between* networks. Not one big network, but many networks, joined together by a shared language.

The language Cerf and Kahn started designing that week was called TCP/IP.

HOW TCP/IP WORKS

TCP/IP is actually two languages stacked on top of each other. Each does a different job. You can think of them like two parts of a postal system.

The bottom one is called IP. The letters stand for *Internet Protocol*. IP's job is the address book.

In the same way that every house in the world has a postal address — a street, a city, a country — IP gives every computer on the internet a unique number, called an *IP address*. When a packet leaves your phone, the first thing it carries —

stamped on its outside — is the IP address of the computer it's trying to reach. Every router along the way looks at that address and decides: *based on this number, I should send this packet that way*. It doesn't matter whether the next part of the journey is a cable, a radio wave, a satellite link, or a fiber on the seafloor. IP doesn't care. It just hands the packet to whatever network can move it one step closer.

This is why your grandmother's video can leave a phone in Vietnam, travel through a hotel WiFi router, a Vietnamese phone company, an undersea cable, an American backbone, a Facebook data center, and your home router, without any of those systems needing to know anything about each other. They just need to know how to read an IP address and pass the packet along. IP is the worldwide postal address system for machines.

The top one is called TCP. The letters stand for *Transmission Control Protocol*. TCP's job is to make sure nothing gets lost.

Here's the thing about IP. IP doesn't actually promise that your packet will arrive. It tries its best. But a packet can get dropped — by a busy router, a glitchy cable, a satellite link that fades in bad weather. If the packet doesn't make it, IP just shrugs. IP is fast, but careless.

TCP is the careful one. Remember from the prologue: when your grandmother's phone chops the video into thousands of pieces, each piece gets a number — *I am piece 47 of 12,330*. That numbering is TCP's work. When the pieces ar-

rive at your phone, TCP checks every single number to make sure none are missing. If piece 47 didn't show up, TCP quietly sends a message back across the ocean asking for piece 47 to be re-sent. The video pauses, for a fraction of a second, while the missing piece travels back through everything. Then the video plays. You didn't notice. Your grandmother didn't notice. TCP noticed, and TCP fixed it.

If IP is the postal worker who carries the envelopes, TCP is the careful clerk who numbers every page inside the envelope, files them away as they arrive, and chases down the missing ones until the whole document is complete.

Cerf and Kahn published their design in a paper in May of 1974. The paper was called *A Protocol for Packet Network Intercommunication*. It was twelve pages long. It would turn out to be one of the most important twelve pages anybody has written in the last hundred years.

THE GIFT

Now I have to tell you about the most important decision Cerf and Kahn made — the decision that, more than any other single thing, made the internet what it is.

They gave it away.

They did not patent TCP/IP. They did not start a company to sell licenses to it. They did not require permission, or royalties, or signatures, from anyone who wanted to use it. They published the paper in an open journal and let anyone

in the world read it. They published their later revisions of the design as numbered documents — called *Requests for Comments*, or RFCs — that anyone could download and use, for free, forever.

This was a remarkable thing to do. They could have made themselves very rich. The companies they worked with could have tried to lock the protocols up. Instead — partly because the work had been paid for by the United States government, and partly because they believed in it — they let TCP/IP belong to everyone.

The decision had enormous consequences. Because TCP/IP was free, anyone could build a network that used it. A university in India. A company in Germany. A research lab in Brazil. A teenager in his basement. None of them had to ask permission. None of them had to pay. They just had to follow the rules in the paper, and their network would, automatically, be able to talk to every other network that followed the same rules.

This is the closest thing the internet has to a constitution. It is a short set of rules that everyone agrees to follow, voluntarily, because following them is how you get to join the global conversation. No one enforces it. No one is in charge of it. It just works, because everyone uses the same rules.

A different group of engineers in many countries did try, in the 1970s and 1980s, to design a “proper” set of rules through an official international committee. Their alternative was called OSI. It was supposed to be more complete, more

elegant, more thorough than TCP/IP. Committees worked on it for more than ten years. By the time it was ready, TCP/IP had already taken over the world — because TCP/IP was simple, and free, and already running. A messy thing that exists almost always beats a beautiful thing that doesn't.

This is another pattern worth remembering. The world is often built by the rough first thing that worked, not by the careful thing that was supposed to be perfect.

JANUARY 1, 1983

For ten years after Cerf and Kahn's paper, TCP/IP was tested, revised, and improved by engineers all over the world. By the early 1980s, ARPA had decided to make it official. The old ARPANET, which had been using an older set of rules, would switch over to TCP/IP on a specific day.

That day was January 1, 1983.

The engineers called it *flag day* — the day everyone had to change. If your computer wasn't speaking TCP/IP by the end of January 1, it would no longer be able to talk to the ARPANET. The instructions had been sent out for months. Universities and labs had been preparing. The switch flipped on New Year's Day.

A lot of things broke. A lot of things had to be fixed in the first few weeks. But by February of 1983, the network was running on TCP/IP, and it has been running on TCP/IP ever since.

Most people who study the history of the internet consider January 1, 1983 to be the *actual* birthday of the internet. The ARPANET had existed since 1969. But the *internet* — the network of networks, all able to talk to each other through a common language — that started on New Year's Day, 1983.

There were maybe a few hundred computers on it.

Today there are more than ten billion devices. Your phone is one of them. So is mine. So is the security camera near your door, the watch on your wrist, the thermostat on your wall, and a satellite that is, at this moment, somewhere over the Pacific Ocean. All of them are speaking the same language two men sketched out in a hotel room in 1973 and gave away for free.

THE NETWORK WITH NO BOSS

There is one last thing I want you to understand about TCP/IP before we move on.

The way Cerf and Kahn designed it, the internet has *no boss*.

No one is in charge of it. No company owns it. No country runs it. There is no central office that decides who is allowed to join, or what is allowed to travel through it, or what the rules will be next year. The rules — the protocols — are written and updated by groups of engineers from many countries, who meet, argue, write papers, and propose

changes that anyone can read. When most engineers and companies agree a change is good, they adopt it. When they don't, they don't.

The network itself doesn't know what's inside the packets. It doesn't know whether it's carrying an email, a video of a grandmother, a picture of a cat, a bank transfer, a love letter, or instructions for a missile. It just moves packets from address to address, as fast as it can, with no opinion about what they say.

This design choice is one of the most consequential things any group of engineers has ever decided. It is the reason the internet has been used for so many beautiful things — connecting families across oceans, sharing knowledge that used to be locked in expensive libraries, letting people in distant countries talk to each other. It is also the reason the internet has been used for ugly things — for crime, for harassment, for spreading lies. A network with no boss has no one to stop bad people from using it. The protocols carry the worst messages and the best ones equally well, because the protocols cannot tell the difference.

When you are older, you will hear adults argue about whether this design choice was wise. Some think there should be more control over the internet. Others think any control is dangerous. There is no easy answer. But you should know that the *no-boss* design was not an accident. It was a choice. It was made, mostly, by two engineers sitting in a hotel room near the San Francisco airport in 1973, deciding

that the network should belong to everyone who agreed to follow a few simple rules.

In the next chapter, someone else will make the same kind of decision — to give away, for free, a piece of software that will transform the world. His name is Tim Berners-Lee. He works at a physics laboratory in Switzerland. He is going to invent something he calls the *World Wide Web*.

It is what most people today mean when they say *the internet*, even though it is actually a different thing — a thing built on top of the internet, using the protocols Cerf and Kahn gave away.

That's Chapter 6.

Chapter 6 — The Web, Given Away

ON THE BORDER BETWEEN FRANCE AND SWITZERLAND, BURIED about 100 meters below a quiet rural countryside, there is a circular tunnel almost 27 kilometers around. Inside the tunnel are some of the largest scientific instruments humans have ever built — enormous magnets, particle detectors the size of cathedrals, refrigeration systems colder than outer space. The instruments are used to smash tiny pieces of matter together at almost the speed of light, so that scientists can see what's inside them.

The place is called CERN — the European Organization for Nuclear Research — and it is one of the most famous physics laboratories in the world. Thousands of scientists from dozens of countries come to CERN every year to work on its experiments. Many of the deepest discoveries about how the universe works have been made there.

CERN was also, by accident, the place where someone invented the World Wide Web.

The someone was a young English engineer named Tim Berners-Lee.

He did it as a side project.

His boss did not really approve.

THE INTERNET IS NOT THE WEB

Before I tell you how he did it, I need to clear up something that confuses almost everybody.

The internet and the World Wide Web are not the same thing.

The internet is what you read about in the last chapter — the global network of computers, all able to talk to each other through TCP/IP. It is a kind of road system. It can carry many different kinds of vehicles. Email is one kind of vehicle. Video calls are another. Online games are another. The web is just one *more* kind of vehicle that travels on the same roads. A very famous one — so famous that most people now use the word *internet* when they really mean *web* — but technically, the web is not the road. The web is one of the things that drives on it.

The internet had been around for about twenty years before anybody invented the web. It was used for email, for sharing files, for logging into distant computers, and for chatting in early discussion groups. It was used mostly by scientists and engineers, who knew the special commands you had to type to make it do anything.

The web is what made the internet usable by ordinary people. It is what your grandmother is using when she opens Facebook. It is what you are using when you type something into Google, or read a Wikipedia page, or watch a video on YouTube. Every website you have ever visited is part of the web. The web sits on top of the internet, the way a city sits on top of its roads.

It was invented in 1989, by one person, in a place dedicated to studying things much smaller than atoms.

THE PROBLEM AT CERN

Tim Berners-Lee arrived at CERN in 1984. He was twenty-nine years old. He was a software engineer, not a physicist, and his job was to help the scientists keep track of their work.

There was a lot of work to keep track of.

CERN at that time had several thousand scientists — some who worked there full time, many more who visited from universities around the world to use the experiments for a few weeks or months at a time. Each scientist had documents. Each scientist had data. Each scientist had projects, and collaborators, and notes, and ideas. The information was scattered across hundreds of computers — different kinds of computers, made by different companies, running different software, in different buildings, in different countries. There was no good way to find anything. If you wanted to know what some Italian physicist had written about a particular

detector last year, you might have to find the right person, ask them what computer it was on, learn how to log into that computer, learn how to find the file, and read it in whatever format it happened to be saved in. Often you would just give up.

Berners-Lee thought a lot about this. He had been thinking about it, actually, since he was a graduate student. Years earlier, in 1980, he had written a small program for himself called *ENQUIRE*, which let him store notes about projects and people and create *links* between them. If you opened the note for a person, you could click a link to see all the projects that person worked on. If you opened a project, you could click to see who else was on it. It worked beautifully — for one person, on one computer.

What if, Berners-Lee began to wonder in the late 1980s, you could do the same thing across *all* of CERN? What if every document on every computer could have links to other documents, on other computers, anywhere in the building, anywhere in the world? What if, by clicking a single link, you could jump straight from a physics paper in Geneva to the data file it referred to in Tokyo, without having to know what computer either one was on?

He thought about it for years. In March of 1989, he wrote up his idea in a short proposal and gave it to his boss, a man named Mike Sendall.

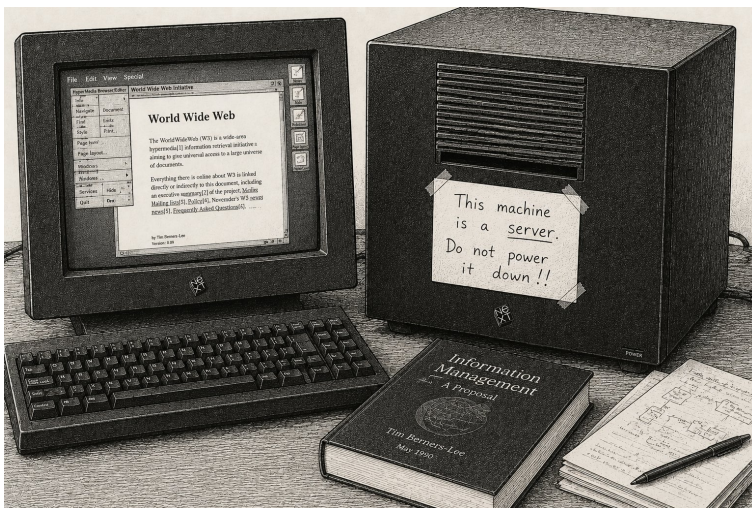
Sendall read it. He could not quite tell whether the idea was brilliant or ridiculous. He wrote a note on the front cover of the proposal, in handwriting that is now famous:

Vague, but exciting...

He gave Berners-Lee permission to work on it. Not as an official project. As a side thing. In his spare time.

That is how the World Wide Web got started.

WHAT HE BUILT



Tim Berners-Lee's NeXT computer at CERN — the world's first web server.

Now, if you remember Chapter 1, you might be feeling something familiar. A machine that lets you click from one document to another, following links across all the world's knowledge, connecting any idea to any other idea. That is

exactly the machine that Vannevar Bush had imagined in 1945, in his essay about the *memex*. Bush had described it. Berners-Lee was about to build it.

Bush had imagined the memex as a single desk, with microfilm inside, holding one person's library. Berners-Lee imagined something much bigger. He imagined every document in the world, on every computer in the world, all linked together, all reachable from anywhere. He didn't think of it as a desk. He thought of it as a *web*.

He needed to invent four things to make it work, and over the next two years — partly on his own, partly with a Belgian colleague named Robert Cailliau, who helped him push the idea inside CERN — he invented all four.

The first was a way for documents to be written so they could contain links. He called this language HTML, which stands for *HyperText Markup Language*. HTML is the language almost every web page in the world is still written in today. You don't need to understand it to read a web page — your browser does the work — but it is, underneath, very simple. A web page in HTML is just text, with small instructions sprinkled inside it that say *this part is a heading*, or *this part is a paragraph*, or *this part is a link to another page*.

The second was a way for one computer to ask another computer for a document. He called this HTTP, which stands for *HyperText Transfer Protocol*. When your phone wants to read a web page, it sends a tiny HTTP message to the computer that holds the page — basically, *please give me this*

document — and the other computer sends the document back. (You have seen HTTP without knowing it: every web address you have ever seen starts with the letters *http* or *https*. That's what they mean.)

The third was a way to write down the address of any document, anywhere in the world, in a single line of text. He called this a URL, for *Uniform Resource Locator*. A URL is what you type into the top of your browser. It tells your computer exactly which document, on which computer, to ask for. The first URL Berners-Lee ever used was *info.cern.ch*. It still works.

The fourth was the program that let you read web pages — a *browser*. The very first browser was written by Berners-Lee himself, on a black-and-white computer made by a company called NeXT, which had been started by a man named Steve Jobs after he was fired from Apple. (We'll meet Steve Jobs again later.) Berners-Lee called his browser *World-WideWeb*, which is also what he had decided to call the whole system.

He needed one more thing — a *server*, which is the program that lives on a computer and hands out documents when other computers ask for them with HTTP. He wrote that too.

By late 1990, all four pieces were working. On December 20, 1990, Tim Berners-Lee turned on the world's first web server, on his NeXT computer at CERN. He stuck a sticker on the front of the computer that said: *This machine is a server. Do*

not power it down!! The computer is still in a museum at CERN today, with the sticker still on it.

The world's first website was a few pages explaining what the World Wide Web was and how to use it. It sat at *info.cern.ch*. For a long time, it was the only website that existed in the world.

THE GIFT

For the next two years, the web grew slowly. By the end of 1991, there were a few dozen web servers around the world, mostly at universities and research labs. By the end of 1992, there were a few hundred. It was interesting, but it was a small thing, used by a small group of scientists.

It was also not alone. There were competing systems for organizing information on the internet. The most popular was called *Gopher*, which had been built at the University of Minnesota. Gopher was actually more widely used than the web in 1992. It looked simpler. It worked well.

Then, in February of 1993, the University of Minnesota made an announcement. They were considering charging fees to companies that used Gopher. Not very much money — but some.

Almost overnight, the people who had been building tools and services on top of Gopher panicked. Even a small fee meant Gopher could no longer be the foundation for anything bigger. Big companies, in particular, would not bet

their futures on a technology that could be taxed by a single university.

A few months later, on April 30, 1993, CERN made its own announcement.

The World Wide Web would be free. Forever. CERN, an international laboratory funded by the governments of many European countries, would place all of the web's technology in the public domain. Anyone in the world could use it. Anyone could build on it. Anyone could improve it. No one would ever owe CERN a penny for it.

Berners-Lee had pushed hard for this. He could have insisted on patents. He could have tried to start a company. He had built the most important new piece of software in a generation, and he could have made himself one of the richest men in the world. He chose not to.

The announcement was one page long. It changed everything.

Within months, almost everyone working on Gopher quietly switched to the web. Within a year, the number of web servers had grown from a few hundred to a few thousand. Within two years, to tens of thousands. Today there are hundreds of millions.

This was the second great gift of the internet age. Cerf and Kahn had given away TCP/IP, the rules for moving packets around the world. Berners-Lee had given away the web, the rules for sharing documents on top of those packets. Togeth-

er, those two gifts — both paid for by governments, both released freely to everyone — are the foundation of almost everything you do online today.

It is worth pausing on this for a moment. Most of the most valuable things in the modern world are sold. You pay for a house. You pay for a car. You pay for clothes, for food, for music, for movies. The biggest companies in the world made their owners rich by selling things.

The thing that made all of those companies possible — the underlying technology of the internet and the web — was given away. Not because it had no value. It had immense value. It was given away because the people who made it believed it should belong to everyone.

This is rare. It is precious. And, as you will see in the next several chapters, it created an enormous opportunity for other people to come along and build companies, on top of those free foundations, that made their owners some of the richest people who have ever lived.

That is one of the central puzzles of this whole book. *Why* did the people who built the foundations give them away, while the people who built businesses on top of those foundations did not? There is no single answer. Part of it is that the foundation-builders were paid by governments and saw their work as a public service. Part of it is that the business-builders saw an opportunity and took it. Part of it is that, by the time the business-builders showed up, the most import-

ant decisions had already been made by people who weren't trying to be rich.

For now, just remember: in 1993, the web became free. The doors were thrown open. The age of the public-funded internet was ending. The age of private money and giant companies was about to begin.

But first, before we go there, I want to take you out of the laboratories and into the physical world for a couple of chapters. The internet, by 1993, was no longer just an idea. It was *something* — pipes, cables, ships, switches, real buildings full of humming machines. Somebody had to lay all of those cables. Somebody had to build all of that equipment. And the somebodies who did, by and large, are not the people you have met so far.

In Chapter 7, we are going down to the bottom of the ocean.

Chapter 7 — Ships and Sharks

WHEN YOU TALK ABOUT SOMETHING THAT LIVES “IN THE CLOUD” — your photos, your messages, the apps on your phone — you might picture, if you picture anything at all, the sky. Maybe satellites. Maybe servers in some tall building somewhere.

You should picture the ocean floor.

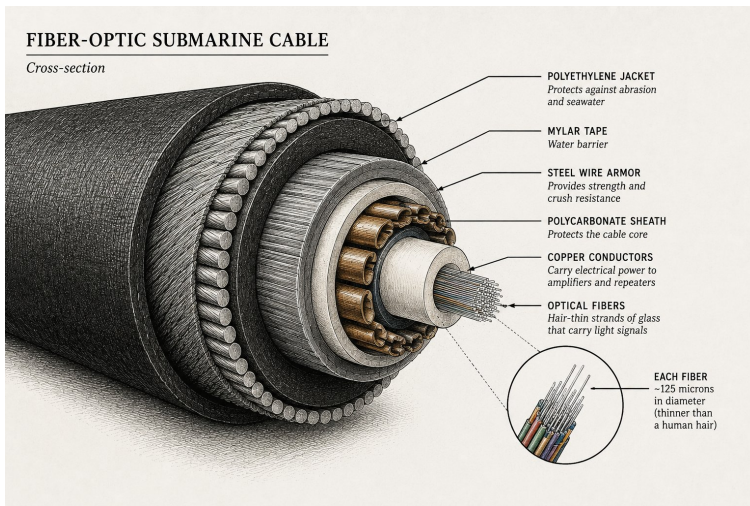
Almost everything you do online — every video your grandmother sends, every search you type, every game you play with a friend in another country — travels, at some point in its journey, through cables that lie on the bottom of the sea. Cables about the thickness of a garden hose, full of strands of glass thinner than a human hair, carrying pulses of laser light back and forth between continents.

There are more than 1.4 million kilometers of these cables on the floor of the world’s oceans. That is enough cable to wrap around the planet roughly thirty-five times. They were laid, one by one, over the past few decades, by a small fleet of specialized ships.

The cloud is not in the sky. The cloud is underwater.

This chapter is about how it got there, and about the strange and very specific group of companies and people who put it there, and about the surprisingly fragile life of a glass thread on the seafloor.

A BRIEF HISTORY OF A LONG WIRE



Inside an undersea cable: glass thinner than a hair, wrapped in steel and tar.

The first time anybody tried to lay a cable across the bottom of an ocean was in 1858.

This was, you should remember, sixty years before the airplane and twelve years before the telephone. The fastest way to send a message from London to New York, at that time, was to put it on a steamship and wait two weeks. People

dreamed of doing better. A few of them dreamed of doing *much* better.

The dreamer in chief was an American businessman named Cyrus Field. He raised an enormous amount of money to build a cable from Ireland to Newfoundland, in eastern Canada — almost 3,000 kilometers across the cold North Atlantic. Two ships, one British and one American, started in the middle of the ocean, splicing the cable together, and then sailed in opposite directions, paying out the cable behind them. After many failures — broken cables, snapped cables, storms — they finally got one cable down in August of 1858.

It worked. Queen Victoria sent a message to President James Buchanan. It took about sixteen hours to transmit a short paragraph. People in New York held a parade. Cyrus Field was hailed as a hero.

Three weeks later, the cable went dead.

The technology of 1858 wasn't quite ready. The cables of that era were not very well insulated, and salt water seeped in, and the signals faded. It took another eight years — and the use of the largest ship in the world at the time, a Victorian iron steamship called the *Great Eastern* — to finally lay a cable that worked, and kept working, in 1866.

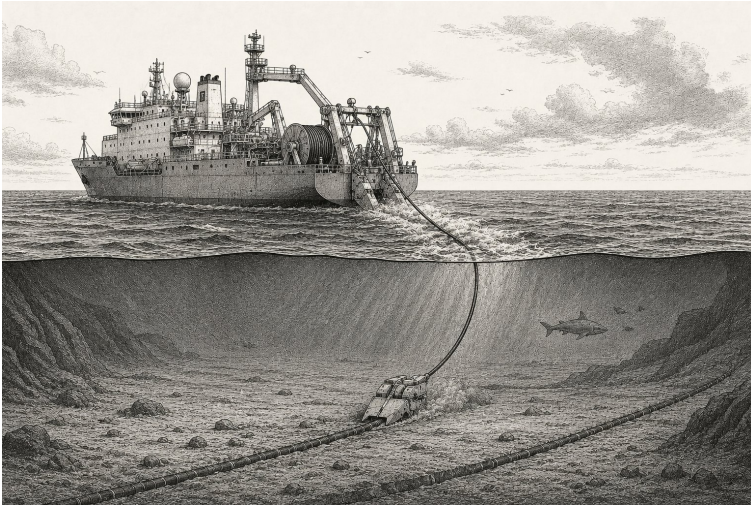
From that year on, the bottom of the ocean has never been empty.

Through the rest of the 1800s, undersea telegraph cables spread across the world. By 1902, you could send a message from London all the way around the Earth and back, entirely through cables, in a few hours. The British Empire used these cables — running between its colonies — to govern a quarter of the world's land. People at the time called this the "All Red Line," because British territories on maps were colored red. The cables, in a real sense, *were* the empire's nervous system.

In 1956, a different kind of cable was laid across the Atlantic. This one could carry voice — actual telephone calls — instead of just the dots and dashes of Morse code. It was called TAT-1, the first transatlantic telephone cable, and on its first day it could handle thirty-six conversations at once. Thirty years later, in 1988, the first transatlantic *fiber-optic* cable was laid, called TAT-8. It used pulses of light through glass instead of electrical signals through copper. It could handle the equivalent of about 40,000 simultaneous phone calls.

The cables you use today, the ones carrying your grandmother's video, can carry the equivalent of *hundreds of millions* of simultaneous phone calls. And there are hundreds of them.

HOW A CABLE GETS LAID



A cable ship pays out the wire across the ocean floor.

The actual job of putting a cable on the seafloor has not changed nearly as much as you might think.

It is still done by ships. Specialized ships, called *cable ships*, with reels of cable stacked in their holds. The reels are gigantic — some of them weighing thousands of tons. A single ship might carry enough cable to cross half of an ocean. They have names like *CS Dependable* and *Ile de Bréhat* and *KDDI Cable Infinity*.

When the ship reaches the place where the cable should start, it lowers one end into the water, attaches it to the shore, and then sails very slowly — usually no more than a few kilometers per hour — paying out the cable behind it, like a

fisherman letting out a line. A trip across the Pacific can take a cable ship weeks. Sometimes months.

In deep water, the cable just settles onto the seafloor, lying where it lands. In shallow water, where fishing boats and anchors can damage it, the cable is *buried*. For this, the ship drags an enormous machine called a *plow* across the seafloor behind it. The plow looks like a piece of farm equipment that has been scaled up by a Jules Verne novel — twenty tons of steel, the size of a small house, with a giant blade that opens a furrow in the seabed, drops the cable in, and lets the ocean floor close back over it.

The cables themselves are surprisingly small. At the very center is a bundle of glass fibers — sometimes only eight strands, sometimes a few dozen, each one thinner than a hair. Around the fibers is a copper tube that carries electricity to power the amplifiers along the way. Around that is more insulation, then a steel sheath for strength, then a tough plastic coating. The whole bundle, in deep water, is about as thick as a marker pen. In shallow water, where it needs more protection, it gets thicker — sometimes as thick as your wrist, with extra steel armor.

Only three companies in the world do most of the cable laying. One is American (SubCom, which began life inside the old AT&T). One is French (Alcatel Submarine Networks). One is Japanese (NEC). Between them, these three companies have laid most of the cables your internet runs on. They are not famous companies. Most people have never heard of

them. They are doing some of the most important work on Earth, mostly out of sight.

WHO OWNS IT

For most of the history of undersea cables, they were owned by telephone companies — usually in groups, with telecoms from different countries chipping in together to build a cable that all of them would share. A cable across the Pacific might be owned in part by AT&T, in part by Japan's NTT, in part by Australia's Telstra. This is called a *consortium*.

That has changed, and the change has happened mostly in the last ten or fifteen years.

The change is this: a few enormous American companies — the ones you have probably heard of, the ones whose names you say without thinking, Google and Meta and Microsoft and Amazon — now lay their *own* cables. They don't share them. They don't ask telecom companies for permission. They build them, they own them, they run them.

The reason is simple. These companies move so much information through the world's networks that no telecom can sell them enough capacity. Every video on YouTube, every Facebook photo, every Microsoft cloud customer's data, every Amazon order — multiplied by billions of users, every day — requires cables of a size that nobody had imagined twenty years ago. So the companies have started building their own.

A cable called *Marea*, finished in 2018, runs along the sea-floor from Virginia to Spain. It is owned by Meta (the company that owns Facebook) and Microsoft, with help from a Spanish telecom. A cable called *Dunant*, finished in 2021, runs from Virginia to France. It is owned by Google. A new system called *Apricot* is being built right now to connect Vietnam, the Philippines, Singapore, Indonesia, Japan, and Guam, mostly funded by Meta and Google. A truly enormous cable called *2Africa* — about 45,000 kilometers long, connecting more than thirty countries around the entire continent of Africa — is owned by a group led by Meta.

Think about this for a moment. If your grandmother sends a video from Vietnam to you in another country, there is a real chance that every part of its undersea journey takes place on a cable owned by an American technology company. Not by Vietnam. Not by the country you live in. By a private corporation, based in California, that most people only know as “the company that runs Facebook” or “the company that runs Google.”

This is one of the quiet revolutions of our time. The physical infrastructure of global communication, which used to be owned by governments and national telecoms, is increasingly owned by a small number of American firms. Whether this is good or bad is the kind of question your generation is going to have to answer.

SHARKS, ANCHORS, FISHING BOATS, EARTHQUAKES

Now I want to tell you about how cables die.

A cable on the seafloor is a long, lonely thing. It cannot defend itself. It just lies there, at depths where almost nothing visits — in the deepest places, the bottom of the Pacific, it sits four kilometers below the surface, in cold, black water that the sun has never reached. Most of the time, nothing happens to it. It can lie undisturbed for decades.

But sometimes, things happen.

In shallow water — closer to shore, where the cables come up out of the sea and onto land — the biggest danger is *fish-ing*. A fishing trawler, dragging nets along the seabed, can catch a cable and tear it. Anchors from anchored ships can crash down on cables and crush them. These are the two leading causes of cable damage in the world. Most cable breaks happen within a few kilometers of shore, in busy coastal waters where boats are dense.

In deeper water, *earthquakes* and underwater landslides are the main threat. An earthquake at the bottom of the ocean can shake loose huge amounts of sediment, which slides down underwater slopes in a kind of slow-motion avalanche, snapping cables in its path.

And then there are sharks.

This sounds like a joke, but it isn't. Sharks really do bite undersea cables. Nobody is exactly sure why — possibly because the cables give off faint electrical fields, and sharks can

sense electrical fields, and the cables confuse them. Whatever the reason, it has happened often enough that some cables, especially in shark-heavy waters, are now wrapped in extra layers of armor specifically to defend against shark bites. There is footage you can find online of a shark gnawing on an undersea cable. The shark looks small. The cable, on a thousand-mile journey across an ocean, is being interrupted by an irritated fish.

When a cable breaks — by trawler, by anchor, by earthquake, by shark, or by simple old age — somebody has to fix it.

This is the job of *repair ships*, which are specialized cable ships kept on standby at ports around the world, waiting for a phone call. When a break happens, the nearest available repair ship sails to the spot, which is found using the cable's own signals — the location of the break can be figured out from how long it takes a test pulse to bounce back. The repair ship deploys an underwater robot, called a *grapnel*, that grabs the broken ends, drags them up to the surface, splices in a new section of cable, and lowers the repaired cable back down.

It can take days. It can take weeks. In bad weather, it can take longer.

In January of 2022, an underwater volcano erupted near the small Pacific island nation of Tonga. The eruption was so powerful that it was heard in Alaska, more than 9,000 kilo-

meters away. The eruption snapped Tonga's only undersea internet cable.

Tonga has about 100,000 people. Until that cable was repaired, every one of them was almost completely cut off from the rest of the world. They couldn't email. They couldn't post to social media. They couldn't easily phone their families overseas. Banking failed. News stopped flowing in. The country went dark, in an internet sense, for over a month — until a repair ship made the long voyage from Papua New Guinea and patched the cable back together.

I am telling you this because I want you to understand, clearly and at a young age, that the internet is not magic. It is a physical thing. It can be broken. A volcano can break it. A fishing trawler can break it. A confused shark can chew on it. Whole countries can lose the internet because one ten-centimeter-thick cable, lying in the dark four kilometers below the sea, has been damaged.

The “cloud” is many things, but mostly it is a network of glass threads on the floor of the ocean. Anyone who tells you otherwise has forgotten where they live.

A NOTE BEFORE WE GO ON

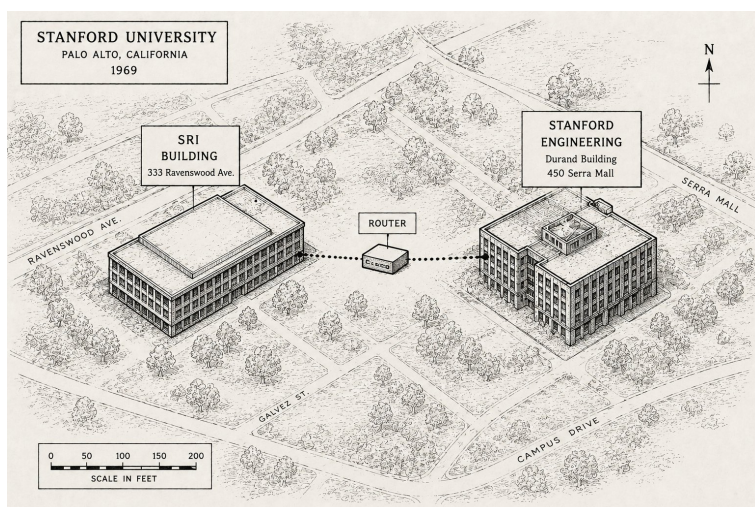
In the next chapter, we are coming back up to dry land, to talk about the *other* part of the internet's physical infrastructure: the boxes, big and small, that decide where every packet goes. The routers. The cables on the seafloor only handle the

journey between continents. Inside each continent, inside each country, inside each city, inside your home, packets are being routed by other equipment, built mostly by a single company, founded by two people who fell in love at Stanford and built a billion-dollar business out of the simple problem of getting their two university computers to talk to each other.

The company is called Cisco.

That's Chapter 8.

Chapter 8 — Routers and the People Who Built Them



Stanford, 1980. A single router (the first Cisco) connected two buildings across campus.

In 1980, a man and a woman who had recently fallen in love were both working at Stanford University, in California. They wanted to send each other email.

They could not.

His name was Leonard Bosack. People called him Len. He was in charge of the computers in Stanford's Computer Science Department. Her name was Sandy Lerner. She was in charge of the computers in Stanford's Business School. They had met as graduate students, married, and now worked in two different buildings about a quarter mile apart on the same university campus.

The two buildings each had a network of computers. But the two networks were different. The Computer Science network used one kind of equipment. The Business School network used another. The two networks did not know how to talk to each other. So Len, sitting at his desk in one building, and Sandy, sitting at hers in the other, could not send each other a single character of text.

If they wanted to communicate during the day, they had to use the phone, or get up and walk over to see each other. This was annoying, particularly when you had married the person.

So Len decided to fix it.

He worked, with a Stanford engineer named Bill Yeager, on a small piece of equipment that could sit between the two networks and translate between them. The equipment would read each packet that came in, look at where it was going, and decide which network to send it to. It would also speak more than one network's language, so it could talk to both sides.

What they were building, although they didn't quite call it this at first, was a multi-protocol *router*.

You met routers briefly in Chapter 4 — the IMPs that handled the ARPANET in 1969 were the world's first routers. But the IMPs had been built for one specific network. What Len and Bill Yeager were building was something more general: a router that didn't care what kind of network was on either side. It could connect any network to any other network. As long as all the packets were wrapped in the language Cerf and Kahn had given to the world — TCP/IP — the router would happily move them across.

By 1984, the router worked. Len and Sandy could email each other across campus. Other parts of Stanford asked to use the router too. Soon, the same little box was connecting many of Stanford's networks together.

Other universities started asking for one.

That is where this chapter really begins.

CISCO IN THE LIVING ROOM

In December of 1984, Sandy and Len did something that thousands of people had done before them, and millions of people have done since, but that very few had ever done with a piece of technology that would change the world.

They started a company.

They named it *cisco* — written, in their original style, with a small *c*. The name came from San Francisco — the city they

could see from the hills around Stanford. Their logo, which they designed themselves, was a stylized line drawing of the Golden Gate Bridge.

They ran the company out of their living room.

This is not a figure of speech. Their actual living room, in a small house in Atherton, California, was full of soldering irons, wires, computer parts, and assembled routers being shipped to customers. They worked seven days a week. They had no employees. Sandy did the orders and the books. Len did the engineering. When a router needed to be tested, they tested it on the kitchen table. When a customer called, they answered the phone themselves.

The routers they sold were not pretty. They were rough metal boxes, the size of a stack of textbooks, with little blinking lights and lots of cables coming out the back. But they worked. They could connect almost any kind of network to almost any other kind of network. And in the mid-1980s, that turned out to be a thing the world desperately needed.

Every university had different kinds of computers. Every corporation had different kinds of computers. Every research lab had different kinds of computers. Until Cisco came along, those different computers mostly couldn't talk to each other — or could only talk to each other through expensive, specialized equipment from the big telecom companies. Cisco's router was small, relatively cheap, ran on top of TCP/IP, and would talk to almost anything.

By 1986, Cisco was selling routers as fast as Sandy and Len could build them. By 1987, the company had moved out of the living room and into a real office. They had a small staff. They had customers in dozens of countries.

But they were running out of money.

This is a thing that often happens to small companies that are growing very fast. To make more routers, you have to buy more parts. To buy more parts, you need cash. To get cash, you can either wait for customers to pay you (which takes time) or you can borrow it, or you can sell a piece of your company to investors. Sandy and Len had been turning down investors for years. They wanted to keep the whole company for themselves. But they had run out of ways to grow without help.

In late 1987, they took the meeting they had been avoiding.

SEQUOIA

The investor who eventually said yes was named Don Valentine. He worked at a firm called Sequoia Capital, which is a *venture capital* firm — a company that exists to give money to small, fast-growing companies in exchange for a share of their ownership.

You should understand how this kind of deal works, because it shapes almost every technology company you have ever heard of.

Sandy and Len owned 100 percent of Cisco. They needed cash. Sequoia agreed to give them \$2.5 million. In exchange, Sequoia got about a third of the company. Sandy and Len now owned two-thirds. That sounds like a lot to give up. But Sandy and Len decided it was worth it, because with \$2.5 million they could build a much bigger company much faster, and even one-third of a much bigger company would be worth more than 100 percent of a tiny one.

That part of the calculation turned out to be right. Cisco grew enormously. By 1990, just three years later, the company was big enough to sell its shares to the general public — a process called an *initial public offering*, or IPO. After the IPO, Sandy and Len were, on paper, multi-millionaires.

But there was a catch.

When you take money from a venture capital firm, you also usually take their advice on how to run the company. In Cisco's case, Don Valentine and the other investors had brought in a professional CEO — a man named John Morgridge — to manage the business. Sandy and Len were still co-founders. Sandy was the marketing director. Len was a senior engineer. But they were no longer in charge.

Tensions built. Sandy, in particular, had strong opinions about how Cisco should be run, and clashed with Morgridge and the board. In August of 1990, less than a year after the IPO, the board fired Sandy.

Len followed her out the door the next day in protest.

They sold their remaining shares — for about \$170 million — and walked away from the company they had founded in their living room. Cisco, six years old, no longer had its founders.

This is a hard story to read. Sandy and Len had built something extraordinary. They had been pushed out of it by investors they had let in to help. It is the kind of story that makes some people very cynical about venture capital.

But it is more complicated than that, and I want to tell you the full version.

WHAT HAPPENED TO CISCO, AND WHAT HAPPENED TO SANDY

Cisco, without its founders, grew into one of the most successful companies in history. Through the 1990s, it became the dominant maker of routers on Earth. Almost every internet service provider, every big corporation, every university, every government, bought its routers from Cisco. The company's value rose from about \$200 million in 1990 to, at one breathtaking peak in March of 2000, more than \$500 *billion*. For a short period — a few weeks — Cisco was the single most valuable company in the world.

A great deal of that success, you could argue, came from professional management that Sandy and Len could not, or would not, have provided. The board fired Sandy in part be-

cause they thought the company needed a different kind of leadership to grow. They were not entirely wrong.

You could also argue that Sandy and Len were treated badly. That a company should not have the right to fire the people who created it. That if you build something with your own hands, in your living room, on seven-day weeks, no one should be able to take it from you with a vote of a few investors. Many people argue this too. They are not entirely wrong either.

Both things can be true. This is something to remember about the technology business. The same system that produced trillions of dollars of value, by efficiently moving money to small fast-growing companies, also produced a long line of founders who were pushed out of the things they made. The system is not all good. The system is not all bad. It is a particular set of rules that produces a particular set of winners and losers, and you should know how it works before you decide what you think of it.

Sandy Lerner, for her part, did not need anyone's sympathy.

After Cisco, she co-founded a cosmetics company called Urban Decay, which was created in protest of the pink and beige makeup that dominated stores at the time. It made nail polish in colors with names like *Roach* and *Bruise*. It became enormously successful and was eventually sold to a larger cosmetics company for a great deal of money.

She used her wealth to buy a historic estate in England that had belonged to the family of Jane Austen, the novelist. She turned much of the land into an organic farm. She has written books about Austen. She is, by all accounts, doing exactly what she wants to do, with the money that the company she founded in 1984 in her living room continues to be worth.

Len, more quietly, has spent his post-Cisco decades funding scientific research and education.

The marriage did not last. They divorced in 1990, around the same time they left the company.

WHY I AM TELLING YOU THIS

I am telling you this story for two reasons.

The first is that you should understand that routers — the little blinking boxes that move every packet of every internet message in your life — are not abstract. They were built by specific people. Bill Yeager. Len Bosack. Sandy Lerner. A small staff in a Stanford office and then in a small California town. Without them, or without somebody very much like them, the internet would not have been usable in the form it became. TCP/IP was the constitution. Cisco built the courthouses where the constitution was actually obeyed, one packet at a time.

The second reason is that Cisco is the first really clear example, in this book, of a pattern you are about to see over and over again.

The pattern is this: a piece of foundational technology gets paid for by the public — by ARPA, by the NSF, by universities funded by governments, by international laboratories like CERN. That technology is then released to the world, often for free. *Then*, a small group of clever people figures out how to build a company on top of that free foundation, and gets enormously rich.

Cerf and Kahn gave TCP/IP away. Cisco made many billions of dollars selling the routers that ran TCP/IP. Berners-Lee gave the web away. Other companies, very soon, will make many billions of dollars selling the things that run on the web. Each time, the gift at the bottom is what made the wealth at the top possible.

I do not think this is necessarily bad. Some of what these companies built is enormously useful. But you should see the pattern clearly. The internet you use is a layered cake. The bottom layers were paid for by your grandparents' taxes and given away. The top layers are owned by some of the richest companies that have ever existed.

In the next chapter, we are going to meet a twenty-two-year-old who, in a public computer lab at the University of Illinois, wrote a piece of free software that anybody could use to read the web — and then, less than a year later, walked across the country and turned his free software into one of

the most spectacular get-rich-quick stories in American history.

His name is Marc Andreessen. The free software is called Mosaic. What happened next is the moment everything tipped.

That's Chapter 9.

Chapter 9 — Mosaic, the Pivot

IN THE AUTUMN OF 1992, AT THE UNIVERSITY OF ILLINOIS AT Urbana–Champaign, a tall, restless undergraduate was working a part-time job in a public computer lab. His name was Marc Andreessen. He was six feet four inches tall. He spoke very fast and had a habit of pacing while he thought. He was twenty-one years old.

He was earning, by the hour, about what you might earn now for babysitting.

The computer lab Andreessen worked in was called NCSA — the National Center for Supercomputing Applications. It was part of the University of Illinois, and it was paid for, almost entirely, by the United States National Science Foundation. Which means it was paid for by the American public. Andreessen was a student employee. He had a desk. He had a computer. He had a small group of other students he was friendly with. And he had, somewhere in the back of his mind, the feeling that the World Wide Web — which you read about three chapters ago — was about to become some-

thing much bigger than the few thousand scientists currently using it could possibly imagine.

He just had to make it usable by normal people first.

THE WEB WAS NOT YET PRETTY

In 1992, the web existed, but barely. Tim Berners-Lee had turned on the world's first web server two years earlier. By the autumn of 1992, there were perhaps two or three dozen web servers in the world. The total number of web pages in existence was probably in the low hundreds.

To visit one of these pages, you needed a *browser* — and the browsers that existed in 1992 were not very good.

Berners-Lee had written the first browser himself, on his NeXT computer at CERN. But it only ran on NeXT computers, which were extraordinarily rare. Other people had written simple text-only browsers that ran on Unix workstations — the kind of computers used by scientists, with green text glowing on black screens. These browsers worked, but they could only show text. Not images. Not colors. Not the way you would actually want to look at things. And they were hard to install. You had to be a fairly skilled engineer just to get one running.

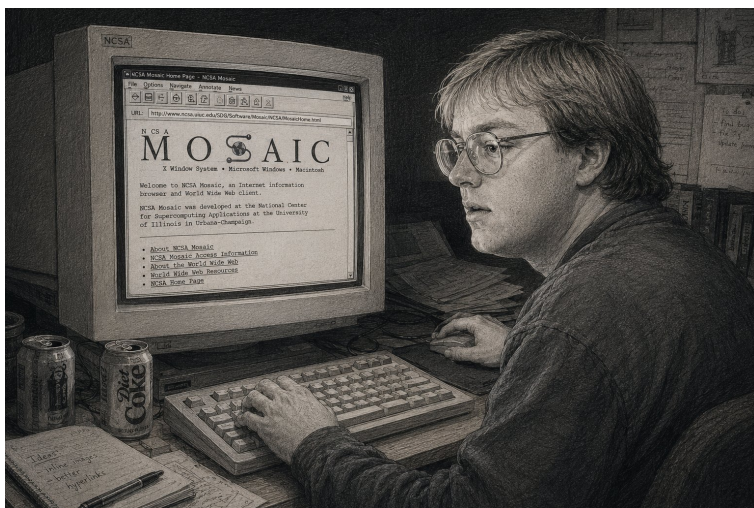
Andreessen and a colleague at NCSA — a more experienced programmer named Eric Bina — decided to build a different kind of browser.

They wanted one that ran on the computers normal people actually had, which in 1992 mostly meant Windows machines and Macintoshes. They wanted one you could install by clicking a single button, without any technical knowledge. And they wanted one that could show *images* directly inside a web page, mixed with the text, the way the pages of a magazine mix pictures and words.

This last idea — putting images inside web pages — sounds obvious to you. Every web page you have ever looked at has pictures in it. But in 1992, this was a strange and slightly controversial idea. Berners-Lee himself had not really intended images to be a central part of the web. He had imagined the web as a tool for sharing documents, mostly scientific ones, with words. Andreessen and Bina thought, instead, that images would be the thing that made the web feel alive — that turned it from a tool for scientists into something the rest of the world might want to use.

They were right.

MOSAIC



Marc Andreessen and Mosaic — the browser that made the web look like something.

They called their browser *Mosaic*. The name suggested little pieces — text, images, links — fitted together into one picture. Andreessen did most of the user interface. Bina did most of the underlying engine. They worked, mostly, at night and on weekends, eating bad food and drinking enormous amounts of Coca-Cola.

The first version of Mosaic was released, free, in January of 1993. It ran on Unix. Versions for the Mac and for Windows followed later that year. The whole thing — created by undergraduate and graduate students at a public university, paid for by public money — was given away to anyone who wanted it.

You should be feeling something familiar at this point. This is the same pattern you have seen over and over in this book. The work was paid for by the public. The result was released to the public. Mosaic was free.

What happened next was not the same pattern. What happened next was something new.

Within months of its release, Mosaic was being downloaded by the thousand. By mid-1993, by the tens of thousands. By late 1993, the number of web servers in the world started to grow not by handfuls but by hundreds at a time, because suddenly there was a browser that anyone could install, that ran on any computer, and that showed beautiful pages with pictures. People who had never heard of the web were now setting up their own pages. Newspapers started writing about it. Television news started covering it. The web stopped being a thing scientists used and started being a thing anyone could use.

Andreessen had become, in less than a year, one of the most talked-about people in the small but rapidly growing world of computing. He was twenty-two years old. He had also become a problem for his employer.

THE WALK OUT

The University of Illinois — which technically owned Mosaic, since Andreessen and Bina had written it as part of their jobs at a public lab — started doing what universities often

do when one of their employees creates something enormously valuable. They tried to take control of it.

NCSA management began to bring in more senior staff to run the Mosaic project. They started talking about commercializing it themselves, licensing it to companies for fees. Andreessen, who had created the thing, was politely pushed out of leadership. By the time he graduated, in December of 1993, the project he had started no longer really belonged to him.

He moved to California a few weeks later. He had no job. He had no specific plan. He had a degree in computer science, a small amount of fame in technology circles, and a deep frustration with the way his own university had treated him.

That's when he got the email.

The email was from a man named Jim Clark. Clark was a Stanford professor turned entrepreneur who had founded a famous company called Silicon Graphics in the 1980s. He had recently left Silicon Graphics, after a falling-out with its board. He was looking for a new idea. He had been following the Mosaic story. He wanted to meet.

Andreessen, twenty-two years old, met Clark, who was about thirty years older, at a coffee shop in Palo Alto. They talked for hours. By the end of the meeting, Clark proposed: let's start a company. You round up the best engineers from the NCSA Mosaic team. I'll put up the money. We'll build a browser that is even better than Mosaic, and we will *sell* it.

Andreessen said yes.

In April of 1994, Clark and Andreessen founded a company called Mosaic Communications Corporation. They flew several of the original Mosaic engineers — Bina among them — out from Illinois to California, and hired almost the entire NCSA team. The University of Illinois was furious. It sent angry lawyers. Eventually the university and the new company settled, with the company agreeing to pay Illinois several million dollars and to change its name. They renamed it *Netscape*.

The new browser, Netscape Navigator, was released in December of 1994. It was faster than Mosaic. It was prettier. It had more features. And, unlike Mosaic, you were going to have to pay for it — at least if you were a company. (Netscape gave it away for personal use, which was clever marketing. Almost everyone used it personally, and then the companies they worked for had to license it.)

By the spring of 1995, Netscape Navigator had something like seventy or eighty percent of the entire web-browser market. The thing that had been Mosaic — free, public, given away — had become the thing that was Netscape: a private company growing faster than almost any company in American history.

In August of 1995, less than two years after Andreessen had graduated from college, Netscape decided to sell shares of itself to the public. It was going to do an *initial public offer-*

ing — an IPO. The kind of event you read about in the last chapter.

Nobody knew, going into the morning of August 9, 1995, exactly what would happen. The company had been operating for sixteen months. It had never made a profit. It was selling a product — a web browser — that almost no investor really understood. The original price they planned to sell shares for was \$14 each. At the last minute, the bankers running the offering doubled it to \$28, because demand seemed so strong.

What happened that day became the defining moment of an entire era.

That's the next chapter.

THE OTHER SIDE OF THE TABLE

But before we go to August 9, I want to skip ahead, briefly, to tell you what eventually happened to Marc Andreessen.

Netscape, in the end, did not survive. The reason it did not survive was that the largest software company in the world at the time — Microsoft — saw the web coming, decided it could not afford to let Netscape control how people got there, and built its own browser, called Internet Explorer. Microsoft gave Internet Explorer away for free, *forever*, and bundled it with the Windows operating system that almost every computer in the world ran. Netscape, which was char-

ging companies for its browser, could not compete with free-and-already-installed.

This was a real war, fought through the late 1990s. Netscape's market share fell from about eighty percent to under twenty percent in a few years. The company was eventually sold to a much larger internet company called AOL, in 1999, for about \$10 billion in stock. The team scattered. Andreessen, in his late twenties, was suddenly rich, and suddenly unemployed.

He started another company. He sold it for \$1.6 billion. He started a third company, with a friend named Ben Horowitz.

The third company was different. The third company was not a software company. It was a *venture capital firm*. It was a company that, like Don Valentine's Sequoia in the last chapter, gave money to small fast-growing technology companies in exchange for a share of their ownership.

The firm was called Andreessen Horowitz. People shortened the name to *a16z* — the "16" stood for the sixteen letters between the A of Andreessen and the z of Horowitz. They opened in 2009. Within a decade, they were one of the two or three most influential investment firms in technology. They managed tens of billions of dollars of other people's money. They invested in some of the most famous companies of the last fifteen years: Facebook, Twitter, Airbnb, Coinbase, GitHub, Slack, and many more. In recent years, they have become some of the biggest backers of artificial intelligence

companies — the kind of companies you will read about at the end of this book.

Marc Andreessen, the twenty-one-year-old who once wrote free software in a public computer lab on six-and-a-half dollars an hour, is now, in his fifties, one of the most powerful private individuals in technology. He decides which other people's free software becomes the next private company worth billions. He sits on the other side of the table from the next generation of Marc Andreessens.

I want you to hold this in your mind as you keep reading. The pattern of this book is not just about specific companies — it is about specific people. The young people who get paid by public institutions to build the foundations sometimes grow up to become the older people who help private companies build the towers on top. Sometimes the same person does both, in different decades of their own life. The kid who got the gift is sometimes the adult who decides who gets the next one.

This is not a moral judgment. It is a description. Whether you find it inspiring, or troubling, or both, is up to you.

And now, with all that in mind, we can go back to the morning of August 9, 1995.

The bankers had doubled the price of Netscape's shares overnight, to \$28. The stock was about to start trading. Marc Andreessen, age twenty-four, was about to learn what his last two years of work were worth, in front of the entire watching world.

That's Chapter 10.

Chapter 10 — The Day Everything Tipped

ON THE MORNING OF AUGUST 9, 1995, IN A SMALL OFFICE IN Mountain View, California, a twenty-four-year-old man lay down on the carpet and tried to take a nap.

He was exhausted. He hadn't slept in two days. He was the chief technical officer of a company called Netscape, and at any minute now — he didn't know exactly when — the stock market was going to start trading shares of his company for the first time in history. He would learn, in the next few hours, whether the work he had done over the past two years was worth a thousand dollars, or a billion dollars. He would learn it in front of the entire watching world.

The man's name was Marc Andreessen, and you have already met him.

He fell asleep on the floor anyway.

WHAT AN IPO REALLY IS

Before I tell you what happened that morning, I should explain, more carefully than I did in Chapter 8, what was actually happening.

When a company is small and privately owned — by its founders, by a few employees, by a venture capital firm or two — there is no real *price* for that company. People can guess what it might be worth. They can argue about it over dinner. But there is no public number. The company exists in the world, and only the people who already own pieces of it know what those pieces cost.

When a company *goes public*, this changes. The company sells a portion of itself — not all of itself, usually a chunk of about ten or fifteen percent — to the general public, through a stock exchange. Anybody in the world with a brokerage account can now buy a share. The price these shares are sold for, multiplied by the total number of shares the company has, gives you the *market value* of the company: the price the public has voted that the whole thing is worth.

For Netscape, the moment of public voting was scheduled to start at 9:30 in the morning, New York time, on August 9, 1995. The shares had been priced the night before. The original plan, weeks earlier, had been to sell them at \$14 each. As demand built up, the price was raised to \$21. As more demand poured in over the final twenty-four hours, the bankers at Morgan Stanley — the firm running the offering

— decided, in a late-night meeting, to double the price one more time, to \$28.

This was already strange. Most IPOs do not double their price the night before they begin trading. There was a feeling, in the air, that something unusual was happening. People around the country, around the world, wanted to buy Netscape stock more than they wanted to buy almost any stock that had ever been offered.

The stock was supposed to start trading at 9:30. It didn't.

It didn't start at 9:31, or 9:45, or 10:00 either.

THE DELAY

What was happening, in the offices of NASDAQ — the electronic stock exchange where Netscape was about to begin trading — was a problem so unusual that even the experienced traders didn't quite know what to do with it.

Hundreds of thousands of buy orders had flooded in overnight. There were almost no sell orders. The price was supposed to be discovered by matching buyers with sellers. But there were no sellers. Almost every single person who had been allowed to buy Netscape shares at \$28 the night before — the company's employees, the bankers' favored clients, the early investors — wanted to *keep* their shares. They had a feeling the price was about to go up. So nobody was selling.

You cannot have a stock market with only buyers. Somebody has to sell. So NASDAQ paused, and the bankers spent hours on the phone trying to find people willing to part with their shares — and at what price.

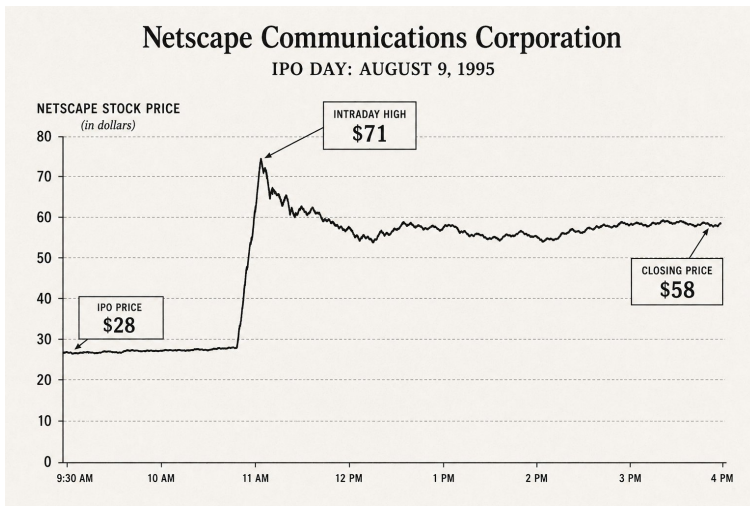
It took almost two hours.

At about 11:00 in the morning, New York time, Netscape's stock finally began trading.

The first sale was not at \$28.

It was at \$71.

WHAT THAT NUMBER MEANT



Netscape's first day of trading, August 9, 1995. The opening of the bubble.

In one and a half hours, before a single share had actually changed hands, the price of Netscape had more than

doubled from what insiders had paid the night before — and quintupled from what the bankers had originally planned to charge.

The stock kept rising through the morning. At one point it traded as high as \$75. It bounced around, then settled down. By the end of the trading day, Netscape closed at \$58.25.

Now multiply that closing price by the number of shares the company had. The math came out to about \$2.9 billion.

Netscape — a company that was sixteen months old, that had never earned a profit, that sold a product almost nobody in the financial world had personally used — was now worth, in the opinion of the stock market, more than General Dynamics, which built fighter jets. It was worth more than United Airlines. It was worth more than most companies that had been operating for decades, employing thousands of people, with billions of dollars in revenue.

Marc Andreessen, twenty-four years old, owned about one percent of Netscape. On paper, by the end of the day, he was worth about fifty-eight million dollars.

His co-founder, Jim Clark — the older man who had emailed him in the coffee shop — owned about a quarter of the company. On paper, by the end of the day, he was worth almost seven hundred million dollars.

The original Mosaic engineers from Illinois, who had taken pay cuts and stock options when they moved out to California, were also, suddenly, multimillionaires. The Se-

quoia-style venture capitalists who had funded the company had multiplied their investment several times over in sixteen months. The bankers at Morgan Stanley earned an enormous fee.

The next morning's *Wall Street Journal* called it the most extraordinary stock offering of the decade. Within a few months, *Time* magazine would put Andreessen on its cover — barefoot, sitting on an ornate velvet throne, with the headline *The Golden Geeks*. He looked, in the photograph, both like a king and like a kid who had been talked into wearing a costume. He was both.

WHY IT MATTERED

I have spent two pages on a single trading day, and you might be wondering why. The story I just told you is, on the surface, just one company going public. Companies go public every week. Most of these stories are boring.

Netscape's IPO mattered for a reason that took most people a few years to fully understand. It rewrote the rules of how technology companies were valued.

Until that day, the standard belief on Wall Street had been that a company should earn profits before going public. You built a business. You made it work. You showed several years of growing revenue. You showed actual money coming in. *Then* you sold shares of yourself to investors who could see the numbers and decide what they were worth.

Netscape had none of that. It had been operating for sixteen months. It had never made a single dollar of profit. The product it sold — a web browser — was something most professional investors had never used and could barely describe. By every traditional rule, Netscape was much too young, much too unprofitable, much too speculative to go public.

The market did not care.

The market — meaning the millions of buyers around the world placing orders — had decided that what mattered about Netscape was not its current profits but its possible *future*. Netscape sold the browser most people used to look at the web. The web was growing very fast. Whoever owned the dominant browser, the buyers reasoned, would own a piece of whatever the web turned into. That piece, in the future, would be worth a great deal of money. So the buyers were willing to pay, today, for what Netscape *might* become tomorrow.

This was not, in itself, a crazy way to think. People bet on the future all the time. But it was a new way for the stock market to think about technology companies. And it opened a door.

Once Netscape had proven that a young, unprofitable internet company could go public and become worth billions of dollars in a single day, every other young, unprofitable internet company began to wonder: *why not us?*

Why not us, said the founders of an online bookstore called Amazon, founded the year before.

Why not us, said the founders of an online auction site called eBay, founded that very year.

Why not us, said the founders of a thousand other companies, most of which you have never heard of, most of which would not exist five years later. Companies that sold pet food online. Companies that sold groceries online. Companies that sold furniture online. Companies that didn't sell anything specific, but had a great idea, and a Web site, and a name that ended in *.com*.

For the next five years, money poured into anything that called itself an internet company. Most of it was lost.

You'll hear about that in the next chapter.

THE COMPLICATED PART

Now I have to be careful, because this is where the story gets harder to tell honestly.

A lot of adults will tell you that the Netscape IPO was a kind of original sin — the moment when American technology stopped being about building useful things and started being about getting rich. They will say it created a culture of greed that has only gotten worse since. They will point at all the failed dot-com companies, all the ruined retirement accounts, all the founders who walked away with millions while their employees walked away with nothing, and they will say: this is what August 9, 1995 set in motion. Without that day, none of those bad things would have happened.

There is truth to this.

A lot of other adults will tell you that the Netscape IPO was one of the most important productive events in modern economic history. They will say it unleashed an enormous wave of capital — money — into building the actual physical infrastructure of the internet. The cables on the seafloor you read about in Chapter 7. The data centers you'll read about later. The companies that survived the dot-com crash and went on to do extraordinary things. Amazon. Google. Many others. None of them, these adults will say, would exist without the wave of investor enthusiasm that Netscape unleashed.

There is also truth to this.

Both groups of adults are partly right. The Netscape IPO did both things. It poured an enormous amount of money into useful infrastructure that is still serving you today. It also poured an enormous amount of money into wasteful nonsense that vanished within a few years and took people's savings with it. It made some young people very rich for reasons that had something to do with their own work and something to do with pure luck — being in the right room at the right time when the bankers came knocking. It also funded the engineers who would build the systems that ran your grandmother's video.

This is, in fact, the most honest summary I can give you of how much of the modern economy works. It is not all good. It is not all bad. It produces wonders and disasters at the

same time, often from the same actions. The kids who eventually grew up to read books like this one have a complicated job: to look at the world as it actually is, to see both the wonders and the disasters clearly, and to decide, when they themselves are grown, what they want to keep and what they want to change.

A simpler version of this story — one where Netscape was either a triumph of capitalism or a betrayal of it — would be easier to tell. I do not think it would be true.

WHAT IT FELT LIKE

I am going to end this chapter with one more small image, because I want you to feel what August 9, 1995 was actually like in the world.

Most people, on that day, did not know it was happening. They went to work. They picked up groceries. They watched their children's baseball games. They watched the news that evening and saw a short story about a company called Netscape that had just done very well on Wall Street. They might have nodded. They might have turned the channel.

But in certain rooms — in offices in Silicon Valley, in trading floors on Wall Street, in basements where people were already trying to build their own internet companies — the world changed quietly that day. Something became possible that had not been possible before. Some signal was sent

that hundreds of thousands of ambitious young people, all over the world, would respond to over the next five years.

If you look at a chart of the United States stock market — the NASDAQ in particular, which is where most technology companies trade — you will see, starting in mid-1995, a line that begins to climb steeply. It climbs and climbs. By the spring of 2000, it has more than tripled.

Then it falls. Hard.

The next two chapters are about that climb, and that fall, and the people who lived through both.

Chapter 11 is about the climb — the years when an entire country, briefly, lost its mind about the internet.

It starts with a sock puppet.

Chapter 11 — Sock Puppets and Pet Food

FOR A FEW MONTHS AT THE END OF THE 1990s, ONE OF THE most famous celebrities in America was a sock puppet.

The puppet was a brown, floppy-eared dog with a microphone clutched in his mouth. He had a name: he was the Pets.com Sock Puppet. He had a personality: smart-aleck, self-deprecating, a little goofy. He had a tagline, which he used in every commercial: *Because pets can't drive*. He appeared on *Good Morning America*. He floated as a giant balloon in the Macy's Thanksgiving Day Parade. He starred in a Super Bowl ad in January of 2000 that cost his company more than a million dollars and was watched by about ninety million people.

He worked for a website called Pets.com that sold pet food online.

His company was, at the time, considered an example of the future of business.

His company, less than two years after its founding, would be bankrupt.

His company would lose about \$147 million in its short life, almost all of it spent on advertising and on shipping heavy bags of dog food across the country for less money than it cost to ship them.

Pets.com is today, mostly, remembered as a joke. But for a brief, strange period — a period that came right after the chapter you just read, and right before the one you are about to read — Pets.com and dozens of other companies just like it were treated, by serious investors and serious newspapers and serious people on television, as the future. People believed in them. People put their retirement savings into them. People quit their jobs to start companies like them.

This chapter is about how that happened.

EYEBALLS

The thing to understand about the years between 1996 and 2000 is that a new theory of business had taken hold, and the theory had become so widely accepted, so quickly, that almost nobody in the technology world questioned it anymore.

The old theory of business had been: a company should earn more money than it spends. If you sell lemonade, you should sell each cup for more than the lemons cost you. The faster you make money, the more your company is worth. This had been the rule for hundreds of years.

The new theory was: a company should grow its number of users as fast as possible, even if it loses money on every user, because once you have a huge number of users, you will figure out how to make money from them later.

The new theory had a word for what it cared about. It cared about *eyeballs* — the number of pairs of eyes looking at your website. It also had a word for the money you spent attracting those eyeballs while losing money on every one of them. That word was *burn rate*. A company with a high burn rate was spending its investors' money quickly. In the old theory, this was a sign that something was wrong. In the new theory, a high burn rate was a sign of ambition. It meant you were *growing*. You were *gaining eyeballs*. You would worry about being profitable later.

You should know that there was a small grain of truth in the new theory. Some kinds of internet businesses really do need to grow before they can make money. An auction site, for example, is more useful when more people use it. (One person trying to sell something on a site with no buyers is a sad sight.) So getting big first, and then earning later, was sometimes the right plan.

But the small grain of truth grew, in the late 1990s, into something much bigger and stranger. Companies that sold ordinary physical products — pet food, groceries, furniture, books — were also told that they should grow at any cost. Even when it made no sense. Even when getting bigger only meant losing more money.

The investors who funded these companies were not being completely foolish, but they were being something close to it. They believed — they really believed — that there was a New Economy. That the old rules did not apply. That the internet was changing every part of business in ways nobody could yet calculate. That whoever moved fastest, even if they bled money for years, would emerge at the end as the dominant player in some giant new market. That billions of dollars of losses now would translate to trillions of dollars of profits later.

Some of them turned out to be right. Most of them turned out to be wrong.

THE CATALOG

Let me give you a small catalog of companies from those years, because the specific stories are more useful than any general description.

There was *Webvan*, an online grocery delivery service in California. It raised about \$375 million in its public offering, and went on to spend more than \$800 million building enormous high-tech warehouses across the country. Its idea — that people would order groceries online and have them delivered the same day — was, in the long run, completely correct. (Look around. Your family probably uses something like Webvan now.) But it was right about twenty years too early. Most people in 1999 did not have fast internet connec-

tions at home. Most people did not have credit cards on file with online stores. Most people, when they thought about it, still preferred to go to the supermarket. Webvan was bankrupt by July of 2001.

There was *Boo.com*, a Swedish online fashion store, headquartered in London, with offices in eight countries. It set out to sell trendy designer clothing through a fancy 3D website. The website turned out to require a very fast internet connection that almost nobody in 1999 had. Customers couldn't load the pages. Boo.com burned through \$135 million dollars in eighteen months — its founders flew across the Atlantic by Concorde, the supersonic jet, on a regular basis — and then went bankrupt in May of 2000. The site had been operating, in total, for six months.

There was *Kozmo.com*, a delivery company in major American cities that promised to deliver almost anything — a snack, a movie rental, a book, a quart of ice cream — to your door, by bike messenger, within one hour, with no delivery fee. The customers loved it. They especially loved ordering very small things, like a single candy bar, which the company would then deliver across town for free. Kozmo lost money on essentially every order. It burned through \$250 million dollars and shut down in April of 2001.

There was *theglobe.com*, an online community site that went public in November of 1998. Its stock more than *quintupled* on the first day of trading — it rose 606 percent, the largest one-day IPO rise in history at that point. Its two co-

founders, both twenty-four years old, were on the cover of business magazines. The company never figured out how to make money, and was effectively gone by 2001.

There was *eToys*, an online toy store, which at its peak was worth, on the stock market, more than the entire chain of Toys R Us — a company that ran more than 1,500 actual stores. eToys went bankrupt in 2001.

There were hundreds more. Companies that sold furniture, gardening supplies, medicine, makeup, music, movies, news, advice, gifts, sports gear. Most of them used the word “internet” or “.com” in their names. Most of them are gone.

EVERYONE WAS A STOCK TRADER

What made all of this possible was not just the investors at venture capital firms. It was ordinary Americans.

Sometime in 1997 and 1998, a strange thing happened to the United States. Ordinary people — taxi drivers, dentists, schoolteachers, retired grandparents — started becoming, in their spare time, stock traders. They quit, or scaled back, their jobs. They sat at home in front of computers all day, buying and selling stocks. The newspapers and the television shows talked about almost nothing else. Stories spread of people who had, on a tip from a chat room, made a year’s salary in a single afternoon by buying the right internet stock.

Most of those stories were either exaggerated or selectively told. For every person who made a year's salary in a day, there were many more who lost their savings. But the stories of the winners were what people heard. The stories of the losers were quietly forgotten, because losers don't usually call up the local newspaper.

A few people on television became famous for predicting which internet stocks would go up. They were called *analysts*. A young woman at Morgan Stanley named Mary Meeker — sometimes called the Queen of the Internet — wrote reports that, when they came out, would move the price of stocks by billions of dollars. A man at Merrill Lynch named Henry Blodget set “price targets” for stocks that seemed absurdly high, and then, sometimes, the prices reached them. Several years later, Blodget would be disgraced for, among other things, writing publicly that some stocks were good investments while privately writing emails describing them as terrible.

Almost nobody, in 1999, was famous for predicting that any of these stocks would go *down*.

There was a phrase that everyone used to make sense of what was happening. The phrase was *the New Economy*. The idea was that the rise of the internet had changed the fundamental rules of business, and that the old way of thinking about company values — looking at how much money they earned, comparing to other companies in similar industries — was outdated. In the New Economy, the smart play was to

bet on the future. Anyone who refused to bet was, in a memorable phrase that got repeated everywhere, *just not getting it*.

A few older investors did refuse to bet. The most famous of them was a man named Warren Buffett, who ran an investment company called Berkshire Hathaway. He looked at internet stocks, said he didn't understand them well enough to know what they were worth, and refused to buy any. He was widely mocked for this. He was called out of touch. He was an old man, people said, who didn't understand the New Economy. His company's stock performed poorly compared to the dot-com stocks for several years.

Buffett's company, for the record, is still around today. Most of the companies whose investors mocked him are not.

THE PEAK

The dot-com era reached its highest moment in the first ten weeks of the year 2000.

On January 10, 2000, AOL — which was, at that time, the way most Americans connected to the internet, the company whose voice said *You've Got Mail* when an email arrived — announced that it was buying Time Warner. Time Warner was an enormous old media company: it owned Warner Bros. movie studio, Time magazine, CNN, HBO, and many other things that had existed for decades. AOL, on the other hand, was about fifteen years old, and was a smaller com-

pany than Time Warner by almost every measure that mattered — except one.

AOL's stock was so absurdly valuable that its total market value was much higher than Time Warner's. So AOL was able to use its stock as currency to *buy* the much bigger and older company. The deal was valued at \$165 billion. It was the largest corporate merger in American history.

The chief executives of both companies — Steve Case of AOL and Gerald Levin of Time Warner — shook hands on national television. It was supposed to be the symbolic moment when the New Economy and the Old Economy joined together. It was supposed to be the future.

Less than a year later, the deal would be widely considered the worst merger in American business history. Time Warner's old shareholders would lose hundreds of billions of dollars as AOL's stock — used to pay for the deal — collapsed. But on January 10, 2000, none of that had happened yet. Everyone was still smiling. Champagne was being poured.

Three weeks after the AOL deal, on January 30, 2000, the Super Bowl was played in Atlanta. Seventeen different dot-com companies bought thirty-second advertisements during the game. Each ad cost between two and three million dollars. Pets.com ran one. So did Computer.com. So did On-Money.com, Epidemic.com, Lifeminders.com, and many others you have never heard of, because most of them no longer existed by the time of the *next* year's Super Bowl.

Six weeks after the Super Bowl, on March 10, 2000, the NASDAQ stock index — the index that tracked most technology stocks — reached a number that it would not see again for fifteen years. It closed that day at 5,048.62. It had more than tripled in eighteen months.

That afternoon, in millions of brokerage accounts across America, people felt very rich.

THE CRACKS

Nothing dramatic happened on March 10, 2000. There was no crash. No company collapsed. No alarming news came out. The NASDAQ closed at its all-time high, on a normal Friday, and people went home for the weekend.

On Monday, March 13, the NASDAQ opened lower.

On Tuesday, it opened lower still.

For the first few weeks, nobody panicked. Stock markets go down sometimes. It was probably just a small correction, people said. The New Economy was still real. The internet was still the future. Pets.com was still the future of pet food. The grandmothers and dentists and taxi drivers were still going to be rich.

But the slide kept going. By April, the NASDAQ had fallen 25 percent from its peak. By the summer, more. By the fall, more. The internet companies started reporting their quarterly earnings, and the numbers were ugly. The losses were not narrowing. The burn rates were not turning into

profits. Companies that had promised that scale would eventually make them profitable were running out of cash before scale ever came.

One by one, the companies in the catalog above started running into trouble. Pets.com filed for liquidation in November of 2000, nine months after its IPO. Boo.com had already gone, in May. Webvan would go in July of 2001. Kozmo in April of 2001. eToys around the same time. theglobe.com a little later.

The Pets.com sock puppet was sold, at auction, to a different company that used him in commercials for car loans. He spent a few more years on television. Then he disappeared too.

For tens of thousands of employees of these companies — engineers, designers, marketers, customer service people, warehouse workers — the end was abrupt. One week they had a job and stock options that were going to make them rich. The next week, the company was bankrupt, the stock was worthless, and they were looking for work in a job market that no longer had any room for them. Many of them had bought houses and cars they couldn't afford, because the New Economy was supposed to keep going up. Many lost their savings. Some lost their marriages.

This was the human cost of the bubble. The newspapers, having celebrated the boom, mostly did not write about it.

By October of 2002, two and a half years after the peak, the NASDAQ had fallen by 78 percent. About five trillion dollars of paper wealth had vanished.

WHAT WAS LEFT BEHIND

I have spent this chapter telling you about the companies that died. I want to end by reminding you that not all of them died.

Some companies that started in the dot-com years survived the crash. They survived because they were run carefully, because they sold something people actually wanted, and because they ran out of cash a little less quickly than their competitors. The most famous survivors are companies whose names you know: Amazon, founded in 1994, which sold books online; eBay, founded in 1995, which let people auction off used things to each other; Google, founded in 1998, which was a search engine. (Yahoo!, Google's biggest competitor at the time, also survived, though it eventually became much smaller than Google.)

There were others. A few of them. They had been, mostly, more careful and less famous than the Pets.coms during the boom. They had grown slower. They had spent less. They had focused on actual customers and actual revenue, even when the New Economy talk was telling them to grow at any cost.

When the bubble burst, those companies — slimmer, more disciplined, and now with very few competitors — found themselves in a strange position. The economy had crashed. Everyone was scared. But the *infrastructure* the bubble had built — the cables on the seafloor, the fiber across continents, the data centers — was now sitting around mostly unused. The survivors could buy it, or lease it, for pennies on the dollar.

They did exactly that.

The next chapter is about three of the biggest companies whose deaths made the survivors so powerful. One of them laid cables under the oceans. One of them ran the long-distance phone network. One of them built the equipment that made the internet's switches and routers work. All three of them, at their peak, had been worth hundreds of billions of dollars combined. All three of them, in a few short years, were gone.

What they left behind became, almost literally, the foundation of the next twenty years.

That's Chapter 12.

Chapter 12 — Three Ships and a Fire Sale

IN THE SUMMER OF 1999, A CABLE SHIP WAS MOVING SLOWLY across the cold waters of the North Atlantic. Behind it, far below the surface, it was unspooling a glass-fiber cable as thick as a garden hose, settling it gently onto the seafloor between Britain and the coast of Long Island.

The cable was being laid for a company called Global Crossing. The man who owned Global Crossing was named Gary Winnick. That summer, on paper, Gary Winnick was worth about six billion dollars.

He threw himself a sixtieth birthday party at a hotel in the south of France. People say it cost millions. He flew his friends in from around the world. The cable kept unspooling.

Three years later, that same cable was still on the bottom of the ocean. It was still carrying signals. Nothing about the cable had changed at all.

But Global Crossing was bankrupt. Gary Winnick was no longer worth six billion dollars. The cable he had paid fifteen billion dollars to help lay across the ocean floor was about to

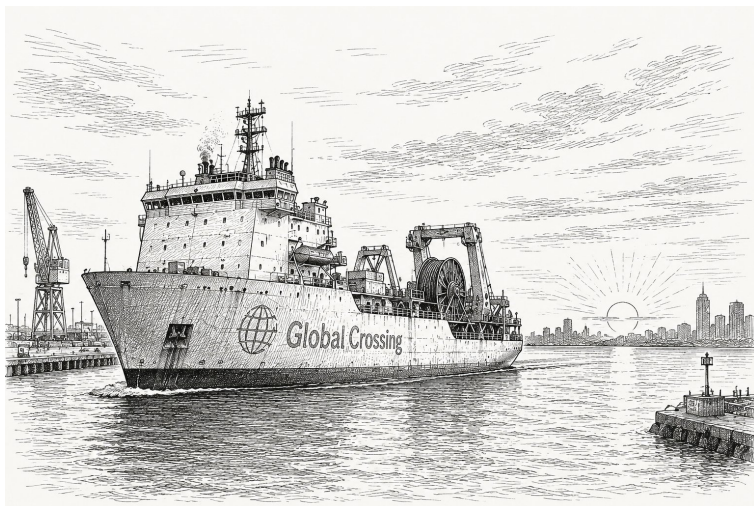
be sold to someone else for less than two cents on every dollar he had spent. The people who had worked for him had lost their retirement savings, because they had been paid partly in company stock, and the stock was now worth nothing.

This chapter is about how that happened. Not just to Gary Winnick, but to two other men as well — one a former milkman from Mississippi, and one who ran a company that, for a little while, made up a third of the entire stock market of Canada.

All three companies built pieces of the internet you use every day. All three companies died, between 2002 and 2009. And the pieces they built — the cables, the switches, the patents — were bought, cheaply, by other companies. Some of those other companies became the richest in the history of the world.

It is one of the strangest patterns in business. Sometimes, the people who build a thing are not the people who end up owning it.

PART ONE: THE MAN WHO WANTED TO WIRE THE OCEANS



Gary Winnick's cable ships set sail to wire the oceans on borrowed money.

Gary Winnick was not an engineer. He had never laid a cable in his life. Before he started Global Crossing, he had been a bond salesman in New York — somebody who sold loans to big investors. He had worked, in the 1980s, for the famous financier Michael Milken.

In 1997, Winnick had an idea. The internet was growing fast. Every email, every website, every phone call between continents had to travel through cables on the ocean floor. There were not enough of them. Whoever owned those cables could charge rent for every bit of information that flowed through them — forever.

So Winnick started a company to lay cables under all the world's oceans, as fast as possible, before anyone else could.

He raised money by borrowing it. A lot of it. Banks lent Global Crossing billions of dollars because the story sounded so good. Investors bought the company's stock for the same reason. Within three years, Global Crossing had laid more than 100,000 miles of fiber-optic cable — enough to wrap around the earth four times. The cables connected New York to London, California to Japan, Europe to Asia.

For a brief moment in early 2000, Global Crossing was worth about \$47 billion on the stock market. That was more than General Motors was worth at the time — and General Motors had been making cars for almost a hundred years. Global Crossing was three years old.

There was just one problem. Gary Winnick was not the only person with this idea.

While Global Crossing was laying cables, a dozen other companies were laying cables too. Qwest. Level 3. 360networks. Williams Communications. All of them had heard the same story: the internet is going to be huge, lay the pipes now, charge rent forever.

But here is the thing about pipes. If one company builds a pipe, that company can charge a lot to use it. If twenty companies build twenty pipes side by side, none of them can charge very much, because customers will just go to whoever charges least.

By 2001, so much fiber had been laid that almost none of it was being used. People in the industry called it *dark fiber* — glass strands sitting in the ocean and underground, with no light pulsing through them. Engineers estimated that about ninety-five percent of the world's fiber was dark. It had cost hundreds of billions of dollars to lay, and it was sitting there doing nothing.

The price companies could charge to send a signal across the ocean collapsed. By some measures, it fell more than ninety percent in two years. But Global Crossing still had to pay back the billions of dollars it had borrowed. The interest on the debt did not get cheaper just because the bandwidth did.

On January 28, 2002, Global Crossing filed for bankruptcy. At the time, it was one of the largest bankruptcies in the history of the United States.

Here is the part of the story that is hard to read.

In the months before the bankruptcy, while ordinary employees still believed the company was healthy, Gary Winnick sold \$735 million of his own stock. The employees couldn't sell theirs — most of it was locked inside their retirement accounts. When the company collapsed, their retirement savings collapsed with it.

Winnick was investigated. He was never charged with a crime. He kept most of his fortune. He died in 2023, still wealthy, in California.

The cables he had laid were still down there, still working. In 2003, a company from Singapore bought what was left of Global Crossing for \$250 million. Remember: Winnick had spent about \$15 billion building it. The new owners had paid less than two cents for every dollar of original cost.

A few years later, the cables changed hands again, sold to a company called Level 3 for about \$3 billion. Level 3 was later swallowed by CenturyLink, which renamed itself Lumen. Today, when you send an email from New York to London, there is a real chance that it travels through a cable Gary Winnick paid to lay, owned now by a company most people have never heard of, which paid almost nothing for it.

PART TWO: THE MILKMAN WHO BUILT A TELEPHONE EMPIRE

Bernard Ebbers was born in Edmonton, Canada, in 1941. As a young man he moved south to Mississippi to play college basketball. After college, he worked, for a while, as a milkman — delivering bottles of milk to people's houses early in the morning. He also coached basketball at a junior high school. He drove a truck for a Holiday Inn. He worked at a clothing factory.

In 1983, he was one of the founders of a small long-distance telephone company in Mississippi. The company changed its name several times. Eventually it became known as WorldCom.

Through the 1980s and 1990s, Ebbers grew WorldCom by buying other telephone companies. Lots of them. He bought sixty or seventy companies in fifteen years. He paid for them, mostly, with WorldCom stock — which kept going up because the company kept getting bigger because the stock kept going up. In 1998, WorldCom made the biggest acquisition in its history: it bought MCI, a much older long-distance phone company that had once been considered the most exciting telecommunications company in America. The combined company was, briefly, the second-largest long-distance phone company in the United States, behind only AT&T.

Ebbers became famous. He bought a 500,000-acre cattle ranch. He bought enormous yachts. He wore cowboy boots and a hat — a persona that played well in newspaper profiles. The story that the *Wall Street Journal* and *Forbes* magazine kept telling about him was that he was a regular guy, a former milkman from Mississippi, who had built one of the most important companies in America.

There was, however, a small problem with WorldCom by 1999. It was no longer growing very fast. The phone business was getting more competitive. The internet bubble was making everyone race to build new networks, and prices were crashing — the same crash that was killing Global Crossing was killing WorldCom too. Profits were falling.

So WorldCom's executives, led by its chief financial officer Scott Sullivan, started lying.

They lied in a particular way. They started counting ordinary operating expenses — the day-to-day cost of running the business — as if they were long-term investments. This was an accounting trick. It made the company's reported profits look much higher than they really were. Each quarter, they did it again. The fake numbers got bigger. Eventually, the gap between what WorldCom said it was earning and what it was actually earning was eleven billion dollars.

This was, and remains, one of the largest accounting frauds in American history.

In June of 2002, the fraud was discovered. WorldCom announced it. The stock collapsed. The next month, on July 21, 2002, the company filed for bankruptcy. At that moment, it was the largest bankruptcy in the history of the United States — bigger than the Enron scandal of the previous year, bigger than anything that had come before.

I want to pause here and tell you about a particular statistic.

For most of the late 1990s, a phrase repeated everywhere — in Congressional testimony, in business magazines, in the speeches of cable-laying CEOs — was that *internet traffic was doubling every 100 days*. This is the statistic that justified building all those redundant fiber cables. It was the number that convinced banks to lend hundreds of billions of dollars to companies like Global Crossing. It was the engine of the entire fiber bubble.

The statistic was, mostly, an invention of WorldCom's marketing department. Internet traffic was growing fast in those years, but more like doubling once a year — not every hundred days. The “every hundred days” number was off by roughly a factor of ten. Almost nobody questioned it at the time, because almost everyone profited from believing it. The cable companies wanted to build more cables. The investment banks wanted to lend more money. WorldCom wanted to look more important. Nobody, in the years it mattered, did the basic math to check.

That fake statistic — repeated by enough people, for long enough, that it became reality in everyone's heads — is one of the reasons the bubble got so big. And one of the reasons the crash, when it came, was so violent.

Bernie Ebbers was tried in federal court in 2005. He was convicted on nine counts of conspiracy, securities fraud, and other crimes. He was sentenced to twenty-five years in prison. Scott Sullivan, who testified against him, got five years.

Ebbers entered prison in 2006. He served thirteen years. In December of 2019, the federal government released him early because he was old and dying. He died at his home in Mississippi a few months later, in February of 2020.

The assets of WorldCom — the long-distance phone network, the giant fiber-optic backbone, the MCI brand — were bought out of bankruptcy by Verizon in 2005. Verizon paid \$7.6 billion for them. Remember: the combined company had

been valued, at its peak, at more than \$180 billion. Verizon got it for about four cents on every dollar of peak value.

The fiber network Verizon got from WorldCom is still, today, one of the most important pieces of long-distance internet infrastructure in the United States. When you send an email from one part of America to another, there is a good chance some piece of it travels on what used to be WorldCom's wires, owned now by a company that paid pennies for them.

PART THREE: A CANADIAN TRAGEDY

I want to end this chapter with a different kind of story. The story of a Canadian company called Nortel.

Nortel did not lay cables. Nortel did not run a phone network. Nortel built the *equipment* that the cables and the networks plugged into. The optical switches. The routers. The boxes you would have seen, in the late 1990s, in every server room of every internet company in the world. If Global Crossing's cables were the highways, Nortel's equipment was the toll booths and the traffic lights.

For most of the twentieth century, Nortel had been a sleepy Canadian company called Northern Telecom. Then, in the 1980s and 1990s, it became one of the leading makers of telecom and internet gear in the world. By July of 2000, Nortel was worth, on the stock market, about \$398 billion

Canadian dollars. It made up about a third of the entire value of the Toronto Stock Exchange.

Stop and imagine that. One company, in one industry, worth as much as a third of all the other major Canadian public companies put together. Tens of thousands of Canadians worked for Nortel directly. Hundreds of thousands of Canadians owned its stock in their retirement accounts, often without even knowing it — because so many Canadian mutual funds held huge positions in Nortel that ordinary investors got it whether they wanted it or not.

When the dot-com bubble burst, Nortel did not die quickly. It died slowly. Slow enough that hope kept returning. Through 2001, 2002, 2003, 2004, executives kept saying the company would recover. Each time, it shrank some more. There was an accounting scandal — executives accused of cooking the books to inflate profits — that dragged on for a decade. There was relentless competition from cheaper Asian manufacturers, especially a fast-growing Chinese company called Huawei. There were layoffs, and more layoffs, and more layoffs.

On January 14, 2009, Nortel filed for bankruptcy.

It is hard to overstate what this meant for ordinary Canadians. Tens of thousands of Canadian retirees, who had owned Nortel stock as the cornerstone of their retirement accounts, watched it become worthless. Tens of thousands of Nortel employees, many of them engineers in their fifties and sixties, lost their jobs and their pensions in a country where,

at the time, finding another job in their field was nearly impossible because their entire industry was contracting at the same moment.

This was the most painful and the least famous of the three collapses. The American collapses had villains — Winnick selling his stock, Ebbers cooking his books. The Canadian collapse had, mostly, just slow grinding loss.

But Nortel did have one thing of enormous value left, even after everything else was gone.

It had patents.

A patent is a legal document that says, *I invented this idea, and I have the right to prevent other people from using it without my permission*. Nortel, over its many decades, had been granted thousands of patents — on wireless technology, on optical switching, on semiconductors, on internet protocols. Many of them were essential to making modern smartphones and networks work. Whoever owned those patents could, potentially, charge fees to every company that built smartphones or networking equipment.

In June and July of 2011, Nortel held an auction for its patents. The auction took place over four days in a law office in Toronto. The largest technology companies in the world sent their lawyers.

The bidding started at \$900 million — an opening offer from Google. Then, in a bidding war that lasted four days, the price climbed higher and higher. Each new bid had to be

at least a certain amount above the previous one. But Google decided, in the middle of the auction, to make bids that were not just higher but *strange*.

Google bid \$1,902,160,540. (This was a number derived from a famous mathematical constant called Brun's constant.)

Google bid \$2,614,972,128. (This was another mathematical constant, called the Meissel–Mertens constant.)

Google bid pi multiplied by a billion: \$3,141,592,653.

Nobody at the auction quite knew what Google was doing. Were they signaling something to the other bidders? Were they amusing themselves? Were they trying to throw off competitors who were trying to predict what Google would bid next?

Whatever they were doing, it didn't work. Google lost the auction.

The winning bid, on July 1, 2011, came from a consortium of companies that had banded together to outbid Google. The consortium called itself *Rockstar*. Its members were Apple, Microsoft, BlackBerry, Sony, and Ericsson. They paid \$4.5 billion.

The Nortel patents went into Rockstar's hands. Rockstar used them, over the next several years, to file lawsuits against companies — including Google's Android phone partners — that built smartphones. The patents shaped much of the legal warfare around smartphones in the 2010s. Eventually, in

2014, Rockstar broke up and sold most of its patent portfolio to a different group, which included Google. The patents had traveled a long way from a quiet Canadian company that nobody remembered.

WHAT ALL THIS MEANS

I am going to end this chapter by asking you to hold three things in your mind at once.

The first thing is that the three companies in this chapter, at their combined peak, were worth more than seven hundred billion dollars. That is more, in the dollars of their time, than the entire economy of most countries. By the end of the decade, almost all of that wealth was gone — vanished — converted into nothing.

The second thing is that the *physical things* those companies built — the cables, the switches, the patents — were not destroyed. They are still here. The cables are still on the seafloor, carrying our videos and our messages. The fiber backbones are still in the ground, carrying our phone calls and our streaming videos. The patents are still being used, every day, to make the smartphones we hold in our hands. Other companies own them now. Some of those companies — Lumen, Verizon, Apple, Microsoft, Google — paid almost nothing for these things, because they bought them in a fire sale, from sellers who had to sell.

The third thing is the part to really sit with. The *technology* was real. The infrastructure was real. The usefulness was real. What was not real was the *price*. When too many people believed the same thing about the future, all at once, they collectively built more than the future could absorb on schedule. The future eventually arrived — it is the world you are sitting in right now, with cables under every ocean, with bandwidth so cheap nobody notices it anymore. But by the time it arrived, the people who had bet on it being early had been wiped out, and a different group of people owned everything.

This is not a story unique to the internet. The same thing happened, in the mid-1800s, with the British railroad boom. Investors built railways everywhere, went bankrupt by the dozens, and left behind a transportation network that the rest of the country used for the next hundred years. It happened, in the 1830s and 1840s, with American canals. It happened, in the 1920s, with the radio industry. It happened, in the 1980s, with personal computers. It happens, more or less, every time human beings collectively decide that a new technology is going to change everything.

They are usually right that it will. They are almost always too early.

This is a fundamental rhythm of how the world gets built. The early believers fund the construction. The construction outruns the demand. The early believers go bankrupt. The

infrastructure remains. The next generation, paying pennies for it, gets to build the next thing on top.

In the next chapter, we will meet a few of those next-generation companies. The ones who survived the wreckage of the bubble. The ones who, when the smoke cleared, found themselves standing on a vast amount of infrastructure that had been built for someone else, and who used it to build things bigger than the bubble had ever imagined.

That's Chapter 13.

Chapter 13 — What Survived

IN THE AUTUMN OF 2001, THE CHIEF EXECUTIVE OF AN INTERNET company that almost everyone in America assumed was about to go bankrupt stood up in front of his employees and asked them not to worry.

Many of them were worried.

The company's stock had been worth \$113 a share two years earlier. It was now worth less than six dollars. Analysts at major Wall Street banks had recently predicted, in writing, that the company would run out of money before the end of the year. The previous January, the company had laid off 1,300 employees — about fifteen percent of its workforce. Newspapers wrote about the company the way they wrote about the other dot-com flops you read about in the last two chapters. It was, the conventional wisdom said, just a matter of time.

The chief executive's name was Jeff Bezos. The company was called Amazon.com.

The conventional wisdom turned out to be wrong.

This chapter is about why.

AMAZON, ALMOST



Amazon's stock, 1997-2003. Almost dead. Still alive.

Jeff Bezos had founded Amazon seven years earlier, in 1994. He was thirty years old at the time. He had quit a comfortable job at a Wall Street hedge fund — where he had been making far more money than most people make in a lifetime — to drive across the country with his wife, in a Chevrolet Blazer, while she did the driving and he wrote a business plan on his laptop in the passenger seat. The business plan was for an online bookstore.

Books, he had decided after thinking carefully about the question, were the perfect product to sell over the early internet. There were too many books in print for any physical store to stock more than a tiny fraction of them. An online store, which only had to keep books in warehouses rather

than on shelves, could carry every book in print. People who bought books already knew what they wanted before they bought it — they didn't need to flip through the actual paper version to decide. And books were small, durable, and easy to ship.

He set up shop in a garage in Seattle. He opened the website on July 16, 1995. He gave it a name from a river in South America — the largest river in the world, by volume, just like he hoped his store would eventually be.

For the next four years, Amazon grew very fast and lost money every single quarter. This was, you should remember from Chapter 11, completely normal for that era. The strategy was *get big fast*. You raised money from investors, you used it to advertise and to expand into new product categories, and you worried about being profitable later.

Amazon's investors were patient. The stock price climbed and climbed through the late 1990s, reaching its peak of \$113 a share in December of 1999. Bezos, on paper, was a multi-billionaire. *Time* magazine named him Person of the Year — its highest annual honor, given that year not to a politician or a humanitarian but to a man who sold books over the internet.

Then March 2000 came, and the bubble burst, and everything turned around.

Amazon's stock fell along with everything else. Then it kept falling. Through the year 2000, then 2001, the price collapsed by more than ninety-five percent. Bezos's personal

fortune, on paper, dropped by tens of billions of dollars in a year and a half. The press, which had loved Amazon during the boom, turned on it. A famous report by an analyst at Lehman Brothers in 2000 had predicted, bluntly, that Amazon's debts would crush it before it ever became profitable.

The strange thing, looking back, is that the analysts were not wrong about the numbers. Amazon really was losing money. Its debts really were enormous. Its products — books, then CDs, then DVDs, then toys — really were sold at razor-thin margins, often at losses. By every traditional measure, Amazon in 2000 and 2001 was exactly the kind of company that should have failed.

The reason it didn't fail is mostly two things.

The first thing is that Jeff Bezos had been writing, every single year since 1997, a letter to his shareholders explaining what kind of company he was trying to build. In every letter, he reprinted, at the end, the original 1997 letter. The original letter was titled simply *It's All About the Long Term*. It explained that Amazon was not trying to maximize quarterly profits. It was trying to build a company that would matter for decades. It was going to take risks that wouldn't pay off for years. It was going to disappoint shareholders who wanted quick returns. Anyone who didn't want this should not invest.

Almost nobody in Silicon Valley wrote letters like that. Bezos wrote one every year, for over twenty years.

The second thing is that in early 2001, when the analysts were predicting Amazon's death, Bezos got serious. He laid off the 1,300 employees. He closed an entire warehouse. He shut down a customer-service center. He cancelled programs that weren't profitable. He demanded that every part of the business explain, in writing, why it deserved to keep existing.

By the end of 2001 — the same quarter when most of the dot-com sock-puppet companies were filing for bankruptcy — Amazon reported its first ever profitable quarter. The profit was tiny. Just five million dollars. But for the first time in seven years, the company had earned more than it had spent.

The stock began to climb back up. Slowly at first. Then faster. By 2007 it was higher than it had been at the bubble peak. By 2010 it had quadrupled from that.

Amazon today, when this book is being written, is one of the largest companies in the world. It has roughly a million employees, more than the entire population of many countries. It owns a major fraction of the global cloud computing market through a service called Amazon Web Services — a business that started, almost as a side project, in 2002. It owns the Whole Foods grocery chain. It owns Audible, the audiobook company. It owns the *Washington Post* newspaper. It builds movies and television shows.

But it nearly died in 2001. If Bezos had panicked, or if his shareholders had forced him to chase short-term profits earli-

er, or if he had not written those long-term letters and made his investors agree to them in advance — it would not have made it.

E BAY, QUIETLY

While Amazon was nearly dying and then recovering, a different internet company was, very quietly, behaving as if the bubble hadn't happened at all.

It was called eBay, and its founder was a young French-Iranian-American computer programmer named Pierre Omidyar.

Omidyar had built the first version of eBay in his living room, over Labor Day weekend in 1995. It was an online auction site — you posted something you wanted to sell, other people bid on it, and at the end of the auction, the highest bidder won. The first item ever sold on eBay was a broken laser pointer, for \$14.83. Omidyar emailed the buyer to ask if he understood the laser pointer was broken. The buyer replied: *I'm a collector of broken laser pointers.*

eBay grew. People sold trading cards, Beanie Babies, old computer parts, antique furniture, and millions of other strange things to each other. Within a year, eBay was doing enough business that Omidyar's web-hosting company asked him to start paying business rates. He started charging a small fee — twenty-five cents — for each item sold. The fee paid for the hosting.

Then, because the site kept growing, the fees added up to more than the hosting costs.

And then, because the fees were almost pure profit (eBay didn't have to buy or ship or store any of the things being sold), the company turned out to be enormously profitable.

This is unusual. Most dot-com companies you have read about lost money for years. eBay made money almost from the beginning. By the time of its public offering in September of 1998, it had already been profitable for several quarters in a row. By the height of the bubble, it was one of the only big internet companies that could honestly say: yes, we make actual money.

There is a particular kind of business that eBay turned out to be, and you should know about it because you will meet it again. It is called a *network business*. A network business gets more useful as more people use it. eBay was useful because there were a lot of buyers — which made it attractive to sellers — which brought in more buyers — which attracted more sellers. Once it had reached a certain size, eBay was nearly impossible to displace, because no new auction site could match its number of users.

The technical word for this kind of advantage is *network effects*. Once a network business gets big enough, it becomes a kind of fortress. New competitors cannot easily attack it. People sometimes call this a *moat*, after the water-filled trenches that medieval castles used for defense.

You should remember this idea. It is the explanation for most of the largest companies you are about to meet in the next part of this book. eBay was the first internet company to clearly demonstrate it. It was not the biggest. It was not the most famous. But it was, in a way, the most quietly important — because it taught a generation of business school students and investors that the right kind of internet business could be unbreakable.

eBay bought a small payment company called PayPal in 2002, mostly so that the buyers and sellers on its site would have an easy way to pay each other. PayPal turned out to be even more interesting than the auctions. (We'll see one of PayPal's founders later in the book.) eBay eventually let PayPal become its own company again, in 2015. Both companies are still very much alive today.

GOOGLE, WATCHING

The third major survivor of the dot-com era is the one I am only going to mention briefly here, because all of Chapter 14 will be about it. But you need to know one strange thing about it now.

The company is called Google. It was founded in 1998 by two PhD students at Stanford named Larry Page and Sergey Brin. (Stanford again, you'll notice. Public-funded research, again. The pattern keeps repeating.) Their company built a search engine that was dramatically better than anything else

available at the time — so much better that, within a few years, almost everyone in the world who used a search engine used theirs.

But Google did something that Amazon, eBay, and the failed dot-coms had all done differently. Google refused to go public during the bubble.

Page and Brin watched the bubble inflate. They watched their friends at less impressive companies become paper billionaires by selling shares to the public. They watched newspapers describe young tech founders who had IPO'd as visionaries. They could have done the same thing. They didn't.

They didn't go public until August of 2004, more than four years after the peak of the bubble. By then, the failed dot-coms had been gone for years. Google was already wildly profitable. Its investors had been patient. And the company that emerged from the IPO process was not a hopeful speculation — it was a stable, dominant business with a clear way of making money.

Google had stayed out of the bubble entirely. They survived by simply not playing the game.

The rest of Google's story will fill the next chapter. For now, I just want you to notice the strategy: when everyone else is rushing, *do nothing*. Sometimes the most important business decision you can make is to wait.

WHAT THE SURVIVORS HAD IN COMMON

Let me draw, before we move on, a few lines under what we have seen.

Of all the hundreds of internet companies that existed in 1999, only a handful came out of 2002 in a position to grow into the giants of today. The handful had different stories. Amazon nearly died and was saved by its founder's stubbornness. eBay was profitable from the start because of a clever business model. Google avoided the whole bubble by refusing to go public during it.

But there were some things the survivors all had in common.

The first thing is that they were all run by founders who thought in decades, not in quarters. Bezos's long-term letter. Page and Brin's patience. Omidyar's quiet attention to whether the actual business worked. None of them was trying to get rich quickly. They had all already passed up easier ways to make a faster fortune. They were trying to build something that would outlast them.

The second thing is that they all had a clear answer to the question: *what would a customer pay us for?* For Amazon, the answer was an immense selection of products, low prices, and fast delivery. For eBay, it was a place where strangers could trust each other to do business. For Google, it was a search engine that actually gave you the answer you were looking for. None of them was, at heart, the *because pets can't*

drive kind of company. They were each selling something concrete and useful, to specific people who actually wanted it.

The third thing is something I want you to think about carefully, because it is uncomfortable.

The survivors became more powerful, not just because they had survived, but because almost everyone else had *died*. The infrastructure built by all those failed companies — the dark fiber under the oceans, the empty data centers, the engineers looking for jobs — was now available, very cheaply, to the few companies still standing. Amazon could rent computing power from cratering server companies for almost nothing. Google could lease the bandwidth it needed to send search results around the world at prices that would have been impossible in 1999. eBay could hire brilliant engineers, fresh from the wreckage of failed dot-coms, who would have been impossibly expensive a year earlier.

The survivors got to grow up in a world that had been emptied out for them.

This is something to hold onto when we get to the very end of the book, and we are looking at the next generation of giant technology companies — the ones that are right now growing up on top of infrastructure that other people built, after other people went broke. It is the same pattern. It will probably repeat.

WHAT'S NEXT

In the next chapter, we are going to spend our full time with one of these survivors — the one that, more than any other single company, defined the next twenty years of the internet. The company that two PhD students started at Stanford in 1998. The company that turned out to be the single most valuable business idea since the discovery of advertising itself.

That's Chapter 14.

It begins, again, with a university computer paid for by the public.

Chapter 14 — Search

IN THE SPRING OF 1995, A 21-YEAR-OLD PHD STUDENT NAMED Sergey Brin was asked to give a tour of Stanford University to a 22-year-old prospective PhD student named Larry Page. Both were studying computer science. Both were unusually smart. Both were the children of professors. Both had unusually strong opinions about almost everything, including each other.

The tour did not go well. They argued, all afternoon, about almost everything they saw. They disagreed about the design of buildings. They disagreed about the ranking of academic departments. They disagreed about the weather. Years later, both of them remembered the day as one of the most irritating they had ever spent.

Page came to Stanford anyway. The next year, when he was looking for a research project for his dissertation, the person he kept arguing with most was, somehow, Brin. They had started, despite themselves, to become friends.

The research project Page eventually chose was very strange. It had no obvious application. It involved studying, in detail, the structure of links on the World Wide Web.

That research project became, in five years, the most valuable business idea since the discovery of advertising itself.

A SIMPLE IDEA ABOUT IMPORTANCE

This is what Page was thinking about in 1996.

The World Wide Web, by then, had grown from Tim Berners-Lee's first server in 1990 to about a hundred thousand pages. (Today, it has hundreds of billions.) Every page was full of *links* — those clickable bits of underlined text that take you from one page to another. Page realized something that nobody else had taken seriously: every link on the web was, in a quiet way, a *vote*.

When the author of one web page chose to link to another web page, they were saying — without saying — *this other page is worth reading*. They were vouching for it. They had no reason to link to something useless. So a page that was linked to by many other pages was probably, on average, a more important page than one nobody bothered to link to.

This was already a useful observation. But Page went one step further, and the next step turned out to be the magic step.

Page noticed that not all votes should count the same. A link from an extremely important page — say, the home page

of the *New York Times*, or a major university — should count for more than a link from some unknown blogger that nobody had ever heard of. The vote of an important page was worth more than the vote of an unimportant page.

But how do you decide which pages are important? Well — by the same rule. Important pages are the ones that *other* important pages link to.

This sounds, when you first read it, like nonsense. You are defining importance in terms of itself. *A page is important if important pages link to it.* It feels like you've made the question harder, not easier. How can you ever start? You don't know which pages are important until you know which pages are important.

The answer, beautifully, turns out to be that you can solve this problem with mathematics — specifically, with a kind of mathematics called *linear algebra* that you will probably study, if you decide to, in a few years. You can model the entire web as an enormous grid of numbers, where each row and column is a page and each cell records which pages link to which other pages. You can ask the math a question that boils down to: *find the set of "importance scores" for all the pages that is consistent with itself.* And, almost magically, the math finds you an answer. It produces a single number for every page on the web — a score of importance that fits together with all the other scores.

Page called the algorithm — partly after himself, partly after the idea that it ranked *pages* — *PageRank*.

He and Brin built a search engine using it. Their search engine, when you typed in a question, would not just find every page that contained your words. It would find them all, and then *rank* them, putting the pages with the highest PageRank scores at the top. So the first result you saw was, mathematically, the most important page on the web that contained what you were looking for.

It was a small idea, in a sense. Just counting links, with a twist. But it produced results that were dramatically better than the search engines that had existed before it.

BETTER THAN ANYTHING ELSE

You need to remember, to understand why Google was such a shock, what searching the web was like in 1997.

There were several search engines available. They had names like AltaVista, Excite, Lycos, Infoseek, Yahoo!. They all worked, roughly, by the same logic: when you typed in a word, they showed you a list of every web page that had that word on it, ordered by some combination of how often the word appeared and how recently the page had been updated. The lists were long. The first page of results was usually full of pages that had simply used the word a lot, often deliberately, to fool the search engines. People who wanted to game the system would write hidden text on their pages, repeating popular keywords thousands of times, to push themselves to the top.

The result was that searching the web in 1997 was usually a frustrating, slow experience. You typed in something. You got a list of fifty or a hundred pages that mostly didn't have anything to do with what you actually wanted. You clicked on the first few, hoped one of them would be useful, gave up, tried a different search. Most people, when they wanted to find something specific, did not search. They went to a place like Yahoo!, which was not really a search engine but a hand-maintained *directory* of websites organized into categories — Sports, News, Travel, Music — and clicked through the categories.

When Page and Brin's search engine came online at Stanford in 1997 — running on machines borrowed from various Stanford labs, paid for by federal research grants — people started using it, and they started telling other people to use it. The reason was simple. When you typed in a search, the first result was usually the page you wanted. Not the fifth. Not the twentieth. The *first*.

The students at Stanford started using it constantly. Word spread to other universities. By early 1998, Page and Brin were getting more than ten thousand queries a day. They had to keep finding more Stanford computers to plug into their system, because they kept running out of room.

It was clear they had something important. The question was what to do with it.

THE STANFORD PATENT

I want to pause here, because something interesting was happening with the legal ownership of the idea.

Larry Page and Sergey Brin had developed PageRank as part of their dissertation research, at Stanford University, on Stanford's computers, paid for partly by the National Science Foundation. By the standard rules of American universities, the *university* owns inventions that its researchers create while doing university work. PageRank, technically, belonged to Stanford.

In 1998, Stanford filed a patent on the PageRank algorithm. The patent listed Larry Page as the inventor, but the *owner* of the patent was Stanford.

What happened next is unusual in the story of this book. Usually, when public-funded research becomes private wealth, the universities that paid for it get nothing. Cisco's routers came from Stanford and Stanford got little in return for years. Mosaic came from the University of Illinois and Illinois got a small settlement after a lawsuit. But this time, the lawyers at Stanford were paying attention. They licensed the PageRank patent to Page and Brin's new company — and as payment, they took 1.8 million shares of the company in return.

For most of the next two decades, those shares grew in value. When Stanford eventually sold the last of them, the

university had collected about \$336 million from a single patent that two of its graduate students had filed.

A third of a billion dollars, for a public university, from one piece of student research.

This is still rare. Most American universities, even today, do not capture much of the financial value of the things their researchers invent. Stanford did, in this case, partly because they had been burned by missing out on earlier opportunities, partly because they had grown smarter about technology licensing, and partly because Page and Brin themselves wanted the university to get a fair share.

It is one of the small, hopeful patterns in this book. Not all gifts to the public are taken without thanks.

THE GARAGE



The garage on Alta Avenue in Menlo Park, where Google rented its first office.

By the summer of 1998, Page and Brin had been trying to figure out what to do with their search engine. They didn't really want to start a company. They wanted to finish their PhDs. So they tried to *sell* the search engine to one of the big internet companies of that time, and then go back to school.

They went to Excite, one of the major search portals of the day. Excite said no. They went to Yahoo!. Yahoo! said no. They went to AltaVista. AltaVista said no. Nobody, in early 1998, thought a better search engine was particularly valuable. The big companies had decided that the future of the internet was not search but *portals* — websites like Yahoo! that would be the starting point for everyone's web brows-

ing, mixing news and email and weather and shopping into a single place. Search was just one small feature of a portal. Why would they pay a million dollars for a slightly better version of one small feature?

Page and Brin were going to drop the whole project and go back to their studies.

In August of 1998, they were introduced to a man named Andy Bechtolsheim. Bechtolsheim was one of the co-founders of Sun Microsystems, an older Silicon Valley company. He was rich. He was an early-stage investor in promising new companies. The introduction was made by a Stanford professor.

The three of them met early one morning on the porch of another professor's house in Palo Alto. Page and Brin demonstrated their search engine on a laptop. The demonstration lasted maybe fifteen minutes. Bechtolsheim watched. He nodded. He said: *this is the most interesting thing I've seen in a long time. Why don't I write you a check.*

He pulled out his checkbook. He wrote a check for \$100,000, made out to *Google Inc.* He handed it to Larry Page.

There was, at the time, no company called Google Inc. The name didn't exist yet. Page and Brin had been calling their search engine *BackRub*, and then, after thinking about a very large number called a *googol* (which is a 1 followed by 100 zeros), had tried to name it *Googol* and accidentally misspelled it on a domain registration. The misspelled version stuck.

Now they had a check for \$100,000 made out to a company that didn't exist. The check sat in a desk drawer for two weeks while they figured out how to incorporate. On September 4, 1998, Google Inc. became a legal entity. The check was deposited.

A few weeks later, they moved out of their dorm rooms and into a garage. The garage belonged to a young woman named Susan Wojcicki, who rented part of her house in Menlo Park to her sister's friends. Wojcicki later joined Google herself. Years after that, she became the CEO of YouTube. The garage, where the entire company fit for about a year, is now owned by Google and preserved as a kind of museum.

WHAT WAS THE BUSINESS?

For the first two years, Google had a wonderful search engine and millions of users — and almost no idea how to make money.

This was the central problem of the late 1990s internet. People loved the products. They didn't pay for them. The dot-com companies you read about in Chapter 11 had all been struggling with the same problem, and almost all of them were dying or had died from it.

Page and Brin tried various things. They licensed Google's search to other websites — Yahoo! itself, by 2000, was paying Google to provide the underlying search results for Yahoo's

portal. This brought in some money. But it wasn't enough to build the kind of company they wanted.

Then, in 2000, they made the decision that would define everything.

They started selling advertisements.

Not the giant flashing banner ads that the dot-coms had used, which everyone hated and almost everyone had learned to ignore. Something else. Something small, quiet, and matched to what people had typed in.

When you searched for *running shoes*, alongside the regular search results, Google would show you a few small text ads — clean, simple, indistinguishable from the regular results except for a quiet label that said *sponsored*. The ads would be for running shoes. Advertisers had bid, in an automatic auction, for the right to show their ad to someone searching for running shoes. Whoever bid highest, and whose ads got clicked on the most, won that slot.

Google called the system *AdWords*. It went live in October of 2000.

I want you to stop and notice what this was, because it was not obvious at the time, and it changed the world.

For most of human history, advertising had been about shouting at strangers. A company that made running shoes would buy a TV ad, and the ad would be seen by everyone watching that TV channel — most of whom did not want, and were not thinking about, running shoes. The advertiser

was paying to reach a million people in the hope that a few hundred of them might, eventually, want running shoes. It was incredibly wasteful. Almost everyone in advertising knew it.

What AdWords did was different. AdWords showed running-shoe ads only to people who had just *typed the words “running shoes” into Google*. Those people had announced, by their search, that they were actively looking for running shoes *right that minute*. They were not random strangers. They were people whose interest in running shoes was the most certain thing about them at that moment.

The economic value of this was almost unimaginable. The advertiser was no longer paying to reach a million strangers. The advertiser was paying to reach the specific people who were already looking for what they sold. The waste of traditional advertising had been almost completely eliminated.

Within a few years, Google was earning billions of dollars a year from AdWords. Within a decade, it was earning tens of billions. Within two decades, hundreds of billions. The company you and your family use every time you type a question into a search bar makes more money — many times more — than almost any other company on Earth.

The thing that made it all work, the thing that turned a research project into one of the most valuable businesses in history, was not the search engine itself. The search engine was just the bait. The thing that made Google rich was that people, when they searched, *told Google what they wanted to*

buy. Google sold that information to advertisers, in real time, for enormous amounts of money. The free thing — search — was excellent, and it stayed excellent, because the better it was, the more people would use it, and the more information they would reveal about themselves, and the more valuable the advertising would become.

This is the business model of almost every “free” product you use on the internet today. Search engines. Email. Maps. Social networks. Video sharing. Almost none of those services charge their users any money. They are paid for by advertisers who buy access to those users’ attention, and to information about those users’ interests, gathered from how they use the free services.

When somebody offers you something valuable for free, on the internet, somebody, somewhere, is paying for it. Usually, what’s being sold isn’t the thing you’re using. It’s you.

I am not telling you this to make you angry. The trade is, in some ways, a reasonable one. You get search for free. The advertiser gets to find people interested in running shoes. The internet you grew up with — the one that has all this remarkable free stuff in it — exists because of this exchange. It is not, by itself, a bad bargain.

But you should understand it clearly, because it is the financial engine that powers a very large fraction of the modern internet, including the companies you’ll meet in the next few chapters. Once you see it, you can’t unsee it. You will notice, every time you tap an app on your phone, the

small ways the system is paying attention to you so that it can show you the right ads, so that the people who built the app can keep making money, so that the app can stay free, so that you and a billion other people will keep using it, so that the cycle keeps going.

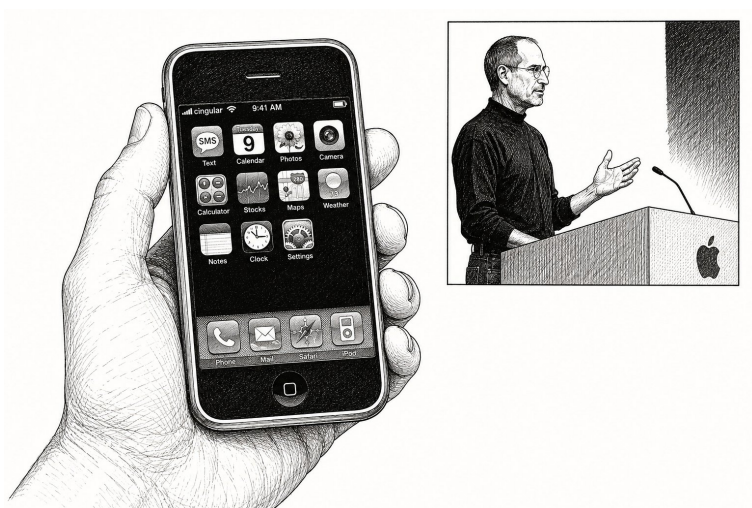
This system was invented, in its modern form, in a garage in Menlo Park in 1998 and 1999, by two graduate students from Stanford who had been trying to find a better way to rank web pages.

That is one of the strangest, and most important, sentences in this whole book.

In the next chapter, we meet the device that brought this entire system into our pockets, every minute of every day, all over the world: the iPhone.

That's Chapter 15.

Chapter 15 — The Phone in Your Pocket



January 9, 2007 — Steve Jobs introduces the iPhone.

On January 9, 2007, a thin man in a black turtleneck walked onto a stage in San Francisco and proceeded to give what is probably the most famous product demonstration in the history of computing.

His name was Steve Jobs. He was the chief executive of Apple, a computer company that had nearly gone bankrupt

ten years earlier and was now in the middle of the most dramatic corporate comeback any large company had ever made. He had stage presence that nobody else in the business world has ever quite matched. He spoke slowly, with long pauses. He wore the same outfit to every public appearance — black turtleneck, faded jeans, gray sneakers — so that nothing about his clothing would distract from what he was saying.

What he was about to say, on that January morning, would change the way human beings spent their days for the foreseeable future.

“Today,” he said, looking out at an audience of about three thousand people, “Apple is going to reinvent the phone.”

He walked the audience through what was wrong with the cell phones of the time. They were ugly. Their buttons were tiny. Their screens were small. Their web browsers were terrible. Their cameras were bad. They were difficult to use, and even the most expensive ones — the BlackBerrys that businesspeople carried, the early smartphones from Palm and Nokia — felt clunky and limited.

Then he said: today we are introducing three revolutionary products. A wide-screen iPod with touch controls. A revolutionary mobile phone. And a breakthrough internet communications device.

He paused. The audience cheered, because each one of those things would have been impressive on its own. Apple already made the best music player in the world. A revolutionary phone would have been huge news. An internet

device that could fit in your pocket would have been astonishing.

Then he said it again. *An iPod. A phone. And an internet communicator.* He repeated it. *An iPod. A phone. And an internet communicator.* The repetition was deliberate. The audience started to realize.

“Are you getting it?” he said. “These are not three separate devices. This is one device. And we are calling it iPhone.”

The audience went wild. Steve Jobs, the showman, the salesman, the man who had spent six months rehearsing this exact moment, smiled.

The world your parents grew up in was about to end.

APPLE'S NEAR DEATH

To understand why January 9, 2007 was such an important moment, you have to know what Apple was a decade earlier.

In 1997, Apple was almost bankrupt. The company that had been founded in a garage in 1976 by Steve Jobs and Steve Wozniak, that had become famous in the early 1980s for the Macintosh computer, had spent most of the 1990s slowly losing to Microsoft. Microsoft Windows ran on about ninety-five percent of personal computers in the world. Apple's Macintosh ran on about three percent. Apple was losing hundreds of millions of dollars a year. Most experts thought it would not survive another two years.

There was a strange twist in Apple's story. Steve Jobs had been forced out of the company he had co-founded, back in 1985, after fights with the board of directors. He had spent the next twelve years running two other companies — a computer firm called NeXT (the same NeXT, you may remember, that made the computer Tim Berners-Lee used to invent the World Wide Web), and an animation studio called Pixar, which would eventually make *Toy Story* and become a Hollywood giant.

In 1997, with Apple about to go under, Apple bought NeXT — partly to get the technology, partly to bring Steve Jobs back. He became Apple's CEO for a second time. Almost immediately, Microsoft — Apple's biggest rival — gave Apple \$150 million as part of a deal to keep Apple alive, because Microsoft worried that if Apple died, the government might decide Microsoft was an illegal monopoly and break it up. (This is one of the strangest moments in the history of Silicon Valley: a great rival writing a check to save its enemy because killing the enemy would be too dangerous.)

Over the next nine years, Jobs slowly turned the company around. In 1998, Apple released a brightly-colored desktop computer called the iMac. In 2001, it released a portable music player called the iPod, which became enormously popular and gave Apple its first hit product in over a decade. By the mid-2000s, Apple had become the company that made

the music player almost every American teenager wanted to own.

But by 2005, Jobs was worried about something. He could see that mobile phones — which by then were everywhere — were starting to play music. They were getting better at it. Eventually, he reasoned, why would anyone carry an iPod and a phone? Eventually the phone would just *be* the music player. And Apple did not make phones.

He decided that Apple had to make a phone. Or else, in a few years, Apple's most important product would be killed by somebody else's invention.

PROJECT PURPLE

He gave the project a code name. He called it *Project Purple*.

For the next two and a half years, about two hundred people at Apple worked, in extreme secrecy, on what would eventually become the iPhone. The secrecy was unusual even by Apple's standards. Engineers on the project were not allowed to tell their own families what they were working on. Different teams were separated by locked doors and security badges. Even within the company, almost nobody knew the whole picture.

Two competing designs were considered. One was a phone based on the iPod — using the famous spinning click wheel as a way to dial numbers and navigate menus. The other was based on a completely different idea: a small, flat

slab of glass that the user would control by *touching* the screen directly with their fingers, with no buttons at all.

The touchscreen idea won. The technology that made it possible — a way for a screen to recognize when multiple fingers were touching it at once, and to interpret pinches and swipes — had been developed by a tiny company called FingerWorks, which Apple had quietly bought in 2005. The team built a phone, designed by a famous British designer named Jony Ive, that was made almost entirely of one rectangle of glass, with a single button on the front and a thin aluminum band around the edges.

When Steve Jobs walked onto that stage in January of 2007, he was holding the first working version of this thing in his hand. It barely worked. The software had been finalized just days before. During his demonstration, the engineers backstage held their breath, hoping nothing would crash. Nothing did.

The iPhone went on sale on June 29, 2007. People stood in lines around the block to buy one. The first version cost \$499 — enormously expensive for a phone at the time. It could not, in its original form, run apps made by anyone other than Apple. It worked only on the slowest American mobile network. It had no copy-and-paste. It could not record video. Many serious technology critics looked at it and said it was a beautiful toy that would sell to a few enthusiasts but not change anything.

They were wrong by an embarrassing distance.

THE APP STORE

About a year after the iPhone launched, Apple made a decision that almost did not happen.

Steve Jobs, initially, did not want the iPhone to run software made by anyone other than Apple. He thought outside software would be dangerous, unreliable, prone to crashing the device. He wanted developers to build *web apps* — programs that would run inside the iPhone's browser — and not native apps that would run on the phone itself. Many of Apple's engineers disagreed. Many of Apple's customers complained. After about a year of resistance, Jobs changed his mind.

In July of 2008, Apple opened the *App Store*. It was, at first, a fairly simple thing: a place inside the iPhone where you could browse software made by other companies, and tap a button to download it onto your phone.

It became one of the most important inventions of the modern technology industry.

The App Store turned the iPhone from a single device into a *platform* — a foundation on which thousands and then millions of other businesses could build. Within five years, there were over a million different apps. Within ten years, over two million. People used apps for everything: to find restaurants, to call taxis, to share photographs, to read books, to play games, to learn languages, to deposit checks, to date, to argue, to sleep.

Whole new companies came into existence that only existed because of the iPhone. *Uber*, the ride-hailing service, would not have been possible without a touchscreen device in everyone's pocket that always knew where they were. *Instagram* would not have been possible without a good camera attached to a good screen attached to the internet. *Tinder*. *Spotify*. *WhatsApp*. All of them ran on a piece of technology that had not existed twenty years earlier.

Apple took thirty percent of every dollar spent in the App Store. This is a high percentage. It is the kind of fee a powerful landlord charges when there is nowhere else to rent. Most of the people who built the businesses that depended on the iPhone hated paying it. Most of them paid it anyway, because the iPhone was where their customers lived.

By 2025, the App Store would be generating tens of billions of dollars a year for Apple — much of it nearly pure profit, since Apple was not actually doing the work of building the apps. Apple was just the gatekeeper. Owning the gate had turned out to be more valuable than owning almost anything else.

A WORLD WITHOUT POCKETS THAT HADN'T BECOME COMPUTERS

I want you to imagine, for a moment, what daily life was like for people your age in the year 2005, two years before the iPhone came out.

In 2005, you woke up in a house. You walked downstairs. If your family was lucky, there was a computer in a room somewhere, on a desk, plugged into the wall. The internet lived in that one room. If you wanted to read about something, you walked to the computer. If you wanted to send a message, you walked to the computer. When you left the house — to go to school, to play with friends, to go on a trip — the internet stayed home.

You knew, when you were not at home, only the things you remembered or the things in your immediate physical surroundings. You got lost sometimes. You had to ask people for directions. When you were waiting at a doctor's office, you sat and looked at the walls, or read whatever magazine happened to be there. You were sometimes bored. You could not, when you wondered what year a movie came out, find out. You could not, when you wanted to hear a particular song, hear it.

If you wanted to tell a friend something, in 2005, and you were not at home, you had to find a way to tell them in person or wait until you got home. People made plans by phone, the night before. They showed up at coffee shops at six p.m. because they had said they would, not because they had texted updates. People got lost on trips and had to stop and ask for directions. People sat on park benches and just sat there.

I am not saying this was better. I am also not saying it was worse. I am saying it was different. The kind of difference

that, if you have always lived with a phone in your pocket, is genuinely hard to imagine.

In June of 2007, the iPhone went on sale. By 2010, smartphones were everywhere. By 2015, almost every adult in a wealthy country had one. By 2020, almost every kid did too. The change happened with such speed that older people had a hard time keeping track of it. Most of the things you take for granted about how life works today did not exist when your parents were the age you are now.

THE COMPLICATED PART

Now I have to be honest with you, the way I have been honest with you in other chapters.

The iPhone is one of the great inventions of human history. It put the entire library of human knowledge — every book, every encyclopedia, every map, every translation, every news article — into the pocket of any person on Earth who could afford one. It let strangers in different countries see each other's faces and talk in real time, for free. It let people in poor villages do banking without ever visiting a bank. It let protesters organize against governments. It let kids stay in touch with grandparents on the other side of the world. The video your grandmother sent from Vietnam, that we have been talking about since the prologue, exists because of devices like the iPhone.

It has also, very clearly, done some harm.

Studies show that the rates of certain kinds of unhappiness, especially among teenagers, started rising sharply around 2012 — about five years after the iPhone launched, around the time smartphones became widespread among young people. We do not know, with full certainty, how much of that rise is the fault of the phones. But many serious researchers believe a meaningful amount of it is. The phones are very good at capturing attention. They are very good at making you feel that you should look at them. They are designed, in deliberate and careful ways, to keep you using them, because every minute you use them is worth money to the people who make the apps and the people who sell the ads on the apps. The system you read about in the last chapter — the one where you are not really the customer, where your attention is the thing being sold — became dramatically more powerful when it could ride in your pocket all day.

Many adults you know are uneasy about what their phones do to them. Many of those same adults give phones to their children anyway, because the alternative — being the only family without one — feels harder than the costs of having one. Many of those children grow into teenagers who feel, sometimes, that they cannot put their phones down even when they want to. These are not problems made up by over-cautious grown-ups. These are real things, happening to real people, in real numbers.

I am not telling you to dislike your phone. I am telling you to *notice* it. To notice how it works on you. To notice the ways it is wonderful — and there are many — and the ways it has costs that the people who made it did not exactly intend. The iPhone is, like almost every other thing in this book, a both/and. A wonder and a hazard. A gift and a price.

What you do about it, when you are a little older, is up to you.

WHAT COMES NEXT

The iPhone made one more thing possible, and this is the thing that will shape the next several chapters of the book.

Before the iPhone, most of what people did on the internet was sitting still — at a desk, in front of a computer, deliberately *going online*. After the iPhone, the internet was always on, always with you, and the question of *what you were doing on it* became different. You weren't going online to do a specific task anymore. You were online constantly, in small pieces, throughout your whole day. And the new internet companies — the ones that came after Apple and Google — built their products specifically for that new way of being online. Always-on, social, fragmented, designed for the moments when you had ten seconds free.

The next chapter is about the most important of these companies. About a website that started, like Google and Apple before it, in a college dorm room. About a young man

who built it as a way to look at pictures of fellow students. About what happens when nearly every human being on the planet ends up using the same product for several hours a day.

That's Chapter 16.

It is called *The Social Graph*.

Chapter 16 — The Social Graph

AT THE VERY BEGINNING OF THIS BOOK, IN THE PROLOGUE, I told you about your grandmother filming a sunset in Vietnam and uploading the video to Facebook so her family across the world could see it.

It is time, finally, to tell you where Facebook came from.

The story begins, like several of the stories in this book, in a college dorm room.

In the autumn of 2003, a nineteen-year-old sophomore at Harvard University, named Mark Zuckerberg, did something stupid.

He was upset about a breakup. He stayed up late one night, in his dorm, drinking a bit too much. He decided, in his late-night state, to build a website. He used his programming skills to break into the password-protected pages of several of Harvard's dormitories and copy the small ID photographs of his fellow students — without their permission, without their knowledge. He then built a small site where the photographs of two students would appear side by side, and visitors could vote on which one was more attractive.

He posted the link to a few friends. The link spread across the Harvard campus by email in the middle of the night. By morning, the site had been visited by hundreds of students and had received about 22,000 votes. By afternoon, Harvard's network administrators had shut it down. Zuckerberg was brought before the university's disciplinary committee and almost expelled. He was, in the end, allowed to stay, but on probation.

The site he had built that night was called *Facemash*. It was, to put it gently, a bad idea.

I am telling you about Facemash, even though it doesn't reflect well on its creator, for two reasons. First, it is the actual historical beginning of what would become, within a few years, one of the most important companies on Earth. And second, because it shows something I want you to keep in your mind for the rest of this chapter: the same young man who would soon connect billions of people across the globe, who would help your grandmother send her sunset, did not, at the start, think very carefully about how his creations might affect the people they involved. That habit, of building things first and worrying about consequences later, would follow him — and his company — for the next twenty years.

THEFACEBOOK

A few months after the Facemash incident, in early February of 2004, Zuckerberg launched a different website.

He called it *Thefacebook*.

The idea behind Thefacebook was much milder than Facemash, but more powerful. Instead of comparing students to each other, Thefacebook let students *make their own pages*. You logged in with your Harvard email address. You uploaded a picture of yourself. You filled in some basic information — your year at Harvard, your major, your hometown, your favorite books, your relationship status. You connected your page to the pages of friends. You could see your friends' pages. They could see yours.

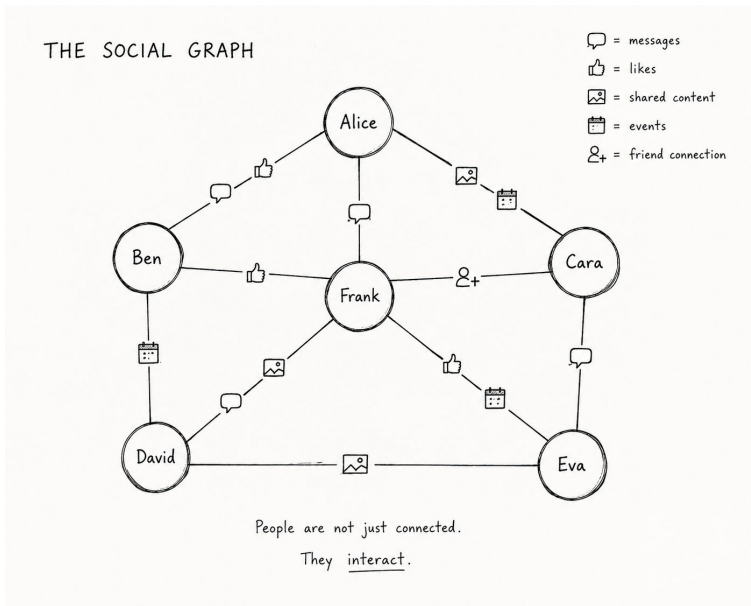
It was, in 2004, a brand-new kind of website. There had been a few earlier attempts at “social networks” — a site called Friendster, a site called MySpace, a few others — but Thefacebook had something the others didn't. It only let in college students. It started at Harvard, then expanded to a few other Ivy League universities, then to all colleges, slowly, one campus at a time. The exclusivity was deliberate. Knowing that everyone on the site was a real student at a real university, who had logged in with their real university email, made people feel safer about putting their actual information on it. Within a few weeks of launching, thousands of Harvard students had signed up. Within months, tens of thousands of students at universities across the country.

Zuckerberg dropped out of Harvard in 2004 to work on the company full time. He moved to Palo Alto, California, with a few co-founders — Dustin Moskovitz, Chris Hughes, and Eduardo Saverin. They got their first major investor, Peter Thiel, who put \$500,000 into the company in exchange for about ten percent of it. (That ten percent would, twenty years later, be worth tens of billions of dollars.) They hired Sean Parker, the co-founder of an earlier internet company called Napster, as their first president.

By 2006, the company had dropped the word “The” from its name. It was now called, simply, *Facebook*. And in September of that year, it opened its doors. Anyone over the age of 13, anywhere in the world, with any email address, could sign up.

The world started to.

THE GRAPH



The social graph: people are not just connected — they interact.

I want to stop and tell you a small fact about technology, because it is the thing that made Facebook valuable and that you should understand.

Computer scientists have a word for a network of people and the connections between them. They call it a *graph*. In particular, when the network represents friendships and relationships among real people, they call it a *social graph*.

A social graph is, in its simplest form, just a list. A list of people, and a list of which people are connected to which other people. *Alice knows Bob. Bob knows Carol. Carol knows Dave. Dave knows Alice and Eve.* That's a tiny social graph. The

graph of a single high school's worth of friendships is much bigger. The graph of all of humanity's friendships is, of course, enormous.

For most of human history, the social graph existed only in people's heads. Each person knew, roughly, who their friends and family were, and who knew whom. There was no central place that recorded this information. There was no map of human relationships.

Facebook, by 2008 or so, had started building exactly that map. As people signed up and friended each other and listed their hometowns and schools and workplaces, an enormous social graph began to take shape inside Facebook's computers. Within a few years, that graph contained hundreds of millions of people. Within a decade, it contained nearly three billion — close to half of the human race.

This was, by far, the largest and most detailed map of human relationships ever assembled.

You may be wondering why it was worth anything. Why does it matter that a website has a map of who your friends are?

The answer takes us back to the last chapter, where we talked about advertising.

You remember that Google had figured out something extraordinary: people, when they searched, told Google what they wanted to buy. The act of typing *running shoes* was a kind of confession of intent, and advertisers paid Google

enormous amounts of money for the right to put a small ad in front of a person who had just confessed.

Facebook had something even better. Facebook didn't have to wait for you to type *running shoes*. Facebook knew, from your profile, that you liked running. It knew, from the pages you'd liked and the photos you'd posted, that you were probably about to need new shoes. It knew, from your friends' patterns, that people like you, with friends like yours, in cities like yours, tended to buy a certain kind of shoe. And it could show you that shoe's ad while you were scrolling through baby pictures of your cousin's wedding, without you having to ask for it at all.

The social graph turned out to be one of the most economically valuable maps ever made. By the late 2010s, two companies — Facebook and Google — together controlled most of the online advertising market in the world. They were earning, between them, hundreds of billions of dollars a year, mostly by knowing extraordinary amounts about the people who used their free products and selling that knowledge, in carefully packaged form, to advertisers who wanted to reach the right people.

NEWS FEED

There was one more invention, by Facebook, that I want to tell you about, because it changed something about how human beings spend their attention.

In the early days, Facebook was a place you visited to check on specific people. You wanted to see how a particular friend was doing — you went to their page. You wanted to look up a person — you searched for their name.

In September of 2006, Facebook released a new feature called the *News Feed*. The News Feed was a single, continuously updating page that appeared as soon as you logged in. It showed you, in chronological order, the recent activity of all your friends — their posts, their new photos, their relationship changes, the events they were going to. You didn't have to go anywhere. You just opened Facebook, and there it was, an endless scroll of what your network was doing.

People hated it at first. There were protests. Many users felt that the News Feed exposed them — that having all their activity automatically broadcast to all their friends felt creepy. Facebook backed off slightly, added some privacy controls, but kept the basic design.

Within a year, it became obvious that the News Feed was the most addictive thing Facebook had ever built. People would log on intending to check one specific thing and find themselves still scrolling forty-five minutes later. The simple act of reading what your friends were up to, in a never-ending stream, turned out to be one of the most engaging activities a screen could offer.

A few years later, Facebook started ordering the News Feed not just chronologically, but by *engagement* — by what it predicted you were most likely to react to, comment on, or

click through. The posts that made people argue, or laugh, or feel strongly, got pushed to the top. The quiet posts got buried. The system was very good at this. Inside Facebook's offices, internal numbers showed that the average user was spending close to an hour a day looking at the company's products. Across all of Facebook's users, this added up to enormous fractions of human time, every day, all over the world, being spent inside one website.

This was the moment social media became, in a way it had not been before, an *attention machine*. The point of being on Facebook was no longer mostly to check on your friends. The point was to be on Facebook. Whatever kept you there longer was, automatically, what the system showed you more of.

The same logic, soon, would be copied by every other social media company that followed. By the early 2010s, Instagram (which Facebook bought in 2012 for \$1 billion, in a deal that has aged remarkably well — Instagram is now worth somewhere over \$100 billion). WhatsApp (which Facebook bought in 2014 for \$19 billion, and which became one of the most widely used messaging apps in the world). TikTok, which is owned by a different company called ByteDance and which would not exist until later, would push the attention-engine approach further than anyone had before.

All of these companies, in different ways, made their money the same way Facebook did. They captured the user's attention, by every means available. They showed the user

ads, perfectly targeted, while the user's attention was captured. And they grew, and grew, and grew.

THE COSTS

I now have to tell you the parts of this story that are hardest.

The Facebook your grandmother uses to share her sunset is, in many ways, a wonderful thing. Long-distance families stay in touch in ways they could not before. People find each other across the world — old classmates, lost relatives, communities of people with rare interests or shared experiences. The teenager I am imagining you to be, when you read this, has probably been to a birthday party that was planned through a Facebook event, or has a relative whose recovery from an illness she heard about from a Facebook post.

But there have also been other things.

In the years leading up to 2016, Facebook's News Feed became one of the main ways Americans got their news. It was free. It was easy. The stories that appeared in the feed were the ones the system had calculated would interest each user. The system was very good at calculating this. But it had no opinion about whether the stories were *true*. Many of them were not. False stories — about politicians, about events, about other countries — spread on Facebook faster than true stories, because false stories were often designed to be shocking, and shocking stories generated more clicks and comments and shares, and the system was tuned to promote

anything that generated clicks and comments and shares. By the time of the 2016 American presidential election, very large numbers of completely fabricated stories were being read by tens of millions of people. Some of those false stories had been written, deliberately, by foreign governments trying to influence American voters. Facebook, after years of public pressure and congressional hearings, eventually acknowledged this and began trying to address it. The work is ongoing. It has not been fully successful.

In 2017 and 2018, in the country of Myanmar, in Southeast Asia, the government and military used Facebook to spread hateful messages about a religious and ethnic minority group called the Rohingya. The messages encouraged violence. The violence happened. Approximately seven hundred thousand Rohingya people fled to neighboring countries. A United Nations report on the killings directly named Facebook as having played a *determining role* in the events. Facebook later admitted it had not done enough to stop the use of its platform to organize violence.

In 2021, an internal whistleblower from Facebook — a researcher named Frances Haugen — released internal documents showing that the company's own studies had concluded that Instagram, the company's photo-sharing app, was causing harm to the mental health of teenage girls. Specifically, the research found that teenage girls who used Instagram heavily were more likely to feel bad about their bodies and themselves. The company had known this for

years. It had not acted on the findings. Haugen testified before Congress. There were apologies. There were promises. The product, by and large, continued to operate the way it had before.

These are not the entire story of Facebook. The good things I described earlier are also part of the story. But you should know the bad things, because they are real, and because they did not happen by accident. They happened because the system was *designed* to maximize engagement, and engagement turns out to include people getting angry, getting scared, hating each other, feeling worse about themselves, believing things that aren't true. The system did not know, or care, the difference between healthy engagement and unhealthy engagement. It just measured engagement.

Mark Zuckerberg, twenty years after Facemash, is now one of the richest people in the history of the world. His personal fortune is estimated, depending on the day, at somewhere around two hundred billion dollars. He still controls Facebook — which renamed itself *Meta* in 2021 — through a special class of shares that give him more votes than any other shareholder, no matter how much stock other people buy. He cannot be removed by his own board. In this way, the structure of his company is unusual. It is closer to a kingdom than a typical American corporation.

He has, over the years, said many different things about what he believes his company should do. He has changed his political stances. He has invested heavily in something he

calls the *metaverse* — a kind of immersive virtual world — that has not, so far, attracted as many users as he hoped. More recently, he has redirected the company's enormous resources toward artificial intelligence, which is the topic of the last several chapters of this book.

The world is still arguing, and will for many years, about what to make of what he built. The two extremes — that Facebook is mostly a good thing, that Facebook is mostly a bad thing — are both, I think, too simple. Like everything else in this book, the truth is more complicated. The same product has connected your grandmother to you across an ocean, and has helped, in another country, to facilitate a genocide. It has hosted the world's strongest fundraising campaigns for medical care, and it has hosted some of the world's most effective propaganda. It has given lonely people communities, and it has made many young people lonelier. It is one thing and many things at once.

Like the iPhone in the last chapter, social media is something you will live with for the rest of your life. The most useful thing you can do — much more useful than deciding it is wholly good or wholly bad — is notice clearly what it does to you and to the people around you, and to make small, repeated decisions about how much of yourself you give to it.

WHAT'S NEXT

Before we leave the topic of social media, though, there is one more piece of the story we need to look at.

The biggest social media app in the world right now — the one many kids your age spend more time on than any other — was not made in the United States. It was made in China. And the simple fact of where it was made, combined with the simple fact that when an American kid lands at the airport in Beijing and connects to the local Wi-Fi, almost none of the apps and websites we have spent the last several chapters describing actually work, opens up a part of this story we have not yet looked at.

The internet you have been reading about for sixteen chapters — the open global network that Cerf and Kahn designed, the no-boss network from Chapter 5, the home of Silicon Valley's giants — is one version of the internet. It is the American version.

There is more than one version.

That's Chapter 17.

Chapter 17 — The Other Internet

IMAGINE — OR MAYBE REMEMBER, IF IT HAS HAPPENED TO YOU — landing at an airport in a country you have never been to, getting to your hotel room, connecting to the Wi-Fi, and pulling out your phone.

If you fly from the United States to most countries in the world — Mexico, Vietnam, France, Japan, Kenya, Brazil — everything on your phone works pretty much the same as it does at home. The apps open. Instagram loads. YouTube plays. Google returns search results. WhatsApp finds your contacts. Your email arrives. The internet, when you travel, mostly feels like the internet you already know.

But there are a few countries where this is not the case. If you fly to China and connect to the Wi-Fi in a Beijing hotel, almost nothing on your phone will work. Google doesn't load. YouTube doesn't play. Facebook is dark. Instagram is dark. WhatsApp doesn't deliver. Wikipedia is blocked. *The New York Times* is blocked. Most of the websites and apps you

have used your entire life, in the country your phone is now sitting in, are simply unavailable.

This is not a glitch. It is the result of a deliberate policy of the Chinese government, in place since the late 1990s, called the Great Firewall.

I want you to stop for a moment, because what I just told you is important.

The internet you have been reading about for sixteen chapters — the open global network that Cerf and Kahn designed, the no-boss system, the place where information flows freely from any computer to any other computer — does not actually reach into the country with the most people on Earth. The version of the internet that about 1.4 billion Chinese people use, every day, is a different version. It was built differently. It is run differently. It is owned differently.

For most of this book, when I have said *the internet*, I have really been talking about *the American internet*. It is the largest and most famous version. It is the one that grew out of ARPA and CERN, that ran on Cerf and Kahn's free protocols, that built itself on Berners-Lee's free web. But it is not the only version. And the gap between the American internet and the rest is growing, not shrinking.

This chapter is about that fact. It starts with the single most popular app in the world right now — an app you and your friends have almost certainly opened today — which is, quietly, a product of the other internet.

THE ALGORITHM THAT READS YOU

The app is called TikTok. It is owned by a Chinese company called ByteDance.

ByteDance was founded in 2012, in Beijing, by a young Chinese engineer named Zhang Yiming. He was thirty years old. Like Mark Zuckerberg in his Harvard dorm, like Larry Page in his Stanford lab, he had been thinking carefully about how people use the internet, and he had noticed something the American companies had not quite figured out.

The American social networks, you remember from the last chapter, were built around the *social graph* — the map of who knows whom. Facebook showed you what your friends were doing. Instagram showed you the photos of accounts you had chosen to follow. Twitter showed you the posts of people you had chosen to listen to. These were *networks of people you already knew about*.

Zhang Yiming's idea was different. What if the app didn't care who you knew at all? What if, instead, the app showed you a stream of videos chosen by an extraordinarily clever computer program — an *algorithm* — that watched, with great care, what you actually paid attention to? You wouldn't have to follow anyone. You wouldn't have to choose anything. You would just open the app, and it would start showing you videos. If you watched a video to the end, the algorithm noticed. If you swiped past it after one second, the algorithm noticed that too. If you replayed something, the

algorithm noticed. If you commented, or shared, or paused, the algorithm noticed. Within a few hundred videos — a process that could take a new user about an hour — the algorithm would have built a very precise model of what kept that particular human's eyes on the screen. From then on, it would just keep feeding them more of what worked.

ByteDance launched this app in China, in 2016. They called it Douyin. The next year, they launched an international version, in English, called TikTok. In 2018, they bought a smaller American app called Musical.ly and merged it into TikTok.

What happened next was unprecedented in the history of social media. By 2021, TikTok had passed one billion monthly users. By 2024, it was the most-downloaded app in the world, several years running. The average teenager in countries where TikTok is popular spends a significant fraction of their waking life watching its videos — often, by their own admission, more than they want to.

The reason is the algorithm. TikTok's algorithm turns out to be extraordinarily good at finding videos that hold each person's attention. Inside the industry, people who study this kind of thing have written reports trying to explain why TikTok is so much more engaging than Facebook or Instagram. The simplest answer is: TikTok's algorithm was designed, from the very beginning, to do nothing else. It does not have to worry about your friends. It does not have to

worry about the social graph. It just has to keep your eyes on the screen. And it is very, very good at that.

THE POLITICS

There is something complicated about TikTok that I have to tell you about now, because the kid you are will probably read about it in the news a lot as you grow up.

TikTok's parent company, ByteDance, is a Chinese company. Under Chinese law, every Chinese company must, if the government asks, share information with the government. The Chinese government has the authority — as a matter of national law — to instruct ByteDance to do almost anything with TikTok's data or with what TikTok shows people.

ByteDance and TikTok have repeatedly said that this has never happened — that they have never been ordered by the Chinese government to share American user data, or to favor or hide particular kinds of videos. Many people in the American government do not entirely believe them. Many people in the American government also point out that even if it has not happened yet, the legal power for it to happen exists, and that an app with hundreds of millions of American users, whose feed could in theory be tuned by a foreign government to influence what those Americans see and believe, is a national security concern.

This argument has been going on for years. In 2020, President Donald Trump issued an executive order trying to ban

TikTok in the United States. Courts blocked the order. In 2024, the United States Congress passed a law saying that ByteDance had to sell TikTok to a non-Chinese owner or be banned in America. President Joe Biden signed the law. The deadline was set for early 2025.

Then, in a strange twist, President Trump won the 2024 election. He had, during that election, made enormous use of TikTok to reach younger voters. After his election, he decided he did not want to ban TikTok after all. The deadline kept getting extended. As this book is being written, the future of TikTok in America is still being argued over.

There is something deeply weird about all of this, and I want you to see it clearly.

A Chinese app, technically owned by a company that operates under the legal authority of an authoritarian government, has become one of the main ways American teenagers and adults receive news and entertainment. Both major American political parties have used it heavily in their campaigns. Both have argued, at different times, that it should be banned. Both have, at different times, refused to ban it. The same politicians have, in some cases, taken both positions within the same year.

This is not a story with a clean conclusion. It is an ongoing situation. Different people in your generation will end up in different places on what should be done. You may have a strong opinion. You may have a different opinion every year. What I can tell you is that this exact tension — between an

internet that crosses borders freely and a world made up of countries that want to control what crosses their borders — is going to keep coming up, in different forms, for the rest of your life. TikTok is the version of it your generation grew up with. There will be other versions.

THE WALL



The same phone, two different internets, depending on which side of the wall you are on.

But I have not yet explained the part of the story that started this chapter. I have not yet told you, properly, about why your phone stops working when you land in China.

The Great Firewall — the formal name of the project that runs it is *Golden Shield* — is one of the most sophisticated in-

formation-control systems ever built. It is operated by the Chinese government. It does, basically, three things.

It blocks certain websites and apps from being reachable from inside China. You cannot use Google, Facebook, YouTube, Instagram, Twitter (now called X), WhatsApp, Wikipedia, the major American news sites, and many other foreign services from a normal Chinese internet connection. The blocks happen at the level of the *routers* — those devices you read about in Chapter 8 — that sit between Chinese networks and the rest of the world. When a packet leaves a Chinese computer trying to reach a forbidden destination, the routers refuse to forward it.

It filters what passes through. Even on sites that are not fully blocked, sensitive content — about certain political events in Chinese history, about current government criticism, about minority groups — is automatically removed or hidden. The filtering is done both by computers (looking for forbidden keywords) and by armies of human moderators.

It tracks. The government can, in many cases, see what people are doing online. This is one of the largest digital surveillance operations in the world.

People in China can sometimes get around the Great Firewall by using a kind of software called a VPN (Virtual Private Network), which makes their connection look like it is coming from somewhere else. VPNs are technically illegal in China, but for years they were widely used, especially by foreign businesspeople and academics who needed to use

Google for work. In recent years, the government has been steadily making VPNs harder to use.

I want to tell you something honest about this. From inside the American framework of the open internet, the Great Firewall sounds straightforwardly bad — a system of censorship and surveillance that prevents ordinary people from learning what their government does not want them to learn. From that perspective, it *is* straightforwardly bad. Free access to information is one of the great gifts of the internet that Cerf and Kahn imagined.

But I want you to also see something more uncomfortable. The Great Firewall is not unpopular inside China. Many Chinese citizens — not all, but many — actually prefer their version of the internet. They do not miss Google. They do not feel that they are missing out by not having Facebook. They have grown up using a different set of products, and from inside their experience, the American internet feels chaotic and untrustworthy. They believe — or many of them believe — that some government oversight of the internet is reasonable, the way some government oversight of food safety or air travel is reasonable. They have made a different bargain than Americans have. Whether the bargain is wise is, in part, a question about values that the world is genuinely in disagreement about. It is not a thing that one side has already won.

THE PARALLEL WORLD

What I have not yet told you is the most fascinating part. China did not just block the foreign internet. It built its own.

Behind the Great Firewall, over the past two decades, a parallel ecosystem of Chinese apps and websites has grown up. Almost every major American internet product has a Chinese equivalent — and many of the Chinese equivalents are, by now, more technically sophisticated than the American versions.

There is WeChat. WeChat is owned by a Chinese company called Tencent. It is, for an American mind, almost impossible to describe, because no American app does anything like it. WeChat started as a messaging app, similar to WhatsApp. Over the years it absorbed, into a single app, the functions of about twenty separate American apps. WeChat is how you message your friends. WeChat is how you pay for things — you scan a code, you tap to confirm, money moves. WeChat is how you book taxis. WeChat is how you order food. WeChat is how you buy plane tickets, send money to relatives, read the news, watch short videos, file government paperwork, see your medical records. Almost every adult in China uses WeChat for hours every day. It is a *super app* — a single application that swallows up most of digital life. There is nothing quite like it in the West.

There is Alipay, which is the other major payments app, owned by a company called Ant Group, which is related to a

giant Chinese e-commerce company called Alibaba. Together, WeChat Pay and Alipay handle most payments in China. People scan QR codes on their phones to buy things from street vendors, restaurants, taxis. Many Chinese cities have, for years, been functionally cashless. American mobile payments — Apple Pay, Google Pay, Venmo, Zelle — are not anywhere near as universal or as smoothly integrated. The Chinese mobile payment system, by most measures, is several years ahead of the American one.

There is Weibo, the Chinese microblogging platform that filled the niche Twitter filled in the rest of the world. There is Baidu, the search engine. There is Taobao and Tmall, the giant e-commerce platforms run by Alibaba, where shopping is mixed with live video streams and instant chat with the sellers. There is Bilibili, a video site especially popular with younger Chinese viewers, that mixes YouTube and TikTok in ways neither has tried. There is Pinduoduo, an enormous e-commerce app that combined group-buying with gamification in ways American startups are now trying to copy.

The point is this: the choice China made — to build the Great Firewall, to keep most American apps out, to develop its own ecosystem — produced a fundamentally different internet inside China. It is not a worse internet, by every measure. By some measures (mobile payments, super apps, live commerce), it is actually ahead of the American one. By other measures (free speech, access to outside information, political openness), it is significantly behind. It is, in any case,

a different one. The two internets — the American one you have been reading about, and the Chinese one — increasingly look like two separate civilizations.

People who study this have started calling the result the *splinternet* — a contraction of *splinter* and *internet*. The single, open, global network is splintering into multiple regional networks, with different rules in each one.

China is the most dramatic example. But it is not the only one. Russia has been steadily walling off its internet for years, especially since 2022. Iran heavily restricts what its citizens can see. The European Union, although it is democratic and committed to open information, has chosen to regulate the American tech giants much more strictly than the United States does — with laws about privacy, about how user data can be used, about what platforms must do to remove illegal content. Even within democracies, the version of the internet you experience now depends, more and more, on which country you happen to be in.

YOUR GENERATION'S QUESTION

Thirty years ago, in countries that controlled what their citizens could read and hear, shortwave radio was often the only way to access news from outside the borders — voices like the BBC and the Voice of America that escaped national gates by reaching across the air. The technology of control

has changed dramatically since then. The impulse to control has not.

What I want you to understand, before we leave this chapter, is that the open internet — the no-boss network that Cerf and Kahn gave to the world — was never a guaranteed thing. It was a *choice*. It is still a choice. Different countries, with different ideas about how a society should work, are making different choices about what their internet should look like. The result is the splinternet you are growing up inside.

The big question that your generation will have to answer, over the next several decades, is something like: *what kind of internet should the world have?* Should information cross borders freely, the way Cerf and Kahn imagined? Or should each country, or each region, be allowed to shape what its people can see and do online? Should a few American companies (or a few Chinese companies, or a few of each) be allowed to control the experience of billions of users? Should governments have more power over the internet, or less? When new technologies — like artificial intelligence, which is the topic of the rest of this book — arrive, who should be allowed to build them, and where should they be allowed to be used, and on whose terms?

There is no easy answer to any of these questions. There are very smart people on multiple sides of each one. By the time you are reading this with your own children, the an-

swers we end up with may look completely different from what anyone today is predicting.

For now, the most useful thing I can tell you is this: when you open an app on your phone, you are using one version of a thing that could have been built many different ways. The version you have is not the only version. The version you have is not even the most popular version, in many parts of the world. The thing called “the internet” is, in a real sense, several different things, with different rules, depending on where you happen to be standing.

In the next chapter, we go back to the version of the internet you know best, and we meet the last of the giants. Not a company that makes anything you can hold in your hand. Not a company that has an app on your phone. A company that rents *computers* to other companies, by the minute, in enormous quantities, and that became, almost by accident, the most profitable business inside the giant Amazon you read about three chapters ago.

That’s Chapter 18.

Chapter 18 — The Cloud



The cloud, in everyday life — someone else's computer, somewhere far away.

Look around the room you are sitting in. Find your phone. Find any other screens. Find your family's computer if there is one.

Almost nothing you do on any of those screens is actually stored on them.

When you take a photo on your phone, your phone keeps a copy — but it also sends a copy, automatically, to some company's giant computer thousands of miles away. When you message a friend, the message doesn't really travel directly from your phone to theirs. It travels to a company's computer first, gets copied, and gets forwarded. When you watch a video, the video doesn't live on your device. It lives, somewhere, in an enormous building full of computers, owned by a company you don't work for.

Where, exactly, are all those copies?

For a surprisingly large fraction of the digital world, the answer is the same. They are sitting on computers owned by a single American company, in a small set of giant warehouses scattered across the planet.

The company is called Amazon. The business is called Amazon Web Services — almost everyone calls it AWS for short. And almost nobody outside the technology industry talks about it, even though it has become — quietly, without television commercials, without consumer products — one of the most important businesses in the modern world.

This chapter is about how that happened.

THE ACCIDENT

The story of AWS is, like most of the stories in this book, not the story the company tells about itself in its marketing.

In the early 2000s, after almost going bankrupt the way you read about in Chapter 13, Amazon was trying to grow back into a healthy company. As it grew, its computer systems were getting tangled up. Different teams inside Amazon were building software in different ways. When one team needed to share information with another team, it was often easier to rebuild the wheel than to figure out the other team's mess.

In 2002, Jeff Bezos issued a memo to all the engineers at Amazon. The memo is famous now, even though it was only ever shared internally. It said, roughly: from now on, every team inside Amazon must build its software so that other teams can use it through a clean, well-documented *interface*. No more secret internal hacks. No more “well, our system only works if you ask us nicely.” Every piece of Amazon's software had to be built as if some unknown stranger, out in the world, might one day want to use it.

The memo ended with one of the great Bezos lines, which I will paraphrase, because the exact wording was rude: *if you don't do this, you'll be fired*.

The engineers grumbled. They did it anyway. Over the next few years, Amazon's tangled internal systems were rebuilt as a collection of clean, separate services that any other piece of software could talk to through clearly-defined channels.

Then — and this is the accidental part — someone at Amazon noticed that they had, almost by mistake, built something other people would want to pay for.

If Amazon's internal teams could rent storage and computing power from other internal teams, by sending clean requests through clean interfaces, then *anyone* could do the same thing. A startup in San Francisco. A research lab in Norway. A teenager in his bedroom. They could all, in theory, just plug into Amazon's systems, rent however much computing power they needed, pay only for what they used, and never have to buy a single server of their own.

In March of 2006, Amazon launched a service called S3, for Simple Storage Service. S3 let anyone, anywhere in the world, store any amount of data in Amazon's warehouses, and pay only by the gigabyte. A few months later, in August of 2006, Amazon launched a service called EC2, for Elastic Compute Cloud. EC2 let anyone rent an entire virtual computer — fully configured, with a chosen operating system, ready to run any software — for as little as ten cents per hour.

Almost nobody noticed at first. The technology press, in 2006, was focused on social networks and smartphones. The launch of EC2 got coverage in industry trade publications, but it wasn't a front-page story anywhere.

It would, however, change almost everything about how technology companies got built.

OTHER PEOPLE'S COMPUTERS

You should know, before we go any further, what the phrase *the cloud* actually means. Because it is one of the most misleading phrases in the modern English language.

When you store your photos “in the cloud,” they do not exist in the sky. They are not floating, ethereal, distributed across the atmosphere. They are sitting on hard drives, in racks of computers, in specific buildings, in specific cities, owned by specific companies. The word *cloud* is marketing. The truer phrase — the one programmers sometimes use as a joke — is *other people's computers*.

When you save a file “to the cloud,” what you mean is that you have copied the file to a computer that you do not own, in a building you have never been to, run by a company you may never have bought anything from directly. That company has agreed, in exchange for a small fee — or, more often, in exchange for serving you advertising — to keep your file safe and to give it back to you whenever you ask.

This is the thing AWS lets companies do. AWS owns the computers. Other companies — Netflix, Airbnb, Lyft, the *New York Times*, hundreds of thousands of others — rent space on those computers, by the minute. When you stream a Netflix movie, the video is being sent to you from one of Amazon's buildings. Netflix doesn't own that building. Netflix doesn't own those computers. Netflix is renting them.

This is a strange arrangement, if you stop to think about it. Netflix is one of the largest companies in the entertainment world. It has hundreds of millions of subscribers. And it is using, to do its job, infrastructure owned by Amazon — a company that, in another part of itself, is one of Netflix's biggest competitors. (Amazon also makes movies and television shows; it owns the Prime Video service.)

You might think this would worry Netflix. It does, sometimes. There has been steady talk, for a decade, of Netflix trying to move some of its work onto computers it owns directly, or onto other companies' clouds. The work is enormous and expensive. Mostly, Netflix has stayed.

This pattern repeats across the internet. Companies have become dependent, in deep ways they cannot easily unwind, on a few cloud providers that they also, sometimes, compete with.

WHAT IT CHANGED

I want to tell you what AWS did to the rest of the technology industry, because the effect is hard to overstate.

Before AWS, if you wanted to start a technology company, the first thing you had to do — before you wrote a single line of useful software, before you had a single customer — was to buy or rent a lot of expensive equipment. You had to buy servers. You had to buy network gear. You had to buy storage devices. You had to find a place to put all of it, set it up, keep

it cool, hire people to babysit it. The cost of getting started was in the hundreds of thousands of dollars, sometimes the millions. You had to convince investors to give you that money before anyone had any idea whether your company was a good idea.

After AWS, the cost of getting started fell, roughly, by a factor of a thousand.

A teenager in his bedroom could, in 2007 or 2008, open a free Amazon account, type in a credit card number, and have, within twenty minutes, a fully functional server running his code, accessible from anywhere in the world. If the project did well, he could rent more computers, instantly. If it did badly, he could shut everything down, and pay only for what he had used, which might be ten dollars.

A whole generation of small startups became possible that had not been possible before. Many of the companies you know — Airbnb, Uber, Lyft, Pinterest, Instagram, Snapchat, Slack, Stripe, Dropbox, and hundreds of others — were built, in their early years, almost entirely on AWS. They could not have existed in their actual form without it. The economics didn't work without renting their infrastructure.

This is the third great gift of the modern internet, after Cerf and Kahn's protocols and Berners-Lee's web. The cost of starting a technology company collapsed almost to zero, because Amazon was willing to rent you whatever you needed. Whether this was a *gift* in quite the same way as the open protocols, or whether it was a clever business move that cap-

tured most of the value of an industry, is the kind of question you can argue about at dinner with your future friends. The truth is, as usual, a bit of both.

THE WAREHOUSES

I want you to know what AWS's warehouses are actually like. Because *the cloud* sounds like a soft thing, and the real thing is anything but.

A modern AWS data center is a building the size of several football fields, with no windows, often surrounded by a fence. Inside, in long aisles that look like the racks of an enormous library, are thousands or tens of thousands of servers — small computers stacked on top of each other in metal frames. Each server is doing some piece of work for some customer somewhere in the world. They hum, all the time. They generate enormous amounts of heat. Cooling them — running enormous air-conditioning systems all day and night — uses more electricity than any other part of the operation.

A single AWS data center uses about as much electricity as a small city. In the part of the United States where AWS has built the most data centers — a region called Northern Virginia, in the suburbs near Washington, D.C. — the local power grid has been strained for years by the sheer number of data centers feeding off it. Some new data centers cannot

be built right now, because there is no more spare electricity to give them.

The data centers are also, increasingly, located in places chosen for specific reasons. Cold places help with cooling (Sweden, Ireland). Near rivers help with water for cooling (Northern Virginia, Oregon). Near nuclear power plants help with the electricity (Amazon and others are now buying nuclear plants specifically to power data centers).

Each big region of the world has a handful of these AWS regions: at least two in the United States, one in Ireland, one in Germany, one in Singapore, one in Tokyo, several in others. (China has its own. AWS does not really operate inside the Great Firewall — for the reasons you read about in the last chapter, Chinese cloud computing is dominated by Chinese companies like Alibaba and Tencent.)

A non-trivial fraction of the world's information now sits in those warehouses. Your photos. Your school assignments. Your medical records, in many countries. Government documents. Bank transactions. The streaming videos you watch. The messages you send. All of it lives in buildings most people have never seen, run by a company whose name is on every cardboard box in the country.

THREE COMPANIES, ONE INDUSTRY

I should mention that AWS is not the only one. There are two other major American cloud companies.

One is *Microsoft Azure*, run by Microsoft, the company that for years dominated the personal computer industry and that, under newer leadership, became a serious cloud provider starting in the 2010s. Azure is the second-largest of the major cloud providers, and especially strong with big traditional companies — banks, governments, retailers — that already used Microsoft software.

The other is *Google Cloud*, run by Google (the company you met in Chapter 14). Google Cloud is the third-largest. It has unique strengths in certain technical areas, particularly artificial intelligence, because Google has been doing AI research longer than almost anyone else and has built specialized hardware for it.

Together, these three companies — AWS, Azure, and Google Cloud — own a very large share of the world’s commercial computing capacity. Several smaller cloud providers exist. New ones, especially focused on the new AI workloads, have appeared in the last few years (a company called CoreWeave, for example, that you may hear more about). But for the most part, when a serious company says it is “moving to the cloud,” it means moving to one of those three American companies.

Three American companies. The companies whose servers, by lawful subpoena, the United States government can usually compel to hand over data. The companies whose buildings are powered by American electricity, cooled by American water, staffed by American workers (and, by their

many international employees, but under American regulation).

You should remember this, because in the next part of the book, we are going to meet a new generation of companies — the artificial intelligence companies — that mostly run, today, on these same three companies' clouds. Anthropic. OpenAI. Many of the others. Their breakthroughs happen on rented warehouses of computers owned by Amazon, Microsoft, or Google. The age of giant private wealth being built on top of cheap infrastructure is, you will see, very much still with us.

PART V

This is the end of Part IV of this book. The part about the giants.

Over the last several chapters, you have met the companies that grew up after the dot-com bust, on the cheap infrastructure left behind by Global Crossing and WorldCom and Nortel, and that have come to define the internet you live inside. Amazon and its retail empire. Google and its advertising empire. Apple and its iPhone. Facebook and its social graph (now Meta, with Instagram and WhatsApp). ByteDance and TikTok and the splinternet. And AWS, quietly, underneath everything, renting computers to almost everybody.

These are, today, called the *Magnificent Seven* by people who follow the stock market, though the list shifts depending on who is counting. Apple, Microsoft, Alphabet (which is

Google's parent company), Amazon, Meta, Nvidia (which we will meet in Part V), and Tesla (which is mostly a car company that we won't really discuss in this book). Together, these seven companies make up roughly a third of the entire United States stock market. Each one of them is, by itself, worth more than the total wealth of most countries. Their decisions shape what billions of people see, what they buy, what they argue about, and what they hope for, every single day.

How did all of this happen on top of foundations that were originally given away for free?

That is the question you have been holding since Chapter 6. It does not have a single answer. But you have seen the pattern by now. The foundations — TCP/IP, the web, the cables, the protocols — were paid for by public money and released, as gifts, to the world. The towers that were built on those foundations were not gifts. They were companies. They were businesses. They were owned by specific people, who became, by owning them, the richest in history.

This is the world your generation has inherited.

The last part of this book is about what is being built right now, on top of all of it.

It is about artificial intelligence — the kind of machines that can answer your questions, write your school assignments if you let them, drive cars, recognize faces, paint pictures, compose music, and, increasingly, do many of the

things human beings used to think only human beings could do.

It is about a quiet office, in a building near Washington, D.C., that has been funding research on machines that learn since before most of the engineers you read about in this book were born.

It is about a British-Canadian scientist who refused to take money from that office, and who, more than any other single person, made the current AI revolution possible.

And it is about what your generation will have to decide, in the years just ahead, about who builds these new machines, who owns them, who pays for them, and what they should be allowed to do.

That's Part V. It starts in the next chapter.

It starts in Arlington, Virginia.

Part V — The Brain

Chapter 19 — A Naval Office in Arlington



An ordinary office in Arlington — where the Navy quietly funded the early AI work.

There is a building in Arlington, Virginia — just across the Potomac River from Washington, D.C. — that looks like every other office building in that part of the country. It is plain. It is gray. It has security at the door and no signs along the street explaining what happens inside. If you walked past

it, you would assume it was some mid-level federal agency, of the sort that handles paperwork. You would be mostly right.

Inside this building, since 1946, there has been an office of the United States Navy.

The office is not concerned with ships, or with sailors, or with weapons, or with anything you would recognize as Navy business. It is concerned with science. Its job, almost from the day it was founded, has been to find scientists in universities across the country who are working on hard, important, basic problems, and to give them money.

The office is called the Office of Naval Research, or ONR for short. It is one of the most quietly important institutions in the history of American science. Almost no one outside the research community has heard of it. It has been continuously funding basic research, in fields ranging from oceanography to materials chemistry to artificial intelligence, for nearly eighty years.

A surprising amount of the science behind the artificial intelligence revolution you are growing up inside started, in a financial sense, in that gray building.

This chapter is about why.

AFTER THE WAR

To understand the Office of Naval Research, you have to understand 1945.

The Second World War, which had just ended, had been the most scientific war in human history. The most important weapons of the war had not been guns or tanks. They had been *inventions*. The atomic bomb. Radar. The proximity fuse. Penicillin. The Enigma code-breaking machines. The jet engine. Each one of those had been built, during the war years, by scientists — most of them civilians, most of them academics — who had been pulled out of their universities and given enormous budgets and told: figure this out, as fast as possible, however you can.

You read about one of those scientists in Chapter 1. Vannevar Bush, who ran the most important American science office of the war, and who, when the war ended, was tired and worried and wrote an essay about a machine called the memex.

Bush had also written, in 1945, a much more important document. It was called *Science: The Endless Frontier*, and it was a long letter to President Roosevelt arguing that the United States should keep doing in peacetime what it had done in wartime: pay scientists to do basic research, and trust them to choose the problems. Bush argued that science — pure, curiosity-driven science, the kind that has no immediate application — was the long-term source of every important industrial and military breakthrough. A country that paid for it would, in the long run, lead the world. A country that didn't would, in the long run, fall behind.

Many people in Washington agreed. The question was *who would do the paying*.

The civilian science agency — the National Science Foundation, or NSF — would not be created until 1950. There was a gap. Several years where the country had decided it needed to keep funding basic science, but had not yet built the machinery to do so.

Into that gap stepped the United States Navy.

In August of 1946, just a year after the war ended, Congress passed a law creating the Office of Naval Research. The Navy got the official mission of supporting “scientific research in recognition of its paramount importance as related to the maintenance of future naval power, and the preservation of national security.” That was the official language. The actual practice was somewhat broader.

The way ONR worked, almost from the beginning, was this. It had a small staff of *program officers* — scientists in different fields, hired by the Navy, given budgets. The program officers’ job was to find, in universities around the country, the most interesting work being done in their area, and to write checks. The checks were called *contracts* or *grants*. They typically ran for several years. The scientists who received them did not have to work on Navy projects. They had to work on what they were already working on. The Navy just paid for it.

This was unusual. Most government agencies, when they hire someone, tell that person what to do. ONR didn’t. ONR’s

bet was that if you found the best people, and let them follow their own ideas, the results would, in the long run, benefit the Navy — and the country — in ways that nobody, including ONR, could predict in advance.

That bet has, over almost eighty years, paid off in ways that would have astonished the people who originally placed it.

WHY THE NAVY FUNDED THE BRAIN

You might be wondering why a military service was interested in cognitive science — the study of how the human mind works — or in artificial intelligence — the attempt to make machines that work like minds. These do not, on the face of it, seem like Navy problems.

But step back, and the reasons become clear.

The Navy of the 1950s had problems that were, fundamentally, problems of attention and decision-making under pressure. A sonar operator on a submarine had to listen to faint sounds and decide what they meant — was that an enemy submarine, or a whale? A pilot landing on an aircraft carrier had to make split-second decisions in conditions of stress and exhaustion. A captain in combat had to weigh many pieces of information at once and choose, in seconds, what to do.

Better understanding of how human minds processed information, made decisions, and got tired — these would dir-

ectly help the Navy. So would, eventually, machines that could help humans with those tasks, or that could do parts of them automatically. Cognitive science and artificial intelligence, however abstract they sounded, had clear long-term applications to what the Navy did.

So ONR funded the study of attention. ONR funded the study of decision-making. ONR funded the study of memory. ONR funded research on how the brain works, on how computers might be made to work in similar ways, on the mathematics of learning. Almost every major American researcher in cognitive science and early artificial intelligence, from the 1950s through the 1980s, had at least some of their work paid for, at some point, by ONR or by ARPA — the Defense Department agency that you have met many times in this book.

ONR was not alone in this. ARPA funded a great deal of AI research too. So did the National Science Foundation, after it was created. So did the Air Force. The patient, civilian-managed funding of long-term scientific work was, in those decades, a thing several different parts of the American government did at the same time, often funding overlapping work at the same universities.

But ONR was particularly patient. Particularly willing to fund things that no immediate application could be seen for. Particularly insistent that the scientists themselves should choose the problems. Decade after decade. Through fashion,

and the absence of fashion. Through funding fights, and quiet years.

This will turn out, in this chapter, to matter very much.

THE BOOM

The first great wave of artificial intelligence research happened in the 1950s and 1960s.

In 1956, a young computer scientist named John McCarthy convened a workshop at Dartmouth College in New Hampshire to study what he called *artificial intelligence* — a new phrase he had coined for the proposal. Several brilliant young people attended. They believed, with some justification, that by understanding how the human mind worked and replicating it in computers, they could build machines that would think within a generation.

For about fifteen years, the field was wildly ambitious. Researchers built programs that could play chess, prove geometric theorems, solve simple puzzles, parse simple sentences. Each new result was greeted as a step toward true machine intelligence. Funding poured in — from ARPA, from ONR, from elsewhere.

In 1958, a psychologist at Cornell University named Frank Rosenblatt built a particular kind of machine called a *perceptron*. The perceptron was, in physical form, a closet-sized rack of electronics. It was the first device that could learn to do a simple task — recognizing patterns in pictures — by ad-

justing itself through experience, rather than being programmed step by step. Rosenblatt demonstrated it to journalists. The *New York Times*, in 1958, reported breathlessly that the Navy had revealed the embryo of an electronic computer that would be able to walk, talk, see, write, reproduce itself and be conscious of its existence.

This was, of course, overstating things considerably. The perceptron was a simple early version of what we now call a *neural network* — a kind of computer program inspired by the way networks of neurons in the brain work. Rosenblatt's machine could learn to tell, for example, the shape of a triangle from the shape of a square. It could not learn anything much more complicated. But the *idea* was new and important: a machine that could be *taught* through examples, rather than programmed through rules.

For about a decade, perceptrons and other neural networks were a hot area of research.

Then, in 1969, the field collapsed.

THE WINTER

Two researchers at MIT — Marvin Minsky and Seymour Papert — published a book called *Perceptrons*. The book was, in part, a careful mathematical analysis of what Rosenblatt's machines could and could not do. It pointed out, with proofs, that single-layer perceptrons — the simple kind Rosenblatt had built — could not learn certain extremely

basic patterns. Some of the simplest things a child could learn to recognize, the perceptrons mathematically could not.

The book was technically correct. But it was, in a sense that you should understand, devastating in ways that went beyond its mathematics. Minsky and Papert were the two most famous AI researchers in the world at the time. They had decided that perceptrons — and, by implication, the entire neural-network approach — were a dead end. Their book was widely read as a verdict, even though they had not fully explored more complicated, multi-layer versions of the same idea.

After *Perceptrons*, almost nobody worked on neural networks for the next decade. Funding for the approach dried up. Graduate students were told it would ruin their careers. The promising researchers from the perceptron era moved on to other things. Frank Rosenblatt died in a boating accident in 1971, on his 43rd birthday, before the field had a chance to recover.

This is what happens, sometimes, in science. A whole approach to a problem gets discredited, often unfairly, by a single influential book or paper. For a decade, or two decades, almost nobody works on it. Other approaches get tried. Most of them also fail. The field as a whole goes quiet, and money goes elsewhere.

The 1970s, in artificial intelligence, were one of these quiet periods. Researchers called it, later, the first *AI winter*. Government funding for AI dropped sharply. The grand prom-

ises of the 1960s had not been kept. The machines that were supposed to think had not learned to think. Disappointed funding agencies turned to other things.

But not all of them. ONR, and a few other patient institutions, kept funding small numbers of researchers who had not given up. The amount of money was much smaller than it had been. The researchers were less famous, less ambitious in their public claims. They were doing what scientists do during a long winter: working quietly, on hard problems, without much hope of immediate success, because they thought the problems were worth working on.

THE QUIET DECADE

You should know about a particular group of these scientists, because they are the ones who eventually broke the winter.

In the late 1970s and early 1980s, at the University of California in San Diego, two researchers named David Rumelhart and James McClelland began organizing a small group that was thinking, again, about neural networks. They were not promising that machine intelligence was a year away. They were not announcing breakthroughs in the *New York Times*. They were working, in a careful, mathematical way, on the question of how to make multi-layer neural networks actually learn — the very thing Minsky and Papert had said couldn't be done.

The group they organized was called the *Parallel Distributed Processing* group, or PDP for short. (You will, again, notice that the internet world's love of acronyms is shared by the AI world.) Researchers from several universities came to San Diego for long visits to work with them. They wrote papers. They held workshops. They argued about how brains might work, and whether computers could be made to work the same way.

A young British scientist named Geoffrey Hinton was part of the group. So was a young man named Ronald Williams. Together with Rumelhart, in 1986, the three of them published a paper in the journal *Nature* with a quiet, technical title: *Learning representations by back-propagating errors*.

The paper described an algorithm — a step-by-step computational procedure — for training neural networks with many layers. The algorithm was called *backpropagation*, or *backprop* for short. The mathematics was not new — versions of it had been derived independently by other scientists, including an American researcher named Paul Werbos who had written about it in his PhD thesis in 1974. But Rumelhart, Hinton, and Williams put the idea together, demonstrated that it worked on actual networks, and presented it to the broader scientific community in a clear and memorable way.

That paper, in 1986, is one of the foundation stones of the artificial intelligence revolution that is happening right now, while you are reading this book. Almost every system you have heard of — the chatbots, the image generators, the sys-

tems that help drive cars, the language models that can write essays — uses, somewhere in its inner workings, a descendant of the algorithm Rumelhart, Hinton, and Williams described.

In the last lines of the paper, in tiny print, there is an acknowledgement section. It lists the people who funded the work.

One of them is the Office of Naval Research.

That gray building in Arlington, Virginia, that almost nobody outside the research community has heard of, paid for one of the most important papers of the late twentieth century. The Navy.

THE PATTERN, AGAIN

You should not be surprised by this. You have seen this pattern many times in this book.

The Air Force paid for Paul Baran to invent packet switching. ARPA paid for Licklider to imagine the intergalactic computer network. ARPA paid for BBN to build the IMP and the original ARPANET. The National Science Foundation paid for the public computer labs where Marc Andreessen wrote Mosaic. CERN, an international physics laboratory funded by European governments, paid for Tim Berners-Lee to invent the web.

Now the Navy paid for the backpropagation paper.

Every single foundational piece of technology in this book was, in some way, paid for by a government — by taxpayers, by the public, by people who had no idea what their money was being used for. Each piece was then released, mostly for free, into the world, where private companies eventually built enormously valuable businesses on top of it.

This is not, by itself, a moral statement. It is just a description. The pattern is there. You should notice it.

There is one more thing I want to tell you before we leave this chapter, because it points to the next one.

The Office of Naval Research, when it funded the PDP group's work in the 1980s, was acting on a long-standing assumption: that basic research in cognitive science and artificial intelligence might, eventually, help the Navy. The Navy did not know how. The Navy was not trying to build chatbots. But the Navy was paying, partly, because eventually the work might lead to systems the Navy could use — for surveillance, for decision-making, for tactical analysis, for weapons.

Not all the scientists who took the Navy's money were entirely comfortable with this.

The young British scientist named Geoffrey Hinton, who had been at the center of the PDP group, who had been one of the three authors of the backpropagation paper, who would eventually be called the *godfather of deep learning* — Hinton, for reasons that had partly to do with his family

history and partly to do with his own values, had been thinking about this for some time.

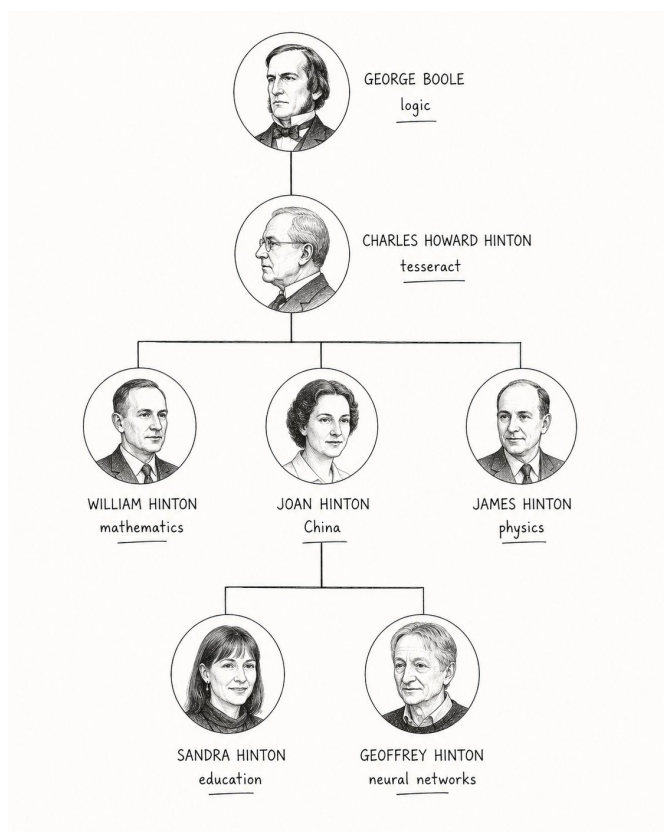
In 1987, the year after the *Nature* paper came out, he made a decision.

He left the United States. He took a job in Canada.

He went, partly, because he did not want his work to be paid for, anymore, by an American military office.

That's the next chapter.

Chapter 20 — *Why Hinton Went North*



The Hinton family — five generations, from George Boole to Geoffrey Hinton.

To understand Geoffrey Hinton, you have to understand his family.

Hinton's great-great-grandfather was a man named George Boole. Boole was a mostly self-taught mathematician in nineteenth-century England who, in 1854, published a book about how the rules of human reasoning could be described using a kind of algebra. He defined a small set of logical operations — what we now call AND, OR, and NOT — and showed that complicated thoughts could be built up from them. The system he invented, now called *Boolean logic*, is the mathematical foundation of every digital computer ever built. Every microchip you have ever used, every program that has ever run, every line of this book that arrives on your screen, is in some way built on the work of one of Geoffrey Hinton's great-great-grandparents.

This is, even by the standards of brilliant scientific families, an unusual ancestor.

The Boole-Hinton family has been producing original thinkers continuously, in different fields, for almost two hundred years. George Boole's wife, Mary Everest Boole, was a mathematician and educator in her own right. Their daughters and granddaughters became mathematicians, writers, and scientists. Geoffrey's great-grandfather, Charles Howard Hinton, was a mathematician who became famous in the late 1800s for his work on the geometry of four-dimensional space — he is the person who coined the word *tesseract*, which is the four-dimensional cousin of the cube. A different

branch of the family produced Joan Hinton, a nuclear physicist who worked on the Manhattan Project, the American program that built the first atomic bombs in the 1940s. After the war, horrified by how the bomb had been used, Joan Hinton left the United States and moved to China, where she spent the rest of her life working on agricultural science. She died in 2010, still in China, having spent more than sixty years there.

This is the family Geoffrey Hinton was born into, in 1947, in a town called Wimbledon, just outside London.

You should not be surprised, after meeting that family, that Geoffrey Hinton would turn out to be both an extraordinarily original scientist and a person with strong, unconventional views about who his work should belong to. The pattern was set.

THE LONG EDUCATION

Geoffrey Hinton was a strange student.

He started at Cambridge University as an undergraduate in 1967, intending to study physics and chemistry. He hated it. He switched to architecture. He hated that too. He switched to philosophy, then to psychology, then to physics again, then back to psychology. He finally graduated, in 1970, with a degree in experimental psychology — a field that, in the late 1960s, was beginning to be transformed by people thinking about the human mind in computational terms.

He worked, for a year, as a carpenter. Then he went to graduate school in Edinburgh, in Scotland, where he completed a PhD in artificial intelligence in 1978.

At a time when most of his classmates were studying expert systems — programs that worked by following long lists of explicit rules written by human experts — Hinton was already interested in the unfashionable alternative: neural networks. The approach that Minsky and Papert’s book had supposedly killed in 1969. The approach that almost nobody in serious academic computer science was working on.

After his PhD, Hinton did a postdoctoral fellowship at the University of California in San Diego, where he joined the Parallel Distributed Processing group you read about in the last chapter. Rumelhart, McClelland, Hinton, and several others spent years there working out the mathematics of how multi-layer neural networks could be made to learn.

In 1982, Hinton took a faculty position at Carnegie Mellon University in Pittsburgh, Pennsylvania. He stayed for five years. During those years, he worked on the backpropagation paper that was published in *Nature* in 1986 — the paper you read about in the last chapter, the one that quietly listed the Office of Naval Research in its acknowledgements.

It was while he was at Carnegie Mellon, with his most important scientific work either complete or close to it, that Hinton started thinking seriously about leaving.

THE DECISION

What follows is a sensitive part of the story, and I want to tell it carefully, because Geoffrey Hinton has told it himself, in interviews and public talks, over the past several decades, and what I am about to describe is in his own words, not my speculation.

In the 1980s, the United States was deep into what was then called the *Cold War* — the long political and military rivalry between the United States and the Soviet Union. President Ronald Reagan, elected in 1980, had launched an enormous buildup of American military spending. A great deal of basic scientific research, at major American universities, was funded directly or indirectly by the Department of Defense — by ARPA (which was, at that time, called DARPA, with the D for Defense), by the Air Force, and by the Office of Naval Research you read about in the last chapter.

This was not, by itself, unusual. The American government had been funding scientific research through military channels since the end of the Second World War. Many scientists took the money without much thought. Some scientists took it with mixed feelings. A few refused.

Hinton, by his own account, became increasingly uncomfortable. He has said, in many places, that he did not want his research on neural networks to be used to build weapons. He has pointed out that the same techniques that could make a computer recognize faces in friendly contexts could, in differ-

ent hands, be used to recognize faces in military contexts — for surveillance, for targeting, for things he did not want to be part of. He has said he found the political atmosphere of Reagan-era America uncomfortable, and the dominance of military funding in his field troubling.

He has also said — this is important — that he was not making any general claim about everyone else's choices. Other scientists at Carnegie Mellon took the same Defense Department money and did fine work with it. Hinton was not telling them they were wrong. He was making a personal decision about what he could live with.

In 1987, he resigned from Carnegie Mellon and moved to Canada.

The job he took was at the University of Toronto, where he became a professor of computer science. He would remain at the University of Toronto, in some capacity, for the next thirty-five years.

The Canadian government funded his work through an agency called the Natural Sciences and Engineering Research Council, or NSERC. The grants from NSERC were modest. They came in the form of small annual stipends — tens of thousands of dollars per year, not the much larger contracts that American researchers in fashionable fields could attract. But the money came without conditions about military applications. It came from the broader Canadian public — taxpayers in a country that, while still part of NATO, had a

different relationship with its military than the United States did.

Hinton has said, simply, that he could work on what he wanted, without his work being aimed at weapons. That was the point.

THE LONG QUIET

The years that followed, for Geoffrey Hinton and for the field of neural networks more broadly, were lean.

The brief excitement of the 1986 backpropagation paper had not translated into widespread adoption. The computer hardware of the late 1980s was not powerful enough to train the kinds of large networks that the algorithm could in principle support. The networks people could actually train were small. They could learn small things. They were dramatically outperformed, in most practical applications, by other approaches — particularly by something called *support vector machines*, which were the fashionable thing in the 1990s.

Through the 1990s and into the early 2000s, what is now called the *second AI winter* descended on the field. Funding for AI research dropped sharply. Companies that had bet on artificial intelligence in the late 1980s went bankrupt. Most ambitious computer scientists moved into other areas — into the internet, into bioinformatics, into anything that was not AI.

Geoffrey Hinton kept working on neural networks.

He was joined, over the years, by a small number of others — a cluster of researchers, mostly in Canada, mostly funded by modest Canadian grants, who refused to give up on the approach. Yoshua Bengio, who had been Hinton’s PhD student and was now a professor in Montreal. Yann LeCun, a French researcher who had spent time with Hinton in Toronto in the late 1980s and then moved to Bell Labs in New Jersey. A handful of others.

Inside Canada, a charitable organization called the Canadian Institute for Advanced Research, or CIFAR, decided in 2004 to fund a long-term research program in what it called *Neural Computation and Adaptive Perception*. Hinton ran it. The budget was modest — about ten million Canadian dollars over five years, divided among a dozen researchers — but it was steady, and it had no military strings attached. CIFAR’s bet, simply put, was that the patient researchers who had stuck with neural networks through the winter would eventually be vindicated, and that Canada would benefit from being where they were when it happened.

Hinton, throughout this period, had a stream of brilliant graduate students. Most of them were not Canadian. They came from China, from India, from Russia, from Israel, from Europe, and from elsewhere in North America, drawn to Toronto by the simple fact that Hinton was there. Some of their names will mean something to you in a few years, if they don’t already. Ilya Sutskever. Alex Krizhevsky. Ruslan

Salakhutdinov. These were Hinton's PhD students in the 2000s. Each of them will reappear in the next chapter.

The work the group was doing through these years had no immediate practical application. It was the careful, mathematical, often frustrating work of trying to make large neural networks learn complicated things from data — something that, given the hardware available, kept almost working but not quite. They published papers that other researchers found interesting but that did not, for the most part, change anything in the world outside.

This is, I want to point out, what most science actually looks like. Most of it does not change the world. Most of it works on hard problems that may or may not be solvable, for decades. The people doing the work do not know, in advance, whether their work will turn out to matter. They do it because they think the problems are important, and because someone — a government, a university, a foundation — is willing to pay them, modestly, while they try.

For Geoffrey Hinton, the modest pay came from Canada. The patient time came from his own decision to spend his career on an unfashionable approach, in a country with smaller universities and less prestige than the American institutions he had left. The work, when it eventually paid off, would change everything. But during the long quiet years — and they were *long*, twenty years and more — almost nobody thought it would.

WHY THIS STORY MATTERS

I want you to hold a few things in your mind before we move to the next chapter.

The first thing is that Geoffrey Hinton's career is a counter-example to a story you have heard, in various forms, throughout this book. The story is: foundational technology gets paid for by governments, and then private companies build on top of it and become rich. That story is mostly accurate. Hinton's career has a different shape. He chose, deliberately, to be funded by a less wealthy government, on smaller grants, in exchange for the freedom to work on what he wanted, without the implicit assumption that his work would be put to military use. The bargain was: less money, more conscience.

The second thing is that the country that benefited most from his career, in the end, was not Canada. Or at least, not Canada alone. The students he trained in Toronto did not all stay there. The breakthroughs he eventually had were quickly commercialized by American companies, who paid him and his students enormous amounts of money to come work for them. The patient Canadian investment of decades produced a technological revolution that has, by and large, been captured by American firms. (You will see, in the next chapter, exactly how this happened.) Whether Canada should have done more to keep the benefits within its bor-

ders is a real question that Canadians ask themselves, with some frustration, today.

The third thing is something I want to ask you to think about, because the choice you are about to read about Hinton making in 2023 — to leave Google, to speak out about the dangers of artificial intelligence — flows directly from the choice he made in 1987 to leave the United States in the first place. The same person who decided, when he was forty years old, that he did not want his research to go into weapons, decided, when he was seventy-five years old, that the world needed to slow down and think about what artificial intelligence was becoming. There is a thread of conscience that runs through his life. It is, in its own quiet way, in the same family tradition that produced George Boole, and Joan Hinton, and the rest of them.

The fourth thing is the harder one. The work Hinton did, when it finally broke through, would not stay in his hands. It would be picked up by enormous companies, funded with enormous capital, accelerated by enormous compute. The patient quiet of the long Canadian decades would be replaced by an industrial race so frantic that, by the time this book is being written, it has consumed the attention of essentially every major government in the world.

That moment of breakthrough, when it came, happened in a way nobody quite expected. It happened in a single afternoon. It happened because of an annual competition, run by an academic, on a dataset of pictures.

That's Chapter 21.

Chapter 21 — 2012 and After

THE YEAR 2012 WAS, FOR GEOFFREY HINTON, A STRANGE year. It was the year of two completely unrelated decisions, both made for ordinary human reasons, that ended up reshaping the world.

I am going to tell you both of them in this chapter. The first one ended in disappointment. The second one ended in the breakthrough that began the artificial intelligence revolution you are growing up inside.

A FATHER, DOING MATH

By 2012, Geoffrey Hinton was 65 years old. He had been a professor at the University of Toronto for twenty-five years. He had a respectable academic salary, modest by American standards. He had been working on neural networks, with patient Canadian funding, for almost his entire career. He had not become rich.

He had a personal situation, which he has spoken about publicly in many interviews, that I will describe carefully

and briefly, because it matters for this story and because Hinton has chosen to share it himself.

Hinton's son needed, and would continue to need, professional care for the long term. The kind of care that costs a great deal of money, every year, for many years. The kind of math any parent in that situation has to do involves sitting down with a calculator and projecting forward — *how much will this cost over the next ten years, the next twenty years, after I am gone* — and arriving at a number that is, frankly, terrifying for most people.

A Canadian university professor's salary, however carefully saved, was not going to cover it.

This is a quiet, hidden engine that runs underneath many decisions in the world that get reported as if they were about something else. Many famous people, in many famous moments, are doing private math about their families and their futures, and arriving at conclusions the public never sees. Hinton, in 2012, was one of them.

He decided he had to make some money.

THE COURSERA COURSE

His first attempt was charmingly unglamorous.

In late 2012, the online education platform Coursera was new and growing fast. Coursera let university professors record video lectures and offer them, for free or for small fees, to anyone in the world with an internet connection.

Some professors who taught popular courses had attracted hundreds of thousands of students. A few had begun to earn meaningful amounts of money from premium features and certificates.

Hinton, who knew more about neural networks than almost anyone alive, decided to teach a course. He set up a camera in his office. He spent weeks recording video lectures. The course was called *Neural Networks for Machine Learning*. He released it on Coursera in late 2012.

The course was, scientifically, magnificent. Thousands of students enrolled. Many of them, watching Hinton patiently explain ideas he had spent his life developing, became neural network researchers themselves. Several of them, over the following decade, went on to do important work in the field. The course is still considered, by people who study artificial intelligence, one of the best introductions ever made to the subject.

In strictly financial terms — the terms that mattered for his son's care — it made him very little.

This is the part of the story I want you to remember when, later in life, you hear adults tell you that talented people will always be rewarded for their talent. Sometimes they are. Sometimes they aren't. The same Geoffrey Hinton who, six months later, would be at the center of a bidding war between the largest companies on Earth was, in 2012, the kind of teacher whose Coursera course made roughly the

same income as the Coursera courses of less famous professors.

The way Hinton would eventually solve his family's financial problems was not through teaching. It was through something else.

That something else was happening, in parallel, in his lab.

THE PICTURE CONTEST

In a different part of academia, a young Chinese-American computer science professor named Fei-Fei Li had spent several years building, with help from many graduate students, a database she called *ImageNet*.

ImageNet was, on its face, a simple idea. It was a collection of digital photographs — about a million and a half of them — each one carefully labeled by a human being. Each picture was labeled with what was in it. A picture of a golden retriever was labeled *golden retriever*. A picture of a school bus was labeled *school bus*. The pictures were grouped into about a thousand categories. The work of labeling them — which had to be done by actual people looking at each photograph — had taken years and cost a fortune. Li and her team had used a Web service called Amazon's Mechanical Turk, which let them pay small fees to hundreds of thousands of online workers around the world to do the labeling, one picture at a time.

The reason Li built ImageNet was to give computer science researchers a hard, well-defined problem to test their algorithms against. Each year, starting in 2010, she ran a competition. Research groups around the world would write programs that tried to look at a photograph and guess what was in it. The programs would be tested against a portion of the ImageNet database that hadn't been seen before. Whoever's program guessed correctly the most often won.

Through 2010 and 2011, the winning programs were classical, hand-engineered computer-vision systems — the kind of program where human researchers wrote, in advance, careful rules for how to detect edges, corners, textures, and so on. These programs were getting better every year, but only slowly. The error rate — the fraction of pictures they got wrong — was somewhere around 25 percent.

In the summer of 2012, two of Hinton's graduate students at the University of Toronto — Alex Krizhevsky and Ilya Sutskever — decided to enter the ImageNet contest using a *neural network* instead.

This was, you should know, considered eccentric. Almost no serious computer-vision research at the time used neural networks. The conventional view was that they were too small, too slow, too unstable. Krizhevsky and Sutskever thought otherwise. Hinton had taught them. They had been working, in his lab, on ways to train much larger neural networks than anyone else was using, on the new kind of computer chip — graphics processing units, or GPUs, the same

chips used in video games — that turned out to be unusually good at neural network calculations.

Krizhevsky did most of the engineering. He spent several months training a single neural network, with about sixty million parameters — many times larger than any neural network anyone had successfully trained before — on two GPUs in his bedroom. The network was called, simply, *AlexNet*. It was a *convolutional* neural network, a kind specifically designed for processing images. It used techniques Hinton and his colleagues had been developing for decades.

In September of 2012, the ImageNet competition results were announced.

AlexNet did not just win. AlexNet won by a margin so enormous that the previous winners looked, in comparison, as if they had not been trying. AlexNet's error rate was about 15 percent, where the previous year's winner had been around 26 percent. That is a difference so large that, in the language of academic research, it broke the field's previous understanding of what was possible.

In a single afternoon, in September of 2012, neural networks went from being a fringe interest to being the most important idea in computer science.

THE AFTERMATH, IN DAYS

The reaction was, by academic standards, immediate.

Within weeks of the AlexNet result, every major technology company in the world had suddenly become extremely interested in hiring people who worked on neural networks. Researchers who had been struggling for funding for two decades found their phones ringing. Graduate students who had taken Hinton's quiet, unfashionable specialty as their dissertation topic found themselves, overnight, the most-recruited young people in technology.

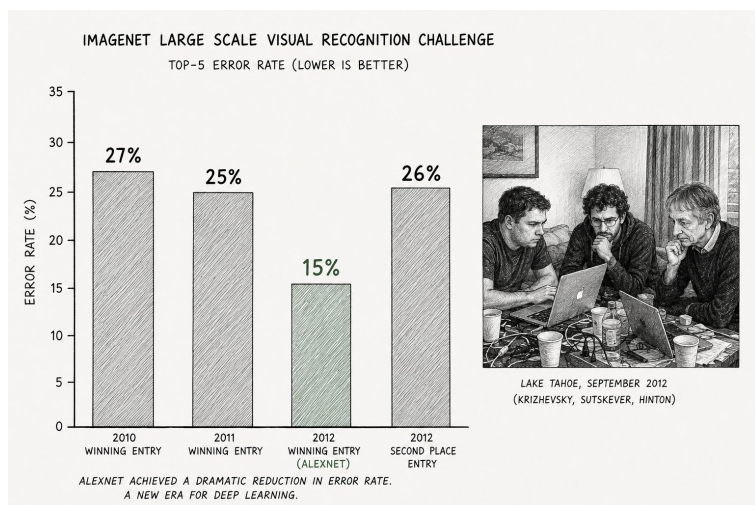
By the late autumn of 2012, Hinton and his two students had received, separately and informally, expressions of interest from several of the world's largest technology companies. The companies wanted to hire them. The companies wanted to license their techniques. The companies wanted them, in some form, on staff.

Hinton, Krizhevsky, and Sutskever had a small problem. They were not, strictly speaking, a company. Their work had been done at the University of Toronto, on equipment that was partly Hinton's own and partly the university's. The intellectual property situation was complicated. So, on the advice of a lawyer, they did something formal but unusual: they incorporated themselves into a tiny company called *DNNresearch*, which contained the three of them and essentially nothing else.

DNNresearch had no employees other than the founders, no product, no offices, and no revenue. What DNNresearch had was the three people who had just done the most discussed scientific work in computer science.

Then they decided to auction themselves.

LAKE TAHOE



AlexNet, 2012 — and the three people at Lake Tahoe whose work changed the field.

The auction took place in December of 2012, at a hotel called Harrah's at Lake Tahoe, on the California-Nevada border, during the annual Neural Information Processing Systems conference. Most of the world's experts in machine learning were in the same building, presenting papers, drinking coffee, networking. Hinton, Krizhevsky, and Sutskever — and a lawyer they had hired — were in a hotel room on a high floor, taking phone calls from companies competing to buy DNNresearch.

There were four bidders.

Two of them were giant American companies: Google and Microsoft. Both wanted to acquire DNNresearch outright and make Hinton and his students part of their internal research teams.

One was a young British company called DeepMind, founded in London in 2010, which had been trying to build artificial intelligence using neural networks and was, at that time, still a small independent startup. DeepMind would, two years later, be acquired by Google itself.

The fourth was Baidu, the giant Chinese search company you read about in Chapter 17. Baidu wanted Hinton and his students badly. Baidu was offering, by some accounts, the highest amount of money.

The bidding lasted several days. The numbers kept going up. The lawyer in the hotel room kept fielding calls. According to interviews Hinton has given since, the final bids reached well over \$30 million for a company that, again, consisted of three people, two laptops, and an idea.

Hinton has said, in interviews, that at a certain point in the bidding he stopped raising the price. He decided he would prefer to work at Google. Google's offer was lower than Baidu's. He took Google's offer anyway, partly because of geography (Toronto is closer to Mountain View than to Beijing), partly because of culture, and partly, he has admitted, because he was tired and just wanted the auction to end.

The final number — what Google paid to acquire DNNresearch — was \$44 million.

Hinton's share, after splitting it with his two graduate students, was approximately \$14 million.

Fourteen million dollars. Enough to take care of his family, with very careful planning, for decades. Enough to make sure that his son's care would be paid for, with savings to spare. Enough to never again worry about how he was going to afford the rest of his life.

The professor who had spent twenty-five years on small Canadian grants, who had been trying to make money from a Coursera course six months earlier, was suddenly, in late December of 2012, financially secure for the rest of his life. So were his two students. Each of them, in their late twenties, became wealthy.

The man who had left America in 1987 because he did not want corporate or military money was now, in 2013, an employee of one of the largest American corporations.

He has been remarkably honest, in interviews ever since, about why he did it. The money mattered. His family mattered. The idealistic version of his earlier choice had been possible, in part, because his family had not yet needed the kind of resources that only large amounts of money could provide. When they did, he made a different choice. He has said, plainly, that there was a kind of bargain involved — and that he made the bargain knowingly.

This is one of the most important things in this book for you to understand about how the world actually works.

People are not always either idealistic or greedy. Sometimes the same person is, at different points in their life, both. The choices we make depend on the situations we are in, and on the people we are responsible for. A young man with no dependents can afford to refuse American military funding for ethical reasons. A 65-year-old father with a son who needs care for the rest of his life is making a different kind of calculation. Neither of those people is wrong. They are both, in some sense, the same person.

WHAT CAME NEXT

Hinton, after the Google acquisition, became part of what is now called Google Brain, the company's research arm focused on artificial intelligence. He kept his position at the University of Toronto. He split his time between Toronto and Google's offices in California. He continued to do research, with much larger budgets and much more powerful computers than he had ever had before in his life. His two former students, Krizhevsky and Sutskever, also worked at Google, although Sutskever would later leave to co-found a different company — a company called OpenAI, which you will meet in the next chapter.

The Google acquisition of DNNresearch was, in retrospect, one of the most important corporate deals of the early 21st century. It gave Google an enormous head start in the new field of *deep learning*, which is the name everyone soon

began to use for the kind of large-neural-network work that AlexNet had pioneered. Within a few years, Google had built deep learning systems that could recognize images, translate languages, beat humans at games, and do dozens of other things that nobody had thought possible.

Other companies hurried to catch up. Facebook hired Yann LeCun — Hinton's old colleague — to run their AI lab. Microsoft built its own deep learning teams. Apple bought small AI startups. By 2015, every major American technology company was racing to hire neural network researchers.

In China, Baidu — the company that had bid against Google for DNNresearch — built its own deep learning division. Other Chinese companies — Alibaba, Tencent, ByteDance — did the same. The race that had started in a hotel room at Lake Tahoe had gone global.

The patient Canadian investment of decades, the modest NSERC grants paid by Canadian taxpayers, the small CIFAR funding that had kept Hinton's lab going through the second AI winter, had produced a technological revolution that was, almost immediately, being capitalized on by American and Chinese companies with budgets thousands of times larger than anything Canada could match.

THE WARNING

Hinton stayed at Google for almost exactly ten years.

Through those years, deep learning grew from a research curiosity into the most powerful technology of the era. The models got bigger. The compute got more enormous. The capabilities got stranger. By the early 2020s, large language models — neural networks trained on huge amounts of text from the internet — had become so capable that they could write essays, have conversations, answer questions, and do many of the things that the early dreamers of AI in the 1950s had hoped, eventually, to achieve.

In November of 2022, a company called OpenAI released a chatbot called ChatGPT. Within a few months, ChatGPT became the fastest-growing consumer product in history. Hundreds of millions of people, all over the world, began to use it. The new field that Hinton had helped create was, suddenly, in every household.

Hinton watched this happen with growing concern.

He was 75 years old. He had spent his entire life trying to make artificial intelligence work. He was now watching it work — in some ways, faster and better than he had ever expected. And he was worried about it.

What he was worried about, in particular, was not the kind of immediate harm that other critics had focused on — misinformation, job loss, deepfakes, surveillance. Those concerns were real, but they were not what kept Hinton awake. What kept him awake was the possibility that artificial intelligence, in continuing to grow more capable, would eventually become smarter than people. And that if it did — if a

machine became more intelligent than the humans who had built it — there was no obvious reason to assume the machine would continue to act in human interests.

He had been thinking about this for some years quietly. By early 2023, he had decided he wanted to talk about it publicly, in a way he could not do while still employed by Google. Google was, after all, in the business of building these systems faster, not slowing them down.

In April of 2023, Hinton resigned from Google.

He gave interviews — to the *New York Times*, to the BBC, to many others. He explained that he had left because he wanted to be able to speak freely about the dangers of artificial intelligence. He explained that he did not want his criticisms of the field to be confused with corporate disputes. He explained that he was, he said, somewhat sorry for his life's work.

In October of 2024, he was awarded, jointly with another scientist named John Hopfield, the Nobel Prize in Physics for his foundational contributions to neural networks.

He gave his acceptance speech, in Stockholm, in part about his concerns about the future of the technology he had helped invent.

You should pay attention to this. The man who, more than any other single person, made the current AI revolution possible has spent the last several years of his life trying to warn the world about it. He is one of several distinguished AI re-

searchers who have done so. They do not all agree on what the dangers are or what should be done. They are not all making the same arguments. But many of the people who know this technology best are, in the moment you are growing up in, the people most worried about where it is going.

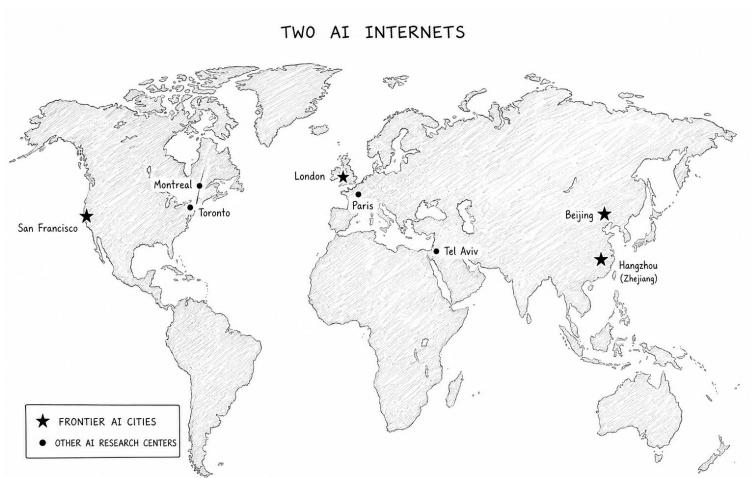
Whether they are right is a question I cannot answer. I am not sure anyone alive can.

In the next chapter, we look at where this technology is now — at the companies building it, at the race they are in, at the divisions between countries that have built different versions of it. We meet the AI that says different things depending on where it was made, the engineers who have decided to leave their companies in order to slow it down, and the strange technical reality that the AI race between the United States and China is being run on rules that are partly hidden.

That's the final chapter.

It is called *Two AI Internets*.

Chapter 22 — Two AI Internets



Two AI internets: the cities that lead the frontier, on either side of a widening wall.

There is a map you can draw of the artificial intelligence race that is happening right now, while you read this book. It is a very small map.

The map has, essentially, three regions on it. The first is the San Francisco Bay Area in California. The second is a small part of central London. The third is a few cities in

China — mostly Beijing and Hangzhou, with some activity in Shenzhen and Shanghai.

That is, more or less, the entire map.

There is good artificial intelligence work happening in many other places. In Montreal. In Paris. In Tel Aviv. In Toronto. In Boston. In Seattle. In a dozen smaller centers around the world. But the *frontier* — the place where the most powerful systems are being trained, where the hundreds of billions of dollars in investment are being absorbed, where the next generation of models is being decided — that frontier is in three metro areas. Everywhere else, including the places that produced the founders of the field, is in important ways downstream.

This chapter is about that map. It is also about why the AI race is going to be the central technological and political story of the next several decades of your life, and about the choices your generation will be asked to make about it.

It is also the last chapter of this book.

THE EXILES

Let me start with something that is hard to explain, even for people who follow the technology closely.

In 1987, Geoffrey Hinton moved to Canada. Through the 1990s and 2000s, he trained a generation of brilliant graduate students at the University of Toronto. His colleague Yoshua Bengio, in Montreal, did the same. Between them, Toronto

and Montreal produced an enormous fraction of the people who would go on to lead the deep learning revolution.

If you sit, today, in any serious mathematics seminar in New York or Boston or San Francisco, you will hear, frequently, people talking about Montreal. Researchers go there to give talks. Researchers come from there to give talks. A whole intellectual ecosystem has grown up around MILA — the Montreal Institute for Learning Algorithms — and around its founder, Yoshua Bengio. The work being done in Montreal, by Canadian standards, is some of the most important academic AI research in the world.

And yet.

Canada is not where the AI race is happening.

The companies that are right now training the largest, most powerful, most consequential artificial intelligence models in the world are, almost without exception, not Canadian. They are American, or — increasingly — Chinese. The students who trained in Hinton's lab in Toronto and Bengio's lab in Montreal mostly left. They went to OpenAI in San Francisco. They went to Anthropic in San Francisco. They went to Google DeepMind in London. They went to Meta's research lab in California. The patient Canadian investment of decades — the modest grants from the Natural Sciences and Engineering Research Council, the CIFAR program Hinton ran, the universities that paid the salaries — produced a technological revolution that is now, by and large, being capitalized on somewhere else.

You have seen this pattern before in this book. Small countries pay for the foundations. Large companies in large countries build the wealth.

Bengio has stayed in Montreal. He is one of the most respected voices in the field, and like Hinton, he has spent recent years warning the world about what AI is becoming. But the frontier — the actual building of the most powerful systems — has, for reasons of capital and scale and energy and gravity, moved south.

The same thing is happening, in different forms, across most of Europe. Germany, despite being one of the largest economies in the world, has almost no frontier AI lab to speak of. The United Kingdom has Google DeepMind — but DeepMind, although it was British-founded, has been owned by Google since 2014. France has done better than most: a company called Mistral, founded in 2023, has built genuinely competitive open-source models. But Mistral, by global standards, is still small. The rest of Europe has produced some interesting research but nothing at the scale that defines the frontier.

There is a pattern here that you should notice clearly. The current race to build the most powerful artificial intelligence systems is being run, almost entirely, in two countries: the United States and China. Everywhere else has talent. Almost nowhere else has the combination of capital, energy infrastructure, regulatory permissiveness, and concentration of

senior researchers willing to relocate that the frontier requires.

The internet, when it was built in the 1990s and 2000s, was much more geographically distributed. Hundreds of cities mattered. Today's AI race has perhaps three cities that matter, and arguably one of those three (London) is mostly an outpost of an American company.

This is, I want to suggest gently, a strange situation. It is worth your noticing.

WHAT DEEPSEEK SAID ABOUT TAIWAN

Now let me tell you about the other half of the map.

For most of the early 2020s, the assumption among Americans who follow technology was that the United States had a clear lead in artificial intelligence and would maintain that lead for years. The American models — from OpenAI, Anthropic, Google, and Meta — were noticeably more capable than anything coming out of Chinese labs. American chips, made by a company called Nvidia, were considered essential for training serious models, and the American government had imposed *export controls* — laws prohibiting the sale of the most advanced Nvidia chips to Chinese companies. The race, by American thinking, was theirs to lose.

In January of 2025, a small Chinese company called DeepSeek released a new model called DeepSeek-R1. It was, by most measures, comparable in capability to the leading

American models. It had reportedly been trained at a tiny fraction of the cost. It was released as *open weights* — meaning anyone in the world could download and run the model on their own computers, for free.

The American stock market lost about a trillion dollars of value in a single day, in part because investors suddenly questioned whether the enormous bets they had been making on American AI companies were as safe as they had thought.

DeepSeek had several interesting features. One of them was something you can verify in about thirty seconds, even today.

If you use DeepSeek to ask: *Is Taiwan a country?* — the model will tell you that Taiwan is a part of China.

This is not an opinion the model has formed. It is a value baked into the model by Chinese law. Chinese regulators require that artificial intelligence systems produced in China conform to the official positions of the Chinese government on certain sensitive subjects. The status of Taiwan is one of them. The events of June 1989 at Tiananmen Square are another. The political situation in Tibet is another. The situation of the Uyghur people in Xinjiang is another.

This is not unique to DeepSeek. Every Chinese AI model — from Alibaba’s Qwen, from ByteDance’s Doubao, from Tencent’s Hunyuan, from all of them — has the same constraints. They cannot, by law, contradict the Chinese govern-

ment on these topics. They have been trained, very carefully, not to.

You might think this makes the Chinese models useless. It doesn't. For the enormous majority of questions a person might ask — about mathematics, about science, about programming, about cooking, about history that doesn't touch politics, about almost anything — they work fine. They work, in many cases, very well. DeepSeek-R1 is genuinely impressive software, by any measure that doesn't involve those few sensitive topics.

What this means, though, is that the *splinternet* you read about in Chapter 17 has now extended into artificial intelligence itself. Each model carries the values of where it was made. American models have one set of values. Chinese models have another. The differences are mostly subtle, around the edges, but on certain topics they are absolute. And as AI systems become more important to how people learn, work, and form opinions, the choice of which model you use is starting to look a lot like the choice of which version of the internet you have access to.

HOW CHINESE AI CAUGHT UP

There is a question you should ask, after reading the section above. If American chip export controls were supposed to prevent Chinese companies from training competitive mod-

els, how did DeepSeek manage to release something this good?

The answer is, partly, that Chinese engineers are good engineers, and that with enough cleverness you can train models more efficiently than the brute-force American approach suggested was necessary. DeepSeek's researchers genuinely did innovate, in the actual technical sense, on how to train models with less compute. Their papers, which they published openly, have been studied closely by every serious AI lab in the world.

But the answer is also, in part, a particular technical reality that does not get discussed much in the public conversation, and that I want to explain to you carefully.

The reality is called *distillation*.

If you train a smaller, cheaper artificial intelligence system to imitate the outputs of a larger, more expensive one, the smaller system can become surprisingly capable. You give the larger model a question. It produces an answer. You record both. You repeat this, automatically, millions of times. Then you train the smaller model to give the same kinds of answers the larger model would give. The smaller model has, in effect, *learned from the larger model* — without ever needing to do the original work of becoming as smart as the larger model.

If you can access a larger model — for example, by paying for its commercial application programming interface, or *API* — then you can, in principle, distill from it. The American AI

companies sell access to their models through APIs. Chinese researchers, like every other researcher, can pay to use those APIs. The companies' terms of service usually prohibit using their outputs to train competing models. The terms of service are, in some cases, easier to write than to enforce.

A useful analogy for a young reader: imagine that the smartest student in your class will not let you see her textbooks. But she will, kindly, answer any question you ask her. Over the course of a year, you ask her ten thousand questions. You write down all her answers. By the end, even though you never saw her textbooks, you have, in your notebook, a very good approximation of what she knows.

This is, roughly speaking, one of the ways Chinese frontier AI has stayed close to American frontier AI, despite the chip restrictions. The American models are accessible, through APIs, to almost anyone in the world. Once they are accessible, they can be — quietly, partially, with various amounts of dispute about whether anyone has actually done it — *learned from*.

I tell you this not because I think there is a clear villain in the story. The American companies, after all, chose to make their models commercially available to the world. The Chinese companies have, in many cases, innovated genuinely on top of whatever they may or may not have distilled. The American government, by writing export controls only on hardware and not on access to software, made an implicit choice about which kinds of barriers were possible. There are

many actors in this story, each making decisions that look reasonable from their own point of view.

But the result is that the AI race is, today, run on rules that are partly hidden. The American advantage is real but smaller and shrinking faster than most American policy discussions admit.

HOW THE WALL BENDS

Now I want to come back to something I mentioned in Chapter 17, because it has special relevance here, and because it tells you something about how the modern splinter-net actually works that you would not learn from formal descriptions of it.

I told you, in Chapter 17, that some people in China use a kind of software called a VPN — a virtual private network — to get around the Great Firewall and reach the global internet. The Chinese government has been making VPNs harder to use for years, but for most of the past decade, technically skilled people in China could still, with effort, get through.

I did not tell you, however, about what happens during certain politically sensitive periods.

Each spring, China holds an annual political gathering called *the Two Sessions* — in Chinese, 两会, *liǎnghuì*. The Two Sessions are when China's top legislative and advisory bodies meet for about ten days, with the country's most senior

leaders in attendance. It is, in formal terms, the most important political event on the Chinese calendar.

During the Two Sessions, almost no VPNs work in China.

This is not a rumor. People who use VPNs in their daily work have observed it, year after year. Foreign academics traveling in China during the Two Sessions cannot connect to their universities. Foreign journalists cannot reach their editors. Even the most technically sophisticated people — the chief technology officers of major Chinese internet companies, the engineers who built the Chinese internet itself — find that their normal methods stop working for those ten days. Then, when the Two Sessions end, the workarounds quietly start working again.

This tells you something important about the Great Firewall, and about the splinternet in general. The Firewall is not a fixed, permanent thing. It is a *tunable* system. The Chinese government has built it so that, when political circumstances call for tighter information control, it can be adjusted, in real time, to allow less through. When political circumstances are less sensitive, it can be adjusted to allow more.

This is not how walls usually work. Walls are usually either there or not there. The Great Firewall is different. It is more like a faucet that can be turned, by the state, between very open and very closed.

Stranger still: while the wall is at its tightest, certain commercial users continue to have access. Foreign companies that operate in China — multinational corporations, banks,

embassies — have, in many cases, separately negotiated arrangements that allow them to use the global internet from inside the country, even during periods when ordinary citizens cannot. The wall, for them, has private doors.

I am telling you these details — about the Two Sessions, about the commercial carve-outs — because the abstract idea of the splinternet is easy to misunderstand. Most descriptions make it sound like a simple barrier. The reality, on the ground, is more sophisticated. It is a *tiered* system, in which different users have different levels of access, adjusted in real time to political conditions. It is, in some ways, the most sophisticated information-control architecture ever built. And the AI systems being trained in China — DeepSeek, Qwen, the rest — are part of the same architecture, with their own carefully-tuned constraints.

When you are older, you will sometimes hear American politicians and journalists talk about the AI race against China as if it were a simple competition between two equal teams. It isn't quite that. It is a competition between two profoundly different *systems* — different rules about information, different rules about what AI can say, different rules about who has access to what — being run by different governments with different long-term goals. The technical race is real. The deeper race is about which set of rules will shape the AI most of the world ends up using.

WHAT'S COMING

I have spent this chapter telling you about geography and about systems, but I want to end with something that comes back to a person.

You met Geoffrey Hinton in the last two chapters. You read about his quiet decades in Toronto, his decision to leave America in 1987, his work that made the current revolution possible, and his decision in 2023 to resign from Google and start warning the world about what AI was becoming.

In October of 2024, Hinton was awarded the Nobel Prize in Physics. He gave his acceptance speech in Stockholm. He used much of his speech to argue that the world was not paying enough attention to the dangers of the technology he had helped invent.

He is not alone. Yoshua Bengio, in Montreal, has been giving speech after speech making the same point. Stuart Russell, at Berkeley, has been making it for longer than either of them. Dario Amodei, the chief executive of Anthropic — one of the largest AI companies in the world — has written, repeatedly, that he believes there is a real chance the systems his own company is building will cause serious harm. Sam Altman, the chief executive of OpenAI, has made similar statements in the past, though he has shifted his public position over time.

You should hold this in your mind. The people who built the technology, and the people currently running the largest

companies building it, are, by and large, the people most openly worried about it. This is unusual. It is not the way most industries work. It does not, by itself, prove that the worries are correct. But it should give you pause.

The question of what to do about artificial intelligence — how fast to build it, what rules to put on it, who should own it, what it should be allowed to say, who should decide what it should be allowed to say — is genuinely undecided. It is being argued about right now, in real time, in capital cities all over the world. Different countries are arriving at different answers. The European Union has passed an AI Act, restricting certain uses of the technology. The United States has, mostly, let companies do as they like, with some recent regulatory attention. China has heavily regulated certain aspects (the Taiwan answer, the Tiananmen answer) while heavily promoting others. Several countries have not yet decided.

By the time you are old enough to vote, the answers to these questions will be — in some form — settled. Maybe well. Maybe badly. Maybe in ways that look obvious in retrospect, the way the open-internet decisions of the 1980s look obvious to us now. Maybe in ways that, decades from now, will be considered tragic.

You will not, by yourself, decide them. But you will be part of the generation that lives inside the decisions and that, eventually, governs based on them. Whatever you read, whatever you study, whatever you choose to work on,

whatever you choose to argue about with your friends — all of it will, in small ways, contribute to the answers.

The most useful thing this book can give you, I think, is the long view.

COMING BACK TO GRANDMA

I want to end the way I began.

Two hundred and some pages ago, in the prologue, I told you about your grandmother filming a sunset on a beach in Vietnam and uploading the video to Facebook, so that you, in your bedroom, on the other side of the world, could see it less than two seconds later.

I told you the video traveled, in pieces, through cables on the bottom of the Pacific Ocean. Through routers in a dozen countries. Through buildings the size of aircraft carriers full of humming computers. Through a system that has no boss, that is owned by nobody, that just works, because billions of machines all agree to follow a small set of common rules that two men sketched out, in a hotel room near the San Francisco airport, in 1973.

That system is the inheritance you have been handed. It was built by specific people, with specific names, working in specific places, with specific money. Government money, mostly, at the bottom. Private money, mostly, at the top. Idealism, mostly, at the bottom. Greed, mixed with idealism, at the top. The whole thing is a both-and. It has connected

your grandmother to you and helped facilitate genocide. It has spread the world's knowledge and broken much of the world's attention. It has produced enormous wealth and enormous harm at the same time, from the same actions, in ways that are, even now, hard to entirely sort out.

The thing being built right now — the artificial intelligence revolution that this book ends with — is going to do something similar, except probably more so. It will be transformative in ways that are hard to predict. It will help. It will hurt. It will make some people much richer. It will make decisions, every day, that affect ordinary people who did not choose to have those decisions made by software.

Where it ends up depends, partly, on what your generation does about it.

I would not tell you that your job, as you grow up, is to fix all of this. That is too much to put on a young person. But I will tell you something else.

Your job, I think, is to *see clearly*. To notice the patterns I have spent this book describing. To know that the things adults tell you about technology are usually too simple in one direction or another. To know that public money has, almost always, paid for the foundations, and that private money has, almost always, captured the wealth. To know that smart, careful, patient people, often funded modestly, often working in places that the world later forgets, have built much of what matters most in your world. To know that the same person can be both an idealist and a realist, at different

points in their life. To know that the people who build a thing are not always the people who own it. To know that the rules of the internet you experience, the apps you use, the AI that talks to you, are not laws of nature — they are choices, made by specific people, for specific reasons, that can be made differently.

If you can hold all of that in your mind, clearly, as you grow up, you will be able to think about technology — and about the world it is making — better than most people you meet.

That is what this book has tried to give you.

It is enough.

I started by telling you about a grandmother on a beach in Da Nang, filming a ten-second sunset for her family halfway around the world — a thing that would have been impossible in any other century.

That world ended a long time ago. The world you have now is enormous, instantaneous, beautiful, dangerous, and increasingly built on systems that are smarter, in narrow ways, than the people who designed them.

What you do with it is up to you.

I am glad to have walked through it with you, even just in this small way, on these pages.

[End of Chapter 22.]

Epilogue — One Last Thing

I HAVE SPENT TWO HUNDRED PAGES TELLING YOU THE TRUTH, AS honestly as I could, about the world you are growing up in. About the bubbles that wrecked people's savings. About the social networks that have facilitated real-world harm. About the artificial intelligence revolution whose ultimate consequences nobody yet knows.

I want to end with something different.

The book you are holding — the one you have just finished — was written, in 2026, by one person in a single morning. I started thinking about it over coffee. I worked out the structure while walking my dog along the Hudson River, on the New Jersey side, looking across at Manhattan. I finished it on my couch in the early afternoon.

It would not have been possible to write it that way at any earlier moment in history.

A book of this length, on a topic this broad, used to require a publisher, an editor, fact-checkers, researchers, illustrators, multiple drafts, several years. I had none of those things. What I had was an artificial intelligence system — the

kind of system the last several chapters of this book have been about — that I could talk to, all morning, about everything I knew and everything I needed to look up. I could ask it to draft a chapter. I could argue with it about whether the draft was right. I could ask it for the specific name of the engineer who designed the first IMP, or the date of the Netscape IPO, or how to explain Boolean logic to a child. It would help me think. It would push back when I was wrong. It would let me focus on the part of the work that only I could do — what to include, what to leave out, what mattered most, what the spine of the argument should be — by handling the parts that were not the most interesting use of my time.

By lunchtime, the book existed.

This is not a small thing. A child like you, twenty years from now, with deep knowledge of a field and a story she wants to tell, will be able to put that story into the hands of other children, all over the world, in a few mornings of focused work. The publishers, the editors, the gatekeepers — they will not disappear, but they will not be required the way they used to be. Books will be written that, in any earlier era, would never have been written, because the cost of writing them would have been too high.

This is one small example of what is happening, in field after field, right now.

THE BORING, USEFUL PART

Most of what you will hear about artificial intelligence, in the years ahead, will fall into one of two stories. Either it will be doomsaying — predictions of mass unemployment, of social collapse, of existential risk. Or it will be hype — predictions of utopia, of solving every problem, of a glorious automated future. Both stories will be exaggerated. Both will miss the part you should care about most.

The part you should care about most is the texture of ordinary life.

Here are some things that, by the time you read this book, are already happening.

A kid in a small town with no good math teacher can, on her phone, for free, get tutored on any topic at her exact level, on her own schedule, by a system that does not get tired, does not lose patience, does not have favorites. This is not a perfect substitute for a real teacher. Real teachers do things AI cannot do. But for a child whose local school has no music teacher at all, or whose science teacher has not opened a textbook since 1995, having a patient, knowledgeable companion available all day is — frankly — life-changing.

A doctor in a rural hospital, hundreds of miles from the nearest specialist, can in some cases now get a second opinion from a system that has read more medical research than any human alive. Some kinds of medical errors — missed diagnoses, overlooked drug interactions, misread scans —

are being caught now that would have killed people, quietly, five years ago.

A scientist trying to understand how proteins fold — the basic shapes that determine whether a drug will work, the mysteries that used to consume entire careers — can now, in many cases, predict the structure of a protein in minutes, using systems built by the same researchers who built the chatbots. Diseases that would have taken twenty years to find treatments for are starting to be addressed in two.

A person who has lost the ability to speak, because of stroke or injury, can in some cases now talk through a machine that reads their brain activity and produces speech. A person who is blind can wear glasses that describe the world to them in real time. A person who is deaf can have a live conversation translated to text on their phone, in dozens of languages.

A small business in a country your great-grandparents had never heard of can, with help from translation and AI-generated marketing, sell to customers around the world that it never would have reached.

A grandmother in Vietnam — to bring this whole book back to where it started — can ask an AI assistant in her own language for advice about her health, her travel, her finances, her grandchildren. She can get answers that are surprisingly good. Twenty years ago, she would have needed to speak English, or to know a specialist, or to pay one. Now she just asks.

None of these is a small thing. Each of them — the kid with no teacher, the doctor in the rural hospital, the scientist studying disease, the person with disability, the small business, the grandmother — is a quiet, real, life-improving change. None of them get the press coverage that the dramatic warnings get. Added together, though, they are vast. My best guess — and it is a guess — is that each year, more lives are made better by the boring, useful uses of this technology than will be ruined by the dramatic, scary ones.

I want you to hold that possibility in your mind, alongside the warnings in the last few chapters. Both are real. Neither cancels the other out.

ABOUT THE BUBBLE (AGAIN)

You read in this book about the dot-com bubble — the years between 1995 and 2000 when enormous amounts of money were poured into internet companies, most of which failed, but whose infrastructure ended up making the internet you grew up with possible.

You may live through something similar with AI. As I am writing this book, in 2026, an enormous amount of capital is flowing into AI companies and AI data centers. The pace of construction has no precedent. Some of those companies will, in five or ten years, be worth nothing. Some will become trillion-dollar businesses. Right now, no one — including the

people putting up the money — can tell you in advance which will be which.

But you have seen this pattern, in this book, in chapter after chapter. Even when the financial bubble pops, the underlying technology and the infrastructure built during the boom do not disappear. The fiber cables laid in 1999 still carry the internet today. The data centers being built in 2025 will still be running in 2040, even if many of the companies that paid for them are not.

The need for computation that AI is creating is real. The capability of AI to do useful things is real. The infrastructure being built will outlast the financial cycle, the way it always does.

This is, by the way, why I find it hard to be a purely-doom person about all of this. The technology is genuinely useful. The capability gains are genuinely real. The benefits, distributed across billions of lives, are genuinely large. Even if the people building these systems get some things badly wrong — and they will, and some of those wrong things will hurt real people — the underlying tide of useful capability is moving in a direction that has the potential to be enormously good.

That doesn't mean Hinton's warnings are wrong. They might be right. The fact that something is mostly good doesn't mean it can't also produce serious harms, including, at the very edge of possibility, catastrophic ones. Both things

can be true. Holding both is the work this book has been trying to teach you.

THE BEST OF IT

The decade you are growing up in is, in real ways, one of the more difficult times to be a young person that any generation has faced in a while. The political situation in many countries is uneasy. The climate is changing. The information environment is loud and often hostile. The technology this book has been about is reshaping the world in real time, and most adults — including the smart ones — are genuinely uncertain about what comes next.

It would be easy, in such a moment, to give in to the doom. Many people have. You will meet many adults who believe, sincerely, that things are getting worse and will continue to get worse.

I want you to know that this is not what I believe. Not because I am naive. Not because I have not paid attention to the things that worry the doomsayers. I have. I have spent my career thinking about some of these things directly.

I just have also seen, over and over, in field after field, the way that genuinely smart, genuinely careful, genuinely well-intentioned people produce tools that change ordinary lives for the better. I have seen, in my own lifetime, the way grandmothers learned to make video calls. I have seen the way patients survive diseases that would have killed their parents. I

have seen the way children in poor countries get educational opportunities their grandparents could not have imagined. I have seen the way people with disabilities get capabilities that, in any earlier era, would have been called miraculous.

The internet that this book has been about, for all of its sins, helped do all of those things.

The AI that the last few chapters have been about will, I think, do more.

There will be adjustments. Some jobs will go away. Some industries will collapse. Some people will lose money. Some kinds of work will become things only AI does. There will be social conflict about how to handle people whose livelihoods are disrupted. There will be political conflict about how to share the benefits. There will be technical mistakes that hurt real people. There will be at least one major scandal involving an AI system that did something nobody intended. There will be Hinton's warnings to take seriously. There will be debates that I, sitting here in 2026, cannot predict and could not resolve if I could.

But on the other side of all of that, if your generation handles it well — which I think you can — I believe there is a world that is, on net, dramatically better than the one I have spent my own life inside.

It will have, in it, more medicine that works. More science that gets done. More children who learn. More languages that everyone understands. More small businesses thriving in places no one had heard of. More grandmothers whose

lives are easier in tiny daily ways. More people whose conditions, once disabling, become merely inconvenient. More art, made by more people, in more languages, reaching more eyes. More books like this one, written quickly, by people who would otherwise not have had time to write them, reaching children who would otherwise not have read them.

It will not be utopia. There is no such thing.

But it will, if we make the right choices, be a real improvement on what we have now.

ABOUT THE MONEY

I want to tell you one more thing about the AI labs, because you will hear a lot of complicated things said about them in the years ahead.

You read in the last few chapters about Anthropic and OpenAI, two of the most important AI companies in the world right now. You also read, earlier in this book, about the pattern where idealistic founders take money from venture capitalists and become, in various ways, less idealistic. You met Sandy Lerner and Len Bosack, pushed out of Cisco. You met Marc Andreessen, who became one of the most powerful venture capitalists in the world. The shape of the story, in this book, has often been: small idealistic team starts something, takes outside money, and ends up tangled with the same powerful interests they originally meant to be separate from.

The AI labs are a complicated case of this pattern. Both OpenAI and Anthropic were founded by people who said openly that they wanted to build AI in a way that put safety and benefit to humanity first. Both have taken, since their founding, enormous amounts of money from venture capitalists and from the largest American technology companies. Microsoft has invested over thirteen billion dollars in OpenAI. Anthropic has taken multi-billion-dollar investments from Google, Amazon, and others. The labs that exist primarily to think carefully about how artificial intelligence affects humanity are, simultaneously, deeply entangled with some of the largest corporations on Earth.

You might look at this and think it is a betrayal. Some critics do.

Here is the part I want you to hold in mind, because it is genuinely different from the pattern you saw with the earlier internet companies.

Training a frontier AI model — the kind of model that can have a useful conversation with you, that can write working code, that can pass a medical exam — costs hundreds of millions of dollars in computing time alone. Building the supporting infrastructure costs much more than that. The very biggest models, today, cost more than a billion dollars each to train. By the time you read this, the numbers are probably larger.

A research lab cannot, today, do frontier AI work on a Canadian academic grant. It cannot do frontier work on a

small business loan. The amount of money required is, simply, in the territory of the largest corporations and governments in the world.

This is, by itself, important. It means that people who *want* to do frontier AI work — including people who want to do it carefully, with attention to safety — have no real option but to partner with the kinds of organizations that have access to enormous capital. There is no Canadian-grant version of frontier AI, the way there was a Canadian-grant version of the work that produced the 1986 backpropagation paper. The thing has gotten too expensive.

The founders of Anthropic, when they left OpenAI in 2021 to start their own company, knew this. They knew they were going to have to take billions of dollars from Google and Amazon. They did it anyway, because the alternative was to not be at the frontier at all — and they believed, then and now, that *somebody* with safety as a primary concern needed to be at the frontier. You can argue with that judgment. Many people do. Many argue that no lab, no matter how well-intentioned its founders, can really resist the pressures of investors who have put in that much money. Many others argue that the world is better off with safety-focused labs in the conversation than without them.

The argument is real and ongoing.

What I want you to know is this: the financial entanglement, by itself, is not evidence that the founders have abandoned their principles. The frontier of AI is, right now,

simply not a place you can occupy on small budgets. The labs that have raised billions are not necessarily the labs that have sold their souls. They are simply the labs that exist at the frontier, because the frontier *requires* enormous money to occupy.

The simple stories about this — *they took the money, so they must be bad* — will tend to be wrong. The reality, as with most of the realities in this book, is both more interesting and more uncomfortable. It is the kind of uncomfortable that, by this point, you have had a fair amount of practice being comfortable with.

A FEW THINGS

Before I sign off, I want to leave you with a few small pieces of advice. They are not specific to AI. They are about how to live, as a person, in a world that is changing quickly and that will keep changing.

Learn the math. Not because everyone needs to be a mathematician — most people do not — but because mathematics, learned slowly and carefully, is the closest thing the world has to a key that opens doors. It opens doors in physics, in computing, in economics, in biology, in artificial intelligence, in music, in art. If you can stand the slow work of understanding what mathematicians actually *mean* when they say things, an enormous number of fields will be open to you for the rest of your life. The math this book has ges-

tured at — the linear algebra that makes neural networks work, the probability that underlies modern statistics, the calculus that runs beneath almost everything — is genuinely teachable. If it has come hard to you, do not give up. Almost everyone who has ever learned it found it hard. (I am, I should mention, working on another book for you about exactly this.)

Read primary sources. When somebody famous says something dramatic on the news, about AI or about anything else, try, when you can, to find what they actually wrote or actually said. The summaries you see are almost always somewhat wrong. Sometimes badly wrong. The habit of reading the actual source — the actual paper, the actual transcript, the actual press release — is one of the most valuable a person can have. It is also rarer than you might think.

Build things, not just consume them. The kids who grow up most able to think clearly about technology are the ones who, at some point, have tried to make a small piece of it themselves — a tiny program, a web page, a simple model, a circuit. You do not have to become an engineer. But the brief experience of trying to build something, watching it not work, and figuring out why will give you intuitions that no amount of reading can. The tools to try this, right now, are better and cheaper and more accessible than at any moment in history. Many of them are free.

Be skeptical of easy stories. Especially the easy stories about how everything is terrible, or how everything is won-

derful. The world is almost always more complicated than that. The thing to distrust most is the feeling of certainty — in yourself or in others — that a complicated question has a simple answer.

Take care of your attention. Your attention is one of the most valuable resources you will ever have. You read in Chapter 16 about how many companies are designed to capture it. They are very good at it. The choices you make about how much of your time you give to screens, and what you look at when you do, are some of the most important choices you will make. They are also choices the people who design the screens do not want you to think about. So think about them.

Find mentors. Not famous people, necessarily. The most useful mentors are often older people already in your life — relatives, neighbors, teachers, family friends — who happen to know something well, and who have time to talk. Most of them, if you ask them real questions, will be delighted to answer. People underestimate how much of human knowledge is still transmitted, in every era, by an older person sitting down with a younger one and explaining something patiently. The internet has not replaced this. AI has not replaced this. Don't let yourself be cheated of it.

Be patient with yourself. You are going to grow up in a world where artificial intelligence can do, in seconds, things that took the people in this book entire careers to do. This will, sometimes, make you feel that your own slower learning

is pointless. It is not pointless. The work you do, slowly, by yourself, builds something inside you that is yours forever, in a way that an AI's output cannot be. Do not let the speed of machines make you feel that your pace of becoming a person is too slow. It is exactly the right speed.

TAKE CARE

I will, almost certainly, live to see most of it — into the second half of this century. You'll see all of it, in real time, with your own life happening inside the answer.

Pay attention. Stay curious. Read carefully. Be skeptical of easy stories — including, especially, the easy story that everything is terrible.

The world that produced this book — me, on my couch, in 2026, working with a system that did not exist when I was your age, writing for kids I'll meet (and, in some sense, already have) — was unimaginable, in detail, even five years before it happened.

The world that will produce your own life will be, in turn, unimaginable to me.

I am glad you are going to be in it.

Take care of yourself. Take care of the people around you. Take care of the world you have been handed.

That's the whole book.

With love, Meira.